Queensland Health

Queensland Community Pharmacy

Acute Exacerbations of Mild Plaque Psoriasis - Clinical Practice Guideline

Service restricted to patients aged 16 years or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
  - Aggravating factors
- Examination
  - Clinical features and severity (PASI & DLQI)

Develop management plan

- Non-pharmacological measures
- Pharmacotherapy
  - Topical treatments

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation

Communicate agreed management plan

- Medication counselling
- General advice and patient information
- When to seek further care and/or treatment
- Communication with other health practitioners

Clinical review

- Response to treatment
- Continue, modify or stop treatment
- Communication with other health practitioners

Refer when

- Unclear diagnosis
- Psoriatic lesions are infected
- Psoriasis is moderate or severe
- Symptoms indicate a type of psoriasis other than mild plaque psoriasis
- The face, scalp, genitals, palms and/or soles are affected
- Inadequate response to optimal topical treatment or the condition reoccurs

Treat and concurrently refer

- The patient has psoriatic comorbidities or risk factors that require management
- The patient has not seen an appropriate healthcare provider for review of their condition in the previous 12 months
- The patient is taking a medicine that can exacerbate psoriasis
- Quality of life is significantly affected
- The patient is planning a pregnancy or is pregnant
- The patient is immunocompromised

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

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Refer when

Refer the patient to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- unclear diagnosis
- psoriatic lesions are infected
- psoriasis is moderate or severe
- symptoms indicate a type of psoriasis other than mild plaque psoriasis
- the face, scalp, genitals, nails, palms and/or soles are affected
- inadequate response to optimal topical treatment.

Treat (if clinically appropriate) and concurrently refer

Provide initial treatment and concurrently refer to an appropriate healthcare provider in the following situations:

- the patient has psoriatic comorbidities or risk factors that require management (e.g., arthritis, risk of venous thromboembolism, depression, increased alcohol consumption, signs of lymphoma, skin cancers and solid tumours)
- the patient has not seen an appropriate healthcare provider for review of their condition in the previous 12 months
- the patient is taking a medicine that can exacerbate psoriasis
- quality of life is significantly affected
- the patient is planning a pregnancy or is pregnant
- the patient is immunocompromised.

Key points

- People presenting with an acute exacerbation of mild plaque psoriasis (psoriasis vulgaris) may receive treatment within the Service. Refer people with all other presentations to an appropriate healthcare provider.
- There is no cure for psoriasis. Treatment aims to clear lesions and manage symptoms [1]. Most cases of mild plaque psoriasis can be treated with topical therapies [2].
- Psoriasis is a systemic autoimmune disease that is associated with an increased risk of hypertension, obesity, elevated lipid levels, heart disease, diabetes, inflammatory bowel disease, lymphoma and depression [3]. Where appropriate, consider screening for cardiovascular risk and possible eligibility for the Queensland Community Pharmacy Cardiovascular Disease Risk Reduction Program.
- Psoriasis can have marked functional, psychological and social impacts, and this does not always correlate with severity [4]. Psoriasis occurring on the face, nails, scalp, genitals, flexures and soles is considered to be severe because these areas are difficult to treat and associated with poor cosmetic outcomes [5].

Patient history and assessment

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- pregnancy and lactation status (if applicable)
- onset, duration, nature, location, severity and extent of the rash and other symptoms
- previous diagnosis of plaque psoriasis and any current or past management plan
- underlying and associated medical conditions, including psoriatic comorbidities such as arthritis (or joint pain) and common comorbidities such as hypertension, hyperlipidaemia, obesity, type 2 diabetes and depression
- response to any previous treatment/s
- impacts on quality of life and psychosocial wellbeing
- exposure to factors that can aggravate psoriasis (see Table 1)
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)
- family history of psoriasis
- allergies/adverse drug reactions.

Table 1. Factors that can aggravate psoriasis [3, 6, 7]

- streptococcal tonsillitis (strep throat) and other infections e.g., viral
- skin trauma and injuries such as cuts, abrasions, sunburn
- sun exposure (although gentle exposure is often beneficial)
- dry skin
- obesity
- metabolic factors (calcium deficiency)
- smoking
- sleep deprivation and emotional stress
- excessive alcohol consumption
- hormonal factors (pregnancy or postpartum)
- medicines such as lithium, beta-blockers, antimalarials, nonsteroidal anti-inflammatories, antibiotics, ACE inhibitors, terbinafine and TNF-α inhibitors
- stopping oral steroids or potent topical corticosteroids (TCS)
- other environmental factors such as a stressful event.

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination

- Physical examination of the patient's skin is required to identify, assess and classify the severity of an acute exacerbation of mild plaque psoriasis.
- In darker skin tones, plaques are generally darker or violet in colour, thicker, and with more obvious scale and itch [4].
- Nail changes including pitting, lifting of the nail (onycholysis) and subungual hyperkeratosis may also be observed, although it may be difficult to distinguish nail changes due to psoriasis from fungal nail infections [1, 4].

Assessment of severity

A Psoriasis Area and Severity Index (PASI) score and Dermatology Life Quality Index (DLQI) score may be used to assess severity and responses to treatment [3, 8, 9]:

- mild, or mild to moderate plaque psoriasis = a PASI ≤10 and DLQI ≤10 [9]
- moderate to severe plaque psoriasis = PASI >10 and/or DLQI >10 [9]
    - The psoriasis is always considered severe when the DLQI score is >10, regardless of the PASI score [9].
    - It may also be considered moderate to severe if it has significant impacts on the patient's quality of life, due to involvement of visible areas, major parts of the scalp, genitals, palms and/or soles, onycholysis of at least 2 fingernails or pruritus leading to excoriation [9, 10].

Refer to *'Pharmacist resources'* section for more information about PASI and DLQI scores.

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Pharmacist management of acute exacerbations of mild plaque psoriasis involves:

- **non-pharmacological measures:**
    - Education and advice regarding lifestyle modification, use of adjunctive agents including emollients/moisturisers, and other measures as per the Therapeutic Guidelines: Dermatology (Psoriasis - Advice for managing psoriasis in primary care) and the Australian Medicines Handbook: Psoriasis [11, 12].
- **pharmacotherapy:**
    - A topical treatment for the management of **acute exacerbations of mild plaque psoriasis** in accordance with the Therapeutic Guidelines: Dermatology (Psoriasis) [11].

Confirm management is appropriate

Consult the *Therapeutic Guidelines, Australian Medicines Handbook* and other resources to confirm that management is appropriate, including for:

- contraindications and precautions

- drug interactions
- pregnancy and lactation.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- the typical cycle of psoriasis and the expectations of treatment
- individual product and medicine
- non-pharmacological measures
- how to manage adverse effects
- when to seek further care and/or treatment, including recognising infection
- when to return for follow-up.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources and information provided to patients (and/or parents/caregivers) and compliance with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

Patient resources

- NPS MedicineWise factsheets:
  - Topical treatments for your plaque psoriasis
  - Plaque psoriasis: my options when topical treatments aren't enough
- Skin Health Institute: Psoriasis patient information

General advice

At the time of initial consultation, patients (and/or parents/caregivers) should be advised to see an appropriate healthcare provider if:

- signs or symptoms worsen
- new signs or symptoms develop (including systemic signs or symptoms)
- the face, scalp, genitals, palms and/or soles become affected
- they do not respond adequately to recommended treatment:
  - a new treatment may take time to work and should be trialled for 2-4 weeks. However, adequate response to optimal topical treatment may take up to 3-6 months to be achieved
- quality of life is significantly affected by symptoms
- the condition reoccurs after initial resolution.

Common adverse effects of TCS including transient burning, stinging or pain on application, can generally be reversed by stopping the medicine. Refer to an appropriate healthcare provider when adverse effects cannot be managed in the pharmacy setting.

Inconsistent use of medication and insufficient duration of treatment is a common reason for treatment failure. The patient should be advised that most treatments take between 6-12 weeks of consistent and correct use to start to be visibly effective [4].

Clinical Review

Clinical review with the pharmacist (or the patient's chosen healthcare provider) should occur in line with recommendations in the *Therapeutic Guidelines*.

Patients should return for a follow-up consultation **2–4 weeks** after initiation of a new therapy and/or changes to the treatment plan. Clinical review should consider:

- response to treatment (and compliance)
- if changes are required to the treatment plan.

If a good response has been achieved, the medication can be reduced or stopped, and the patient can continue using regular moisturiser only. If there has been an inadequate response, the pharmacist should check the patient's compliance with the medication and skin care measures, ensuring enough medication is applied [11].

Pharmacotherapy for the management of psoriasis may be required longer-term; pharmacists should refer the patient to an appropriate healthcare provider for review and ongoing management after the acute flare has been managed. An annual review by an appropriate healthcare provider is recommended to assess for adverse effects from TCS therapy [3, 6].

For patients continuing therapy, follow-up should be individualised to the patient's clinical presentation, response to treatment, and therapeutic goals.

Pharmacists should prescribe a quantity of medicine (including repeats), that aligns with the expected duration of therapy, ensuring sufficient supply until the patient's next review. The total quantity prescribed should not exceed 12 months' supply (including repeats).

Pharmacist resources

- Therapeutic Guidelines: Dermatology
  - Psoriasis
  - Application and quantity of topical corticosteroids
  - Considerations in the use of topical corticosteroids
- Australian Medicines Handbook:
  - Drugs for psoriasis
  - General principles: topical treatment of skin conditions
  - Corticosteroids (skin)
  - Topical steroids – how much do I use?

- Australian Pharmaceutical Formulary and Handbook (APF): Treatment guideline for pharmacists (Psoriasis mild plaque psoriasis)
- DermNet NZ:
    - PASI score
    - Psoriasis
- Australian Government:
    - PASI calculation and body diagram – face, hand and foot form
    - PASI calculation and body diagram – whole body form
- Cardiff University: Dermatology Life Quality Index
- MSD Manual (Professional version): Psoriasis
- Journal of Clinical Medicine: Topographic Differential Diagnosis of Chronic Plaque Psoriasis: Challenges and Tricks
- Skin Deep: An open-access bank of high-quality photographs of medical conditions in a wide range of skin tones for use by both healthcare professionals and the public.
- Mayo Clinic: Types of psoriasis (psoriasis pictures)
- NPS MedicineWise: Plaque psoriasis
- The Australasian College of Dermatologists:
    - Consensus statement: Treatment goals for psoriasis
    - A-Z of skin - Psoriasis
    - Taking care of skin: How to recognise and respond to health issues in Aboriginal and Torres Strait Islander Health Peoples (an online course/ teaching resource for Aboriginal Health Workers).
- Australian Journal of General Practice: The multiple comorbidities of psoriasis.

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Version 1.0	01.02.2024	
Version 2.0	01.07.2025	Updated to reflect current clinical guidance, revised referral criteria, and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy

Acute Minor Wound Management - Clinical Practice Guideline

Service restricted to patients aged 5 years or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
- Wound history, assessment and examination
- Assess need for pathology testing for Gram staining and culture

Develop management plan

- Administration of local anaesthetic (if required)
- Wound care
- Pharmacotherapy:
  - Presumptive antibiotic therapy for bite wounds (including marine animal bites) and clenched-fist injuries
  - Prophylactic antibiotic therapy for significantly contaminated traumatic wounds
  - Empirical antibiotic therapy for localised post-traumatic wound infections, localised bite and clenched-fist wound infections and localised water-immersed wound infections
  - Tetanus post-exposure prophylaxis with tetanus toxoid vaccine

Confirm management is appropriate

- Contraindications, precautions and interactions
- Appropriate antibiotic use

Communicate agreed management plan

- Wound care and general advice
- Medication counselling
- When to seek further care and/or treatment
- Communication with other health practitioners

Clinical review

- Response to treatment
- Dressing changes and suture removal (if necessary)
- Communication with other health practitioners

Refer when

- Burns that require referral to a specialist centre as per the Australian and New Zealand Burn Association (ANZBA) criteria
- Injury has suspected vascular, tendon, joint, bone, muscle or nerve involvement
- Fractures, crush injuries, partial amputations of a digit or penetrating wounds of the torso
- A penetrating wound, including through footwear, or a marine envenomation
- Recent orofacial trauma
- Suspected concussive head injury
- Pain that requires management with treatments not available to pharmacists
- Immunocompromised or at high risk of infection or compromised wound healing
- Signs or symptoms of systemic infection or is generally unwell
- Infection of a surgical or chronic wound
- A tetanus-prone wound that requires tetanus immunoglobulin
- Significant risk of suicide, or self-inflicted injuries (see '*Treat and concurrently refer*')
- Injury sustained in the workplace or as a result of a work-related incident
- The wound history is unclear
- The wound is showing abnormal signs of healing or not progressing through the normal stages of healing
- The wound is suspected to be full thickness
- The wound is to the head (scalp, face, ears, eyes) or neck, or where sustained closure is difficult to achieve
- Anaesthesia (if indicated) cannot be achieved for adequate cleansing or closure

Treat and concurrently refer

- The wound is significantly contaminated, e.g. a foreign body or other contamination that cannot be removed by the pharmacist, significantly contaminated water-immersed wounds
- Localised post-traumatic wound infections, including
  - localised bite and clenched-fist wound infections
  - localised water-immersed wound infections
- Self-inflicted injuries

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Provide first aid and concurrently refer the patient to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- burns require management at a specialist centre as per the Australian and New Zealand Burn Association (ANZBA) criteria [1]
- injury has suspected vascular, tendon, joint, bone, muscle, or nerve involvement
- fractures, crush injuries, partial amputations of a digit or penetrating wounds of the torso
- a penetrating wound, including through footwear, or a marine envenomation (a complex wound at high risk of infection)
- recent orofacial trauma (not yet seen by a dental practitioner)
- suspected concussive head injury
- pain that requires management with treatments not available to pharmacists
- immunocompromised or at high risk of infection or compromised wound healing
- signs or symptoms of systemic infection or is generally unwell
- infection of a surgical or chronic wound
- a tetanus-prone wound that requires tetanus immunoglobulin (e.g. patient has had less than 3 doses of the tetanus toxoid vaccine, or vaccination history is unknown)
- significant risk of suicide (e.g. a current intent, a specific plan, access and means or have had previous suicide attempts), or self-inflicted injuries (see '*Treat and concurrently refer*')
- injury sustained in the workplace or as a result of a work-related incident
- the wound history is unclear (how, where and when the wound was acquired)
- the wound is showing abnormal signs of healing or not progressing through the normal stages of healing (e.g., chronic ulcers or wounds, necrotic or hyper-granulation tissue, wounds older than 4 days that have not begun epithelialisation)
- the wound is suspected to be full thickness (involving epidermis, dermis and subcutaneous tissue)
- the wound is to the head (scalp, face, ears, eyes) or neck, or where sustained closure is difficult to achieve
- anaesthesia (if indicated) cannot be achieved for adequate cleansing or closure.

Treat (if clinically appropriate) and concurrently refer

Provide initial treatment (wound care and antibiotic therapy if indicated) and concurrently refer to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- the wound is significantly contaminated, e.g., a foreign body or other contamination that cannot be removed by the pharmacist, significantly contaminated water-immersed wounds
- localised post-traumatic wound infections, including:
    - localised bite and clenched-fist wound infections
    - localised water-immersed wound infections
- self-inflicted injuries.

Key points

- Comprehensive assessment that includes wound factors (e.g. site, type, size, onset, contamination, infection, drying/maceration) and patient factors (e.g. age, medical conditions, medicines, nutrition, lifestyle) is required to determine appropriate management, including whether referral to an appropriate healthcare provider is required.
- Wound healing is optimised when the wound is cleansed to remove debris and devitalised tissue, kept moist, infection is prevented, and the wound has adequate blood supply [2].
- Topical antibiotics are not recommended for the treatment of wounds and are not included for use as part of this Service. Topical antibiotics do not improve healing potential and may contribute to antimicrobial resistance [2, 3].
- Antibiotic prophylaxis for acute minor wounds is not routinely recommended. Thorough cleansing and dressing are the mainstay of treatment [4].

Patient history and assessment

A comprehensive and holistic wound assessment must consider both patient and wound history [2, 5, 7].

For the purposes of the Service, an acute minor wound is defined as a wound:

- caused by injury or trauma (as opposed to surgical wounds)
- involving the epidermis and upper dermis (superficial wound) or dermis (partial thickness) [3, 8]
- in the inflammation stage (approximately 0–4 days post injury) and expected to follow the normal physiological wound healing process.

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- weight
- pregnancy and lactation status (if applicable)
- current medical conditions, particularly conditions that may increase risk of infection (immunocompromised) or impair wound healing (e.g., vascular disease, diabetes mellitus, autoimmune disorders, malignancy)
- factors that may impact on successful wound healing as per the Therapeutic Guidelines: Ulcer and Wound Management (Factors affecting ulcer and wound healing) [2]
- symptoms that may indicate systemic infection or other injury including fever, lymphadenopathy, nausea and vomiting
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines), including medicines that may delay wound healing
- previous wound healing and scarring

- allergies / adverse drug reactions
- alcohol and drug history
- smoking status
- immunisation status including tetanus vaccination
- risk factors for infection with methicillin-resistant *Staphylococcus aureus* (MRSA)
- psychosocial history, emotional and mental state including level of distress, history of mental health conditions and potential for family violence
- ability to undertake appropriate self-care
- nutritional and hydration status.

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Wound history

Consider:

- mechanism of injury:
    - to assess the potential for retained foreign bodies
    - to assist in determining if presumptive, prophylactic or empirical antibiotic therapy is required
- date and time of injury
- environment where the wound was acquired e.g., fresh or salt water, muddy locations, contaminated soil
- treatments already provided and response e.g., dressings, antiseptics, antibiotics
- whether the injury was sustained in a workplace or related to employment
- whether the injury may be associated with family violence or self-harm (see below).

Injuries associated with family violence or non-accidental injury in children

Indications of injuries associated with family violence may include injuries and/or bruising to the head, neck, face, areas usually hidden from view, bite marks, unusual burns, injuries that do not match the history/explanation given by the patient (and/or parent/caregiver) or no explanation can be provided [9].

If the pharmacist suspects the patient has been subjected to family violence, they should provide appropriate support and assistance to the patient (separate from other family members), including referral to support options. A list of services is provided on the Queensland Government Supports for victims of abuse webpage [10].

Pharmacists are strongly encouraged to voluntarily report suspected child abuse and neglect to Child Safety. Information on the reporting process and other supports for affected families can be found on the Queensland Government Child Safety webpage [11].

Injuries associated with self-harm

The most common form of deliberate self-harm is self-poisoning, followed by self-cutting and other methods such as hanging, burns, bruising and picking at sores [12]. It is often done in secret and on places of the body that may not be seen by others (e.g., thighs). People may try to conceal wounds caused by self-harm by wearing long-sleeve clothing, or high neck shirts [12].

Pharmacist assessment of a patient presenting with self-inflicted injuries, or a suspected self-inflicted injury, should consider:

- the patient's emotional and mental state and level of distress
- whether there is an immediate concern related to the patient's safety
- the need for referral to a mental health service [13].

Self-harm may be a repetitive behavior [12]. People presenting for treatment of self-inflicted injuries must be concurrently referred to an appropriate healthcare provider or mental health service. They may be provided treatment if clinically appropriate.

Self-inflicted injuries are a warning sign that indicate a patient might be at a heightened risk of suicide. Consider key questions and/or other risk factors for suicide risk in accordance with Therapeutic Guidelines: Psychotropic (Suicide risk - Assessing suicide risk) [14].

Patients at significant risk of suicide (e.g., if they have a current intent, a specific plan, access and means or have had previous suicide attempts) require urgent referral to an appropriate healthcare provider and/or acute psychiatric/mental health service.

Helplines and support services for people at risk of suicide include:

- Beyond Blue (1300 224 636)
- for young people, e-headspace (1800 650 890)
- Kids Helpline (1800 551 800)
- Lifeline (13 11 14)
- Suicide Call Back Service (1300 659 467) [14].

Wound assessment and examination

Wound-related factors to be considered and documented during clinical assessment and at clinical review (if applicable) are detailed in Table 1.

Conduct assessment of vital signs for any patients presenting with signs of infection.

The TIME acronym (Tissue, Infection/Inflammation, Moisture balance, Edge and peri-wound area) as detailed in the Therapeutic Guidelines: Ulcer and Wound Management (Assessing patients with an ulcer or wound) may also be used to guide wound examination and documentation of characteristics [2].

Table 1. Clinical acute wound assessment [2, 3, 5, 6, 15]
Anatomical location
• Consider possible damage to joint, tendon, nerve and vascular structures. • Check distal sensation, motor function, range of motion (extensors and flexors) and vascularity.

Size and degree of tissue loss
• Length, circumference and width. • Approximate depth – measured with a damp cotton tip, although generally not required for a minor acute wound. • Tissue loss – superficial, partial or full thickness.
Wound appearance
• Presence of wound contamination – consider if the wound can be adequately cleaned, including with simple debridement, in the community pharmacy setting, or if referral is required. Consider if a foreign body is present and refer if needed. • Stage/phase of healing – inflammation, reconstruction/proliferation, maturation. • Wound bed clinical appearance (if visible) – granulation tissue, epithelial tissue, sloughy, necrotic, hyper-granulation tissue. • Wound edges and surrounding skin (colour, contraction, change in sensation) – signs of normal or abnormal healing. • Exudate – type, amount, colour and odour.
Signs of infection
• Local infection – redness, heat, pain and swelling above normal inflammatory responses, presence of exudate, odour, impaired wound healing, contact bleeding, tissue breakdown, unhealthy granulation tissue. • Spreading infection – increasing wound size, lymphangitis, inflammation and swelling of lymph nodes, wound breakdown, patient malaise or deterioration in health. • Systemic infection – fever, tachycardia, hypotension, other signs of sepsis.
Pain
• Timing, triggers, severity and impact. • Type of pain – background, breakthrough pain, precipitating factors, pain before during and/or after dressing, cleaning and exposure to air. • Response to any medication used to manage the wound.

Pathology testing for Gram stain, culture and sensitivity testing

Infected wounds

Wound swabbing is not routinely recommended for uncomplicated wounds. Pathology testing for microbial Gram stain, culture and antibiotic sensitivity should be considered prior to commencing empirical antibiotic therapy, where:

- the result of testing will help to guide management
- the wound is not responding to empirical treatment
- there is suspicion of unusual or resistant pathogens
- there are concerns regarding local antimicrobial resistant organisms.

If pathology testing is indicated, the patient should be referred to an appropriate healthcare provider to conduct specimen collection. If timely referral is not practicable, then empirical

antibiotic therapy may be provided in accordance with the *Therapeutic Guidelines* with concurrent referral (see *Pharmacotherapy*).

Wounds at high risk of infection

Pathology testing for microbial Gram stain, culture and antibiotic sensitivity is not required before commencing presumptive or prophylactic antibiotic therapy for wounds at high risk of infection [4].

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Pharmacist management of acute minor wounds may involve:

- **administration of local anaesthetic:**
  - Use of ≤1% lidocaine preparations (without adrenaline, maximum of 10ml) if required, in accordance with the current online version of
    - Therapeutic Guidelines: Pain and analgesia (Drugs used for pain - Local anaesthetics for acute pain management) [16]
    - Queensland Health and Royal Flying Doctor Service (Queensland Section) Primary Clinical Care Manual [3]
- **wound cleansing**:
  - Irrigation, simple mechanical debridement (using sterile gauze or non-woven swabs) and use of antiseptics in accordance with the:
    - Therapeutic Guidelines: Ulcer and wound management [2]
- **wound closure:**
  - Closure using adhesive strips, topical skin/tissue adhesives and basic suturing techniques only in accordance with the:
    - Queensland Health and Royal Flying Doctor Service (Queensland Section) Primary Clinical Care Manual [3]
- **wound dressing:**
  - Application and changing of dressings in accordance with the:
    - Therapeutic Guidelines: Ulcer and wound management [2]
- **pharmacotherapy:**
  - Presumptive **oral** antibiotic therapy for bite wounds (including marine animal bites) and clenched-fist injuries at high risk of infection, in accordance with the:
    - Therapeutic Guidelines: Antibiotic (Traumatic wound infections – Bite wound infections, including clenched-fist injury infections) [4]
    - Therapeutic Guidelines: Antibiotic (Traumatic wound infections – Water-immersed wound infections) [4]
  - Prophylactic **oral** antibiotic therapy for significantly contaminated traumatic wounds, in accordance with the:
    - Therapeutic Guidelines: Antibiotic (Traumatic wound infections – Post-traumatic wound infections) [4]
    - Therapeutic Guidelines: Antibiotic (Traumatic wound infections – Water-immersed wound infections) [4]

- Empirical **oral** antibiotic therapy for localised post-traumatic wound infections, localised bite and clenched-fist wound infections, and localised water-immersed wound infections, in accordance with the:
  - Therapeutic Guidelines: Antibiotic (Traumatic wound infections – Post-traumatic wound infections) [4]
  - Therapeutic Guidelines: Antibiotic (Traumatic wound infections – Bite wound infections, including clenched-fist injury infections) [4]
  - Therapeutic Guidelines: Antibiotic (Traumatic wound infections – Water-immersed wound infections) [4]
- Tetanus post-exposure prophylaxis[NB1] with tetanus toxoid vaccine in accordance with the:
  - Therapeutic Guidelines: Antibiotic (Traumatic wound infections – Post-traumatic wound infections) [4]

NB1: Tetanus-toxoid vaccines must be given in accordance with the Australian Immunisation Handbook: Tetanus and recorded on the Australian Immunisation Register [16].

Confirm management is appropriate

Consult the *Therapeutic Guidelines*, *Australian Medicines Handbook*, Queensland Health and Royal Flying Doctor Service (Queensland Section) Primary Clinical Care Manual [3] and other resources to confirm that management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

Reminder

The prescribing and use of antibiotics should be consistent with the principles described in the Therapeutic Guidelines: Antibiotic. Pharmacists should carefully assess whether antibiotic therapy is clinically indicated, and if so, consider local antimicrobial resistance patterns and relevant clinical factors when selecting therapy.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information if required), should be provided to the patient regarding wound care. Relevant information may include:

- schedule for dressing changes and suture removal
- instructions for individual product/dressing and medicine use
- non-pharmacological measures
- how to recognise the signs of local and systemic infection
- when to commence antibiotic therapy and any required follow-up
- measures to prevent further injury
- how to manage adverse effects of pharmacotherapy
- when to return for follow-up.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources and information provided to patients (and/or parents/caregivers) and compliance with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

General advice

At the time of initial consultation, patients (and/or parents/caregivers if applicable) should be advised to see an appropriate healthcare provider if:

- the wound becomes infected (for non-infected wounds) or deteriorates
- the wound is not healing as expected (e.g. no evidence of epithelialisation 4 days post wound event)
- their pain is not well managed with recommended analgesia
- they become systemically unwell.

Reiterate to the patient (and/or parents/caregivers) how to care for the wound and/or dressing at home as appropriate (e.g. when showering, bathing).

Clinical review

Clinical review with the pharmacist (or the patient's chosen healthcare provider), including for dressing changes and suture removal, should occur in line with recommendations in the *Therapeutic Guidelines* and the Queensland Health and Royal Flying Doctor Service (Queensland Section) Primary Clinical Care Manual [3].

Pharmacists should prescribe a quantity of medicine that aligns with the expected duration of therapy. Advise patients (and/or parents/caregivers) to see an appropriate healthcare provider if the condition does not improve or resolve (see *Refer when* and *General advice*).

Pharmacist resources

- Australian and New Zealand Burn Association: ANZBA Referral Criteria
- Therapeutic Guidelines:
  - Ulcer and wound management
  - Pain and analgesia: Local anaesthetics for acute pain management
  - Antibiotic: Traumatic wound infections
  - Antibiotic: Principles of antimicrobial use
- Australian Pharmaceutical Formulary and Handbook (APF):
  - Wound management
- Australian Medicines Handbook:
  - Drugs for local anaesthesia
  - Antibacterials

- DermNet NZ:
  - Wounds
  - Synthetic wound dressings
  - Topical skin adhesives
  - Suturing techniques
  - Surgical wound closure
- Wounds Australia: Australian Standards for Wound Prevention and Management
- Queensland Health and Royal Flying Doctors Service (Queensland branch) - Primary Clinical Care Manual 12th edition 2025
- Australian Immunisation Handbook: Tetanus
- The Royal Children's Hospital Melbourne:
  - Wound assessment and management
  - Burns - Acute management
  - Wound dressings – acute traumatic wounds
  - Lacerations
- International Wound Infection Institute:
  - Wound infection in Clinical Practice 2022
  - Therapeutic wound and skin cleansing 2025
- Australian Prescriber: An update on wound management
- Frontiers in Pharmacology: First-Line Interactive Wound Dressing Update: A Comprehensive Review of the Evidence
- Wound Practice and Research - Wounds Australia Journal: The negative impact of medications on wound healing.
- Royal Australian College of General Practitioners:
  - Selecting appropriate dressings
  - Acute lacerations: Assessment and non-surgical management
  - Surgical management of acute lacerations

Document version number	Date	Comments
Version 1.0	01.02.2024	
Version 1.1	25.03.2024	Updated Pathology testing for Gram stain, culture and sensitivity testing section with requirement to refer patients for pathology testing for Gram stain, culture and sensitivity testing section to a medical practitioner.
Version 1.2	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance, revised referral criteria, and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy

Acute Nausea and Vomiting associated with gastroenteritis - Clinical Practice Guideline

Service restricted to patients aged 18 years or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
  - Consider possible causes of nausea and vomiting
- Examination
  - Vital signs
  - Hydration status
  - Abdominal examination

Develop management plan

- Identify, treat or remove cause (if possible)
- Non-pharmacological measures for gastroenteritis
- Oral pharmacotherapy

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation

Communicate agreed management plan

- Medication counselling
- Supportive care, including the importance of rehydration
- General advice and patient information
- When to seek further care and/or treatment

Clinical review

- Response to treatment
- Communication with other health practitioners

Refer when

- Prolonged vomiting (>24–48 hours)
- Patient looks very unwell or is very drowsy
- Significant recent weight loss
- Abdominal distension or tenderness
- Rectal bleeding
- Green, bile or blood vomit
- Fever, neck stiffness, confusion
- Severe headache, altered or loss of consciousness
- Isolated vomiting without nausea
- History of head injury/trauma
- Increased blood ketone and glucose levels
- Diabetic or is immunocompromised
- Recent overseas travel
- Nausea and vomiting are post-operative, radiation-induced or suspected to be drug or medicine-induced
- Parenteral administration of an antiemetic is indicated
- The nausea and vomiting is chronic (symptoms for >4 weeks)
- The patient has recurring episodes of acute nausea and vomiting
- Inadequate response to optimal treatment
- Signs of severe dehydration
- Severe pain, or nausea and vomiting occur with chest or back pain

Treat and concurrently refer

- The patient is pregnant
- Labyrinthitis is suspected

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Provide supportive management (hydration) and refer the patient to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- prolonged vomiting (>24–48 hours)
- patient looks very unwell or is very drowsy
- significant recent weight loss
- abdominal distension or tenderness
- rectal bleeding
- green, bile or blood (coffee grounds) vomit
- fever, neck stiffness, confusion
- severe headache, altered or loss of consciousness
- isolated vomiting without nausea
- history of head injury/trauma
- increased blood ketone and glucose levels., including if ketones are 0.6 mmol/L or higher
- diabetic or is immunocompromised
- recent overseas travel
- nausea and vomiting are post-operative, radiation-induced or suspected to be drug or medicine-induced
- parenteral administration of an antiemetic is indicated
- nausea and vomiting is chronic (symptoms for >4 weeks) [1]
- the patient has recurring episodes of acute nausea and vomiting
- inadequate response to optimal treatment
- signs of severe dehydration e.g. reduced or no urine output, intense thirst, marked loss of skin turgor, sunken eyes, tachycardia, severe hypotension, increased respiration
- severe pain, or nausea and vomiting occur with chest or back pain.

Treat (if clinically appropriate) and concurrently refer

Provide initial treatment and concurrently refer to an appropriate healthcare provider in the following situations:

- the patient is pregnant
- labyrinthitis is suspected (vertigo, nystagmus with symptoms worsened by motion, tinnitus).

Key points

- It is essential to consider the clinical context when assessing patients with nausea and vomiting, including the possibility of life-threatening causes, before developing a management plan [2,3].

- Causes of acute nausea and vomiting can stem from the gastrointestinal (GI) tract, the central nervous system (CNS), or result from a systemic condition, and can often be attributed to an identifiable cause, most commonly acute gastroenteritis, food poisoning or other toxins, migraine, motion sickness or pregnancy [2, 4].
- An oral antiemetic may be considered for symptom relief if the nausea and vomiting is suspected to be caused by gastroenteritis, although spontaneous improvement of acute nausea and vomiting is likely [2].

Patient history and assessment

The assessment approach focuses on identifying the cause (or excluding significant underlying diseases) with a view to expectant management or directing specific treatment.

Patient history

Obtain sufficient information from the patient to identify the likely cause of the nausea/vomiting and assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- pregnancy and lactation status (if applicable)
- nature, severity and frequency of symptoms
- food and/or drink consumed in the last 24 hours
- onset and duration of symptoms, including timing in relation to eating/drinking
- signs and symptoms of dehydration
- risk factors for dehydration or electrolyte abnormality (e.g. kidney transplantation, malnutrition, frailty)
- ability to adhere to treatment or monitoring for dehydration (e.g. intellectual disability)
- recent head trauma or injury
- similar symptoms in close contacts
- exposure to toxins, poisons, bites or stings
- recent travel
- accompanying symptoms e.g., chest pain or heartburn, headache, dizziness, weight loss, fever, abdominal pain, last bowel motion, diarrhoea, constipation, dysuria, or frequency of urination
- precipitating and relieving factors
- underlying medical conditions e.g., immunocompromised, diabetes or recent surgery
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements, and over-the-counter medicines)
- drug allergies/ adverse drug reactions
- medication tried to treat current symptoms
- alcohol and drug history.

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination

- Where appropriate, conduct assessment of vital signs.
- Assess hydration status as per the Therapeutic Guidelines: Gastrointestinal (Acute gastroenteritis - Assessment of acute gastroenteritis) [5].
- Examine for abdominal tenderness or distention, particularly areas where hernias commonly present (excluding the groin).

Consider possible causes of nausea and vomiting

A non-exhaustive list of possible causes of acute and chronic nausea and vomiting are presented in Appendix 1. When in doubt of the cause of the nausea and vomiting, the pharmacist should refer the patient to an appropriate healthcare provider.

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Before starting treatment, identify, treat or remove the cause of the nausea and vomiting (if possible), and ensure adequate hydration [6].

Pharmacist management of acute nausea and vomiting associated with gastroenteritis includes:

- **non-pharmacological measures:**
    - Oral rehydration therapy to prevent or correct dehydration, in accordance with the Therapeutic Guidelines: Gastrointestinal (Acute gastroenteritis – Rehydration for acute gastroenteritis in adults) [5].
- **pharmacotherapy:**
    - An oral medicine for the treatment of acute nausea and vomiting associated with gastroenteritis mentioned in the Therapeutic Guidelines: Gastrointestinal: Nausea and vomiting (Antiemetic drugs in adults) [2].

Confirm management is appropriate

Consult the *Therapeutic Guidelines*, *Australian Medicines Handbook* and other relevant resources to confirm that management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- individual product and medicine use
- non-pharmacological measures, including the importance of rehydration

- how to manage adverse effects
- the impact of vomiting on the absorption and effectiveness of other regular medications
- when to seek further care and/or treatment, including recognising patient deterioration and dehydration.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources and information provided to patients (and/or parents/caregivers) and to ensure compliance with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

General advice

At the time of initial consultation, patients (and/or parents/caregivers) should be advised to see an appropriate healthcare provider if:

- the condition does not improve or resolve within 24-48 hours of starting treatment
- the patient's condition worsens, including developing systemic signs and symptoms
- the patient develops new, or worsening, signs or symptoms, or signs of severe dehydration
- the patient's symptoms recur after initial resolution
- adverse effects of treatment cannot be managed in the pharmacy setting.

Clinical review

Clinical review is generally not required. Advise patients (and/or parents/caregivers) to see an appropriate healthcare provider if the condition does not improve or resolve (see *Refer when* and *General advice*).

Pharmacist resources

- Therapeutic Guidelines: Gastrointestinal
  - Nausea and vomiting
  - Acute gastroenteritis
- Australian Medicines Handbook:
  - Antiemetics
- MSD Manual (Professional version):
  - Nausea and vomiting
  - Acute abdominal pain
- SOMANZ: Position statement on the Management of Nausea and Vomiting in Pregnancy and Hyperemesis Gravidarum
- Australian Prescriber: Antiemetic drugs: what to prescribe and when

- Queensland Health and Royal Flying Doctor Service (Queensland branch): Primary Clinical Care Manual 12th edition 2025
- Food safety (Queensland):
    - Outbreak management
    - Make a food safety complaint online

Appendix 1 - Possible causes of nausea and vomiting[2, 4, 7-12]

Condition	Presentation and suggestive findings	Action
Gastrointestinal disorders		
Adynamic ileus	• Abdominal distention • Increased risk after surgery (up to 72 hours post-surgery) • Severe illness of electrolyte abnormality	Referral to an appropriate healthcare provider if suspected.
Bowel obstruction	• Abdominal distention, severe constipation and tympany to percussion • Vomiting is often bilious • History of abdominal surgery (visible surgical scars) or hernia	Immediate referral to an appropriate healthcare provider is required if suspected.
Gastroenteritis – viral or bacterial/ food poisoning	• Vomiting accompanied by diarrhoea • Normal abdomen when examined • Food poisoning apparent based on history	Self-limiting – May be managed under this guideline. For suspected food poisoning, consider notification to local public health unit or make a food safety complaint online.
Gastroparesis	• Vomiting of partially digested food within a few hours of consumption • Often occurs in people with diabetes or after abdominal surgery	Referral to an appropriate healthcare provider is required if suspected.
Hepatitis	• Mild to moderate nausea (sometimes vomiting) over many days • Jaundice • Malaise • Slight liver tenderness is sometimes present	Referral to an appropriate healthcare provider is required if suspected.
Peritonitis or other acute abdominal condition e.g., acute mesenteric ischemia, pancreatitis, appendicitis	• Severe abdominal pain that may be vague and nauseating (visceral), sharp and localised (somatic) or pain perceived distant to its source (referred) • Other warning signs may be present	Immediate referral to an appropriate healthcare provider is required if suspected.
Toxic ingestion including alcohol	• Generally apparent based on history	Immediate referral to an appropriate healthcare provider is required if suspected.
Central nervous system disorders		
Cannabis use (cannabis hyperemesis syndrome)	• Ongoing nausea, vomiting and indigestion that occurs with chronic cannabis use	Symptoms resolve after stopping cannabis use and may be relieved with a hot shower. Referral to an appropriate healthcare provider and other services if required.
Closed head injury or concussion	• Generally apparent based on history	Immediate referral to an appropriate healthcare provider is required if suspected.

Condition	Presentation and suggestive findings	Action
CNS haemorrhage	• Sudden onset of headache • Change in mental status • Meningeal signs including photophobia, neck stiffness and seizures may be present	Immediate referral to an appropriate healthcare provider is required if suspected.
CNS infection e.g., meningitis	• Gradual onset of headache • Change in mental status • Meningeal signs including photophobia, neck stiffness and seizures may be present • Petechial rash may be present	Immediate referral to an appropriate healthcare provider is required if suspected.
Increased intracranial pressure	• Headache • Change in mental status • Focal neurological deficits sometimes present	Immediate referral to an appropriate healthcare provider is required if suspected.
Labyrinthitis	• Vertigo, nystagmus with symptoms worsened by motion • Tinnitus sometimes present	Referral to an appropriate healthcare provider is required if suspected.
Migraine	• Headache, with or without aura or photophobia • Patient will usually have a history	Usual pharmacy care and referral to an appropriate healthcare provider.
Motion sickness	• Generally apparent based on history	Usual pharmacy care.
Systemic conditions		
Advanced cancer, chemotherapy or radiation exposure	• Apparent based on history	Referral to an appropriate healthcare provider is required.
Diabetic ketoacidosis (DKA)	• History of diabetes, although may occur without a known history of diabetes • Polyuria, polydipsia, hyperventilation, abdominal pain and/or impaired consciousness • Significant dehydration often present • Patients with diabetes should be advised to check their blood glucose levels and test ketones in blood or urine [9]	Immediate referral to an appropriate healthcare provider is required if this is suspected, including if ketones are 0.6 mmol/L or higher.
Drug adverse effect or toxicity	• Generally apparent based on history • Assess severity and probable cause	If nausea and vomiting is a known adverse effect, referral to the original prescriber is required. If suspected overdose or serious reaction, immediate referral to an appropriate healthcare provider is required.
Liver or renal failure	• Generally apparent based on history • Asterixis • Jaundice in advanced liver disease often present • Uremic odour in renal failure often present	Immediate referral to an appropriate healthcare provider is required if suspected.
Myocardial infarction	• Chest pain, which may radiate to the back, jaw, left arm, shoulders (or all these areas) • Shortness of breath, sweating, dizziness	Immediate referral to an appropriate healthcare provider is required if suspected.

Condition	Presentation and suggestive findings	Action
Severe pain	• Multiple causes and presentations	Immediate referral to an appropriate healthcare provider is required.
Addisonian (acute adrenal) crisis	• History of adrenal insufficiency, trauma, acute stress, illness or recent surgery, or abrupt cessation of glucocorticoid therapy • Symptoms of acute circulatory failure (hypotension, confusion) • Other symptoms may include weakness, loss of consciousness, dizziness, fever, sudden onset of pain in lower back or legs, high heart rate	Immediate referral to an appropriate healthcare provider is required.
Other causes		
Pregnancy	• Can occur at any time of the day or night. May be triggered by food or odours • Patient may not be aware they are pregnant	Treat (if clinically appropriate) and concurrently refer to an appropriate healthcare provider. Usual pharmacy care including non-pharmacological therapies may be recommended for mild to moderate N&V.
SARS-COV-2 infection	• Fever, fatigue, dry cough, shortness of breath • Nasal symptoms (congestion, rhinorrhoea)	Usual pharmacy care and referral to an appropriate healthcare provider.
Urinary Tract Infection	• Generally apparent based on history • Presence of dysuria, urinary urgency and frequency • Suprapubic tenderness or discomfort	Referral to an appropriate healthcare provider or an appropriate pharmacy service is required.

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Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy
Acute Otitis Externa - Clinical Practice Guideline

Service restricted to patients aged 2 years or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
- Examination
  - Ear examination
  - Vital signs
  - Tender or enlarged cervical lymph nodes
  - Mastoid area

Develop management plan

- Non-pharmacological measures
- Pharmacotherapy
  - Otic antibiotics
  - Oral analgesia

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation
- Appropriate antimicrobial use

Communicate agreed management plan

- Medication counselling
- General advice and patient information
- When to seek further care and/or treatment
- Communication with other health practitioners

Clinical review

- Response to treatment
- Communication with other health practitioners

Refer when

- Unclear diagnosis
- Complete occlusion of the ear canal
- Red, swollen and tender behind the ear
- Systemic symptoms
- Recent significant head or ear trauma
- Immunocompromised
- Diabetes
- Tympanic membrane (TM) 'red flags':
  - Crust/scab or granulation in attic region (discharge/odour may be present)
  - Severely retracted (sucked in) TM or retraction pocket in attic area of eardrum
  - Dull, white mass behind TM or chalky patches
  - Perforation or suspected perforation of the TM or near the edge of the TM
- Evidence of infection beyond the ear canal including inflammation to the pinna or folliculitis
- Inadequate response to optimal treatment
- Recurrent symptoms (at least 3 episodes in 6 months, or 4 episodes in 12 months)

Treat and concurrently refer

- The patient has severe symptoms (without complete occlusion of the ear canal) associated with intense pain

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/carer.

Refer when

Refer the patient to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- unclear diagnosis
- complete occlusion of the ear canal
- red, swollen and tender behind the ear (possible acute mastoiditis)
- systemic symptoms including fever, weakness, irritability, loss of appetite
- recent significant head or ear trauma
- immunocompromised
- diabetes
- tympanic membrane (TM) 'red flags':
    - crust/scab or granulation in attic region (discharge/odour may be present)
    - severely retracted (sucked in) TM or retraction pocket in attic area of ear drum
    - dull, white mass behind TM or chalky patches
    - perforation or suspected perforation of the TM or near the edge of the TM.
- evidence of infection beyond the ear canal including inflammation to the pinna or folliculitis
- inadequate response to optimal treatment
- recurrent symptoms (at least 3 episodes in 6 months, or 4 episodes in 12 months).

Treat (if clinically appropriate) and concurrently refer:

Provide initial therapy and concurrently refer to an appropriate healthcare provider (including emergency services, if required) in the following situation:

- patient has severe symptoms (without complete occlusion of the ear canal) associated with intense pain.

Key points

- Acute diffuse otitis externa (AOE) can present with generalised or diffuse inflammation of the external ear canal, often associated with infection [1, 2].
- Most cases of AOE, also known as swimmer's ear, are caused by bacterial infection. Fungal infection occurs less frequently than bacterial infection, often follows the prolonged use of antimicrobial ear drops, and is more common in patients with diabetes or immunocompromise and in humid or tropical areas. Mixed bacterial and fungal infections can occur [1-3].
- AOE can affect people of any age but is more common in children and adolescents aged between 5 and 14 years [4].
- Oral antibiotics should **not** be used for uncomplicated AOE. Patients that require assessment for oral antibiotics due to presence of complications e.g., systemic symptoms or evidence of infection beyond the ear canal, should be referred to an appropriate healthcare provider [1, 2, 5].

- Ear conditions are often very painful, particularly for young children, and it is important to approach with care [6]. If the patient will not allow for otoscopic examination due to pain, advice should be provided on appropriate analgesia and the patient referred to an appropriate healthcare provider.

Patient history and assessment

Diagnosis of AOE is based on clinical history and ear examination using an otoscope [1, 6]. A diagnosis of diffuse AOE requires the rapid onset (generally within 48 hours) of:

1. **symptoms of ear canal inflammation:**
    - ear pain (commonly severe), itching or a feeling of fullness in the ear (can be with or without hearing loss or jaw pain), pain in the ear canal and temporomandibular joint region intensified by jaw motion [5].

    AND

2. **signs of ear canal inflammation:**
    - tenderness of the tragus and/or pinna

    OR

    - diffuse ear canal oedema and/or erythema (with or without otorrhea, regional lymphadenitis, TM erythema or cellulitis of the pinna and adjacent skin) [5].

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- pregnancy and lactation status (if applicable)
- nature, severity and frequency of symptoms including pain, itch, discharge, hearing loss, a feeling of fullness and dizziness
- onset and duration of symptoms, including preceding illness
- history of head or ear trauma
- recent history of upper respiratory tract infection
- other presenting symptoms including temperature >38°C or <35.5°C, rash, increased respiratory rate/distress, dehydration and/or reduced urine output, runny nose, sore throat or cough
- family history of ear conditions and hearing loss, previous hearing tests
- delayed speech or language development
- previous history of AOE, otitis media (OM) or other ear conditions:
    - patient age at first episode of ear conditions
    - under the care of an ENT specialist or audiologist
    - ear surgery to insert tympanostomy tube (grommets)
    - prior treatment for AOE, otitis media or other ear conditions - when and what treatment [6].
- frequency and method of ear cleaning

- precipitating factors e.g., swimming (especially in a dam or creek), humid or tropical weather conditions, local trauma to the ear canal (e.g. vigorous cleaning/scratching), use of topical antimicrobial agents or exposure to other irritants, obstruction of the ear canal (e.g. cerumen, foreign bodies), use of hearing aids, use of ear plugs
- relieving factors
- underlying medical conditions e.g., immunocompromised (including diabetes), cochlear implant, craniofacial abnormalities, persistent hearing loss, dermatological conditions (e.g. psoriasis, dermatitis), herpes zoster (Ramsay Hunt syndrome)
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)
- previous radiation treatment
- drug allergies/adverse drug reactions
- immunisation status as per the *Australian Immunisation Handbook*.

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination

- Conduct assessment of vital signs.
- Examine for tender or enlarged cervical lymph nodes.
- Conduct visual examination and palpation of mastoid area.
- Check for other symptoms if required (nose, throat, cough) [6].

Ear examination using otoscopy

Physical examination including otoscopy of both ears (starting with the least painful ear) is required to differentiate between diffuse AOE, AOM and other ear conditions [1, 7, 8].

Ear examination using simple otoscopy should be conducted in accordance with the Queensland Health and Royal Flying Doctors Primary Clinical Care Manual [6]. An ear differential diagnosis flowchart is included, and features to assist in differentiating between bacterial and fungal infections [6].

If ear discharge is visible, the ear canal should be gently cleaned prior to performing ear examination, to remove any discharge [6, 9]. The Remote Primary Health Care Manual – Clinical Procedures Manual [9] and the Queensland Health and Royal Flying Doctors Primary Clinical Care Manual contain a step by step guide for dry mopping [6].

Visualisation of the TM can be difficult in the presence of extensive otorrhoea or cerumen, or when there is complete occlusion of the canal [1, 7].

In these cases, it can be difficult to distinguish AOE from purulent OM with a TM perforation; pain triggered by gently pulling on the pinna is suggestive of diffuse AOE [10].

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Pharmacist management of AOE involves:

- **non-pharmacological measures:**
    - Advice and education regarding keeping ears dry in accordance with the Therapeutic Guidelines: Antibiotic (Ear, nose and throat infections - Otitis externa) [1] and the Queensland Health and Royal Flying Doctors Primary Clinical Care Manual [6].
- **pharmacotherapy:**
    - Oral analgesia in accordance with the Therapeutic Guidelines: Pain and Analgesia (Pharmacological management of acute pain: Mild, acute nociceptive pain) [10].
    - Otic antimicrobial therapy in accordance with the Therapeutic Guidelines: Antibiotic (Ear, nose and throat infections - Otitis externa) [1].

Confirm management is appropriate

Consult the *Therapeutic Guidelines*, *Australian Medicines Handbook* and other relevant resources to confirm that management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

Reminder

The prescribing and use of antibiotics should be consistent with the principles described in the Therapeutic Guidelines: Antibiotic. Pharmacists should carefully assess whether antibiotic therapy is clinically indicated, and if so, consider local antimicrobial resistance patterns and relevant clinical factors when selecting therapy.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- individual product and medicine use
- non-pharmacological measures
- how to manage adverse effects
- when to seek further care and/or treatment (including recognising patient deterioration).

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources provided to patients (and/or parents/caregivers if applicable) and that they comply with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

Patient resources

- The Royal Children's Hospital Melbourne Fact sheet: Ear infections and glue ear [12].

General advice

AOE symptoms typically improve within 48-72 hours of starting treatment, however full resolution may take up to 2 weeks [13].

At the time of initial consultation, patients (and/or parents/caregivers) should be advised to see an appropriate healthcare provider if:

- there is no improvement in symptoms after 48–72 hours of treatment
- signs or symptoms worsen, or their condition deteriorates
- new signs or symptoms develop, including systemic symptoms
- otorrhea continues for ≥2 weeks.

Patients (and/or parents/caregivers if applicable) should be provided with advice regarding the prevention of AOE including:

- the use of over-the-counter ear drops containing drying agents such as acetic acid and isopropyl alcohol following water exposure (after current infection resolution, only if eardrum is intact) [1, 6, 10, 13]
- keeping the external ear canal dry and free of water e.g. using earplugs or a shower or bathing cap [1, 13]
- avoiding local trauma to the ear canal e.g. use of cotton buds or other implements [10, 13].

Clinical review

Clinical review is generally not required. Pharmacists should prescribe a quantity of medicine that aligns with the expected duration of therapy. Advise patients (and/or parents/caregivers,) to see an appropriate healthcare provider if the condition does not improve or resolve (see *Refer when* and *General advice*).

Pharmacist resources

- Therapeutic Guidelines: Antibiotic
  - Otitis externa
- Therapeutic Guidelines: Pain and analgesia
  - Pharmacological management of acute pain
  - Mild, acute nociceptive pain

- Australian Medicines Handbook
  - Otitis externa
  - Drugs for ear infections
  - NSAIDs
  - Drugs for pain relief
- MSD Manual (Professional version): External otitis (Acute)
- Otolaryngology Journal: Clinical Practice Guideline: Acute Otitis Externa
- Queensland Health and Royal Flying Doctors Service (Queensland branch): Primary Clinical Care Manual 12th edition 2025
- DermNet NZ: Otitis externa
- Remote Primary Health Care Manual: Clinical Procedures Manual
- McGovern Medical School: Ear disease photo book

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Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance, revised referral criteria, and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy
Acute Otitis Media – Clinical Practice Guideline

Service restricted to patients aged 2 years or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
- Examination
  - Vital signs
  - Ear examination
  - Tender or enlarged cervical lymph nodes
  - Visual examination and palpation of mastoid area

Develop management plan

- Non-pharmacological measures
- Pharmacotherapy
  - Oral antibiotics
  - Oral analgesia

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation
- Appropriate antibiotic use

Communicate agreed management plan

- Medication counselling
- General advice and patient information
- When to seek further care and/or treatment
- Communication with other health practitioners

Clinical review

- Response to treatment
- Communication with other health practitioners

Refer when

- Unclear diagnosis
- Recent significant ear or head trauma
- Severe symptoms or rapidly worsening symptoms
- Signs or symptoms of sepsis
- Tympanic membrane (TM) 'red flag' signs:
  - Crust/scab or granulation in attic region (discharge/odour may be present)
  - Severely retracted (sucked in) TM or retraction pocket in attic area of eardrum
  - Dull white mass behind TM or chalky patches
  - Perforation or retraction in attic region, near the edge of the TM, or medium to large in size (refer to Figure 1)
- Otoscopic examination is not possible, or TM cannot be visualised (refer to *Refer when* section)
- TM perforation still evident after 6 weeks following initial consultation

Treat and concurrently refer

- Otic discharge/otorrhea present
- Children presenting with signs and symptoms of a complication, or at a high risk of complications from AOM (refer to Table 1)
- Symptoms that do not improve or worsen after 48 hours of appropriate analgesia and supportive management
- Recurrent symptoms (≥3 episodes in 6 months, or ≥4 episodes in 12 months)
- Persistent otitis media with effusion (glue ear)

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Queensland Government

Refer when

Refer the patient to an appropriate health care provider (including emergency services, if required) in the following situations:

- unclear diagnosis
- recent significant ear or head trauma (may be associated with hearing loss or balance issues)
- severe symptoms or rapidly worsening symptoms, including intense pain, headache or facial paralysis/palsy, balance issues, confusion, focal neurological signs & red, swollen and tender behind the ear
- signs or symptoms of sepsis (e.g. fever, hypothermia, respiratory distress, tachycardia, rash, hypotension, cold extremities), or other significant systemic symptoms including signs of dehydration, weakness, significant irritability, lethargy, paleness, loss of appetite
- tympanic membrane (TM) 'red flag' signs:
  - crust/scab or granulation in attic region (discharge/odour may be present)
  - severely retracted (sucked in) TM or retraction pocket in attic area of ear drum
  - dull white mass behind TM or chalky patches
  - perforation or retraction in attic region, near the edge of the TM, or medium to large in size (refer to Figure 1).
- otoscopic examination is not possible, or TM cannot be visualised**
- TM perforation still evident after 6 weeks following initial consultation.

Treat (if clinically appropriate) and concurrently refer

Provide initial treatment and concurrently refer to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- otic discharge/otorrhea present
- children presenting with signs and symptoms of a complication or at high risk of complications from AOM (refer to Table 1)
- symptoms that do not improve, or worsen after 48 hours of appropriate analgesia and supportive management
- recurrent symptoms (≥3 episodes in 6 months, or ≥4 episodes in 12 months)
- persistent otitis media with effusion (glue ear).

**If the TM cannot be visualised in a paediatric patient at first presentation, and antibiotic therapy is not indicated, they can be advised on appropriate analgesia and asked to return within 48–72 hours for reassessment. They should be advised to seek care from an appropriate healthcare provider if their condition deteriorates.

Key points

- Acute otitis media (AOM) is predominantly a childhood infection, however, it may also occur in adults [1]. The peak age prevalence is 6–18 months [2].
- AOM is commonly viral but may also be bacterial, or a combination of both [3].

- In the majority of paediatric cases, AOM is self-limiting and spontaneously resolves without the requirement for antibiotics [2,3].
- The mainstay of treatment for AOM is conservative management with regular and adequate analgesia. Antibiotics can be safely withheld in most cases and are often inappropriately prescribed [3].
- Pain alone is not sufficient to diagnose AOM; pneumatic otoscopy is required to diagnose and exclude other ear conditions with similar presentations [3].
- Ear conditions are often very painful, particularly for young children, and it is important to approach with care [4]. If the patient will not allow for otoscopic examination due to pain, advice should be provided on appropriate analgesia and the patient referred to an appropriate healthcare provider.

Patient history and assessment

In both children and adults, diagnosis of AOM is based on clinical history and ear examination using a pneumatic otoscope [1,3,4,5].

- Children with AOM may present with ear pain and signs of systemic illness including fever, vomiting, diarrhoea, irritability, poor feeding and ear discharge. Tugging, holding or rubbing the ear may indicate ear pain in younger children [3].
- In children, a diagnosis of AOM requires the rapid onset (within 48 hours) of [6]:
    - pain
    - signs of acute inflammation of the TM:
        - TM is bulging and opaque
        - TM may be red from inflammation or white from pus in the middle ear.
    - signs of middle ear effusion
        - otorrhoea can be indicative of TM perforation and middle ear effusion if acute otitis externa (AOE) has been excluded [2,3,4,6]
        - otitis media with effusion is characterised by middle ear effusion without other signs and symptoms of AOM. When present for longer than 3 months, this is suggestive of persistent otitis media with effusion (glue ear) [3]
        - otorrhea that has been present for ≥2 weeks and a perforation is suggestive of chronic suppurative otitis media (CSOM) [4].
- **A red TM or pain alone is not diagnostic of AOM** [3].
- In adults, unilateral symptoms are most common and include mild to severe ear pain and decreased hearing.
    - An upper respiratory tract infection (URTI) or allergic rhinitis will often precede the onset of AOM [1].
    - Other symptoms in adults include dizziness/vertigo (infrequent), systemic symptoms (particularly fever), partial or complete opacification of the TM, or erythema [1].
    - In adults, the key signs of AOM are bulging and reduced mobility of the TM [1].

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- weight
- pregnancy and lactation status (if applicable)
- whether the patient identifies as Aboriginal and Torres Strait Islander
- nature, severity and frequency of symptoms including pain, itch, discharge, hearing loss, a feeling of fullness and dizziness/vertigo
- onset and duration of symptoms including preceding illnesses
- history of head or ear trauma
- recent history of URTI
- other presenting symptoms including temperature >38 °C or <35.5 °C, rash, increased respiratory rate/distress, dehydration and/or reduced urine output, runny nose, sore throat or cough
- precipitating and relieving factors
- delayed speech or language development
- frequency and method of ear cleaning
- family history of ear conditions and hearing loss
- previous history of AOM, AOE or other ear conditions:
    - hearing loss and recent hearing tests
    - patient age at first episode of ear conditions
    - under the care of an ENT specialist or audiologist
    - ear surgery, current or previous tympanostomy tubes (grommets)
    - prior treatment for AOM, otitis externa or other ear conditions – when and what treatment [4].
- underlying medical conditions, e.g., immunocompromised (including diabetes), cochlear implant, craniofacial abnormalities, persistent hearing loss, that mean the patient is at high risk of experiencing AOM complications (refer to Table 1)
- social history including arrangements to support management at home, ability to access further care and to undertake appropriate self-care
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)
- medication and other strategies tried to treat current symptoms
- drug allergies/adverse drug reactions
- immunisation status as per the *Australian Immunisation Handbook*.

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination

- Conduct assessment of vital signs.
- Examine for tender or enlarged cervical lymph nodes.
- Conduct visual examination and palpation of mastoid area.

Ear examination using otoscopy

Physical examination including otoscopy of both ears is required to differentiate between AOE, AOM and other ear conditions [1,3,5].

Ear examination should be conducted in accordance with the Queensland Health and Royal Flying Doctors Primary Clinical Care Manual [4]. An ear differential diagnosis flowchart is also included [4].

Before performing the ear examination, the ear canal must be gently cleaned to remove any discharge with a tissue rolled up to a point (dry mopping) to visualise the ear drum [4,7]. The Remote Primary Health Care Manual – Clinical Procedure Manual [7] and the Queensland Health and Royal Flying Doctors Primary Clinical Care Manual contain a step by step guide for dry mopping [4].

The TM must be visualised to confirm a diagnosis of AOM, although this can be difficult in the presence of extensive otorrhoea [3]. TM presentations that suggest a serious complication require immediate referral to an appropriate healthcare provider, see *Refer when*.

- If the TM cannot be visualised in paediatric patients when they first present and antibiotics are not indicated, they should be advised on appropriate analgesia and asked to return within 48–72 hours for reassessment. They should seek care from an appropriate healthcare provider if their condition deteriorates [3].
- If the TM cannot be visualised in adult patients, they should be referred to an appropriate healthcare provider.
- Perforation of the TM in children is common and may result in ear discharge and relief of pain [4].
- In AOM with perforation, the pharmacist should document the size and position of the perforation to allow ongoing assessment of progression of the condition and to guide use of therapies [8].

If pain levels allow, consider looking for movement of intact ear drum(s) by pneumatic otoscopy, tympanometry (if no discharge, Type A is normal) or gentle Valsalva (hold nose and blow):

- If the TM moves freely back and forward, perforation and fluid in the middle ear (AOM) can generally be excluded [4].
- Reduced mobility of an intact TM is a good indication of the presence of middle ear fluid [8].

Check for other symptoms if required (nose, throat, cough) [4].

Figure 1. Guide for size of perforation of tympanic membrane [8]



Complications of AOM

Aboriginal and Torres Strait Islander children are generally at higher risk of complications from AOM. Complications in adults and non-Aboriginal and Torres Strait Islander children are less common [4,6].

Children at high risk of complications are outlined in Table 1.

Table 1. Children at high risk of complications from AOM [4]

Aboriginal and Torres Strait Islander children if (any of the following)	**All children if (any of the following)**
• residing in a remote community • <2 years old NB1 • first episode of OM experienced <6 months of age • family history of CSOM • history of (or current) TM perforation. **NB1:** Children <2 years old are ineligible for the Service and must be referred to an appropriate healthcare provider.	• immunocompromised • cochlear implant • craniofacial abnormalities including cleft palate • developmental delay, Down Syndrome • severe visual impairment • current hearing loss.

All children presenting with signs or symptoms of a complication, or at high risk of complications, must be referred (concurrent to commencement of antibiotic therapy where appropriate) to an appropriate healthcare provider for management.

In rare cases, AOM infection may spread from the middle ear (intracranial spread) causing mastoiditis, petrositis or labyrinthitis [6]. Intracranial spread can cause meningitis or other intracranial complications that are slow to resolve, particularly in immunocompromised patients. Signs of intracranial complications include severe headache, confusion, focal neurologic signs, facial paralysis/palsy and dizziness/vertigo [6]. Patients with these signs or symptoms of a complication must be referred for immediate assessment, see *Refer when*.

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Pharmacist management of patients with AOM involves[NB1]:

- **non-pharmacological measures:**
    - Care at home in accordance with The Royal Children's Hospital Melbourne Fact sheet: Ear infections and glue ear [9] and the Australian Medicines Handbook: Drugs for ear infections [10].
    - If discharge is present, advice and education regarding keeping ears dry and the use of dry mopping with tissue spears. The Queensland Health and Royal Flying Doctors Primary Clinical Care Manual contains a step by step guide for dry mopping [4].
- **pharmacotherapy:**
    - Oral analgesia in accordance with the Therapeutic Guidelines: Pain and Analgesia (Pharmacological management of acute pain - Mild, acute nociceptive pain) [11].
    - Oral antibiotic therapy[NB2-3] in accordance with the Therapeutic Guidelines: Antibiotic (Ear, nose and throat infections - Otitis media) [3].
    - Consider immunisation where the patient has not received all routine vaccines in accordance with the Australian Immunisation Handbook.

NB1: There is limited evidence to guide the management of AOM in adult patients. The *Therapeutic Guidelines* states that management of AOM in adults is similar to management in children. Adequate and regular analgesia is the primary treatment [3]. As with paediatric patients, a shared decision-making approach is required to manage patient expectations of antibiotic treatment [3].

NB2: Antibiotic therapy in paediatric patients should only be commenced in accordance with the *Therapeutic Guidelines* and the Otitis Media Guidelines for Aboriginal and Torres Strait Islander Children for children: [3,4,8]

- with discharge/otorrhoea (from a small perforation which is not in the attic region, or near the edge of the TM)
- presenting with signs and symptoms of a complication, or at high risk of complications, from AOM (see Table 1)
- with symptoms that have not improved or have worsened after 48 hours of appropriate analgesia and supportive management
- with recurrent symptoms (≥3 episodes in 6 months or ≥4 episodes in 12 months)
- with persistent otitis media with effusion (glue ear).

NB3: Antibiotic therapy may be commenced if there is uncertainty with access to follow-up [4].

Confirm management is appropriate

Consult the *Therapeutic Guidelines*, *Australian Medicines Handbook* and other relevant resources to confirm that management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

Reminder

The prescribing and use of antibiotics should be consistent with the principles described in the Therapeutic Guidelines: Antibiotic. Pharmacists should carefully assess whether antibiotic therapy is clinically indicated, and if so, consider local antimicrobial resistance patterns and relevant clinical factors when selecting therapy.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- individual product and medicine use
- non-pharmacological measures
- how to manage adverse effects
- when to seek further care and/or treatment (including recognising patient deterioration)
- when to return for follow-up.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources and information provided to patients (and/or parents/caregivers) and to ensure compliance with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

Patient resources

- The Royal Children's Hospital Melbourne Fact sheet: Ear infections and glue ear [9].

General advice

AOM symptoms should start to improve within 48–72 hours after onset, regardless of whether antibiotic treatment is required. However, full resolution of symptoms may take up to 7 days (or longer for effusion) [3].

- Middle ear effusion that persists for several weeks after AOM is common and is not a sign of treatment failure [10] .
- An acute perforation and discharge should resolve within 14 days [4].

Decongestants, antihistamines or corticosteroids are not effective for AOM and should not be recommended [2,10].

Antibiotics have limited benefit in AOM and are not required for most patients [3].

- The use of antibiotics can cause harm, including adverse effects, serious hypersensitivity reactions and can contribute to bacterial resistance [3].
- Pharmacists should address misconceptions regarding antibiotic effectiveness and discuss the relative harms and benefits of treatment [2].

Patients commenced on antibiotics who have been concurrently referred to an appropriate healthcare provider should be advised (and/or parents/caregivers) to seek further urgent care where:

- the response has been inadequate after 48–72 hours of antibiotic treatment
- signs or symptoms worsen, or their condition deteriorates
- new signs or symptoms develop, including systemic symptoms
- otorrhea continues for ≥2 weeks [3,4].

Patients who have **not** been prescribed antibiotic therapy at the initial consultation should be advised (and/or parents/caregivers) to see an appropriate healthcare provider if:

- they develop otic discharge/otorrhoea
- their symptoms do not improve after 48 hours of appropriate analgesia and supportive management
- their signs or symptoms worsen, new signs or symptoms develop, or symptoms are recurrent, see *Refer when* [3].

For all patients, it is important to avoid local trauma to the ear canal, including not using cotton ear buds and dry mopping with a tissue spear instead [4].

Pharmacists should provide advice on the importance of preventing ear infections in children to avoid associated hearing loss, and poor language and social skills [8]. Preventative advice that can be provided includes:

- ensuring vaccinations against *S. pneumoniae*, *H. influenzae* type B, and influenza are up to date
- minimising exposure to smoke e.g., cigarettes and woodfires
- encouraging personal hygiene e.g., frequent hand and face washing and drying, nose blowing techniques [3,8,10,12].

Advise parents/caregivers of children under the age of 5 to seek review by an appropriate healthcare provider if there is any concern regarding the child's speech, language, behaviour or school progress [4,13].

Clinical review

Clinical review with the pharmacist (or the patient's chosen healthcare provider) should occur in line with recommendations in the *Therapeutic Guidelines* and the Queensland Health and Royal Flying Doctors Primary Clinical Care Manual.

- Patients with an intact TM should return for a follow up consultation at **3 months** to check for persistent fluid behind the TM. If fluid is still present or the patient experiences a recurrence of AOM during this time, refer the patient to an appropriate healthcare provider [4].
- Patients with small acute perforations should return for a follow up consultation in **6 weeks** to assess healing. If the perforation is still evident, refer the patient to an appropriate healthcare provider [4].

Decisions regarding the timing and necessity of clinical review should be individualised to the patient's clinical presentation and response to treatment.

Pharmacists should prescribe a quantity of medicine that aligns with the expected duration of therapy. Advise patients (and/or parents/caregivers) to see an appropriate healthcare provider if the condition does not improve or resolve (see *Refer when* and *General advice*).

Pharmacist resources

- Therapeutic Guidelines:
  - Otitis media
  - Pharmacological management of acute pain
  - Mild, acute nociceptive pain
- Australian Medicines Handbook
  - Otitis Media
  - Drug for ear infections
  - NSAIDs
  - Drugs for pain relief
- Australian Immunisation Handbook:
  - Pneumococcal disease
  - Influenza (flu)
  - Haemophilus influenzae type b (Hib)
- MSD Manual (Professional version): Otitis media (Acute)
- Remote Primary Health Care Manuals:
  - Clinical Procedures Manual
  - CARPA Standard Treatment Manual
- Queensland Health and Royal Flying Doctors Service (Queensland branch): Primary Clinical Care Manual 12th edition 2025
- The Royal Children's Hospital - Clinical Practice Guidelines – Acute otitis media
- 2020 Otitis Media Guidelines for Aboriginal and Torres Strait Islander Children

McGovern Medical School - Ear disease photo book

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Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy

Allergic and Nonallergic Rhinitis - Clinical Practice Guideline

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
  - Presenting signs and symptoms
- Examination
  - Examination of nose, throat, eyes and ears
  - Palpation of neck
  - Vital signs
  - Chest examination

Develop management plan

- Allergic rhinitis
  - Treatment plan
  - Non-pharmacological measures
  - Pharmacotherapy
- Nonallergic rhinitis
  - Non-pharmacological measures
  - Pharmacotherapy

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation

Communicate agreed management plan

- Medication counselling
- General advice and patient information
- When to seek further care and/or treatment
- Communication with other health practitioners

Clinical review

- Response to treatment
- Continue, modify or stop treatment
- Communication with other health practitioners

Refer when

- Unclear diagnosis
- Systemic symptoms, including fever, or generally unwell
- Signs or symptoms of infection or infectious rhinosinusitis
- Nasal polyps or other structural abnormalities
- Signs or symptoms of undiagnosed or uncontrolled asthma
- Asthma in patients under the age of 16 years
- Inadequate response to optimal treatment

Treat and concurrently refer

- Using a medicine that can cause or exacerbate rhinitis
- Quality of life is significantly affected

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Refer the patient to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- unclear diagnosis
- systemic symptoms, including fever, or generally unwell
- signs or symptoms of infection or infectious rhinosinusitis (e.g., facial pain, fever, purulent nasal discharge)
- nasal polyps or other structural abnormalities
- signs or symptoms of undiagnosed or uncontrolled asthma
- asthma in patients under the age of 16 years
- inadequate response to optimal treatment.

Treat (if clinically appropriate) and concurrently refer

Provide initial treatment and concurrently refer to an appropriate healthcare provider in the following situations:

- using a medicine that can cause or exacerbate rhinitis
- quality of life is significantly affected.

Key points

- Rhinitis has a broad aetiology and can be classified as allergic, nonallergic (vasomotor), infectious, drug-induced and occupational. Allergic rhinitis (which may co-occur with nonallergic rhinitis) is the most common type [1]. Accurate differentiation of the type of rhinitis is important to inform effective treatment [2-4].
- Rhinitis can have significant impact on quality of life including sleep, cognitive and psychomotor function, social activities and learning impairment in children [4, 5].
- Emerging evidence suggests that allergic rhinitis may be part of a systemic airway disease, as opposed to a localised disorder of the nose and nasal passages [6].
- Rhinitis and asthma commonly co-exist (sometimes referred to as United Airway Disease) [7]. Effective management of allergic rhinitis is important for the management of asthma; all asthma symptoms should be investigated and/or the patient provided with a referral to an appropriate healthcare provider [1, 6].

Patient history and assessment

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- pregnancy and lactation status (if applicable)
- onset, nature, duration, frequency, severity and pattern of respiratory symptoms

- other signs and symptoms suggestive of infection including fever, purulent nasal discharge or facial pain
- respiratory virus testing (PCR or RAT) and results
- presence of asthma symptoms
- co-existing and underlying medical conditions including asthma, nasal polyps or other allergic diseases (e.g., atopic dermatitis)
- family history of nasal polyps or allergic disease (including rhinitis)
- triggering, aggravating and relieving factors (e.g., occupational exposure to cleaning chemicals, foods, pollen, cigarette smoke)
- emotional and social impacts of the condition
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)
- pharmacological and non-pharmacological strategies tried to treat current symptoms, and response, including use of nasal decongestants
- drug allergies/adverse drug reactions
- smoking status.

The symptoms of allergic and nonallergic rhinitis are included in Table 1.

Table 1. Symptoms and clinical signs of rhinitis [1, 2, 5, 8, 9]

Allergic rhinitis	**Non-allergic rhinitis**
Symptoms after exposure to animals, pollen, dust or other environmental factors/allergens: • sneezing • nasal congestion • clear rhinorrhoea • upper airway cough syndrome (post-nasal drip) • mucosal itching of the nose, eyes, ears and palate. Non-specific symptoms: • 'fuzzy' head • tiredness and daytime sleepiness • constant 'colds'. Allergic conjunctivitis frequently co-occurs (eye redness, watering and itching). In children, additional symptoms may include sniffing, blinking and eye rubbing, speech problems, snoring, mouth breathing, and dark undereye circles.	Symptoms without an identifiable infectious or allergic cause: • nasal blockage and/or congestion • clear rhinorrhoea • post-nasal drip. May or may not be present: • sneezing • itchy nose, ears or throat. Symptoms may present sporadically at any time of the year, although they may be exacerbated by environmental factors such as barometric pressure or temperature changes, bright lights or physical irritants.

Reported symptom improvement with second generation antihistamines is strongly suggestive of allergic rhinitis, as is a previous response to intranasal corticosteroids [6].	

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination

- Physical examination of nose, throat, eyes and ears is generally required [5].
- Where appropriate, palpate the neck for lymphadenopathy (more relevant when throat symptoms present).
- Where appropriate, conduct an assessment of vital signs.
- Outward signs of rhinitis upon examination may include:
  - enlarged/swelled, pale, boggy nasal mucosa with thin secretions
  - nasal polyps
  - dark undereye area due to venous congestion (particularly in children)
  - cobblestone throat due to irritation of post-nasal drip
  - mouth breathing [6].
- Patients with symptoms of asthma or lower respiratory disease should have a chest examination in accordance with the clinical protocol for the Queensland Community Pharmacy Improved Asthma Symptom Control Program.

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Allergic rhinitis

Pharmacist management of allergic rhinitis involves:

- **developing a treatment plan:**
  - Based on the Australasian Society of Clinical Immunology and Allergy: Allergic Rhinitis Treatment Plan [10].
- **non-pharmacological measures:**
  - Education regarding allergen avoidance and minimising exposure to known/clinically obvious allergens and irritants if possible, as per the Australian Society of Clinical Immunology and Allergy: Allergic Rhinitis Clinical Update 2024 [4].
  - Nasal irrigation with saline to soothe nasal passages, and clear mucous and irritants [11].

    - Advice to patients that oral and intranasal decongestants have no role in the management of allergic rhinitis. Use of intranasal decongestants for more than 3–5 days causes rebound congestion (rhinitis medicamentosa). Prolonged use of oral decongestants can increase blood pressure and the risk of arrythmias in people with cardiovascular disease [1].
- **pharmacotherapy** [NB1]:
    - A medicine mentioned in the Therapeutic Guidelines: Respiratory (Rhinitis and rhinosinusitis - Allergic rhinitis), including eye drops for allergic conjunctivitis[1].

NB1: Anti-inflammatory eye drops (e.g., ketorolac and corticosteroids) are mentioned within the Therapeutic Guidelines: Respiratory (Rhinitis and rhinosinusitis - Allergic rhinitis) for use under specialist advice only and are not authorised for use by pharmacists; these are not indicated for the initial management of allergic conjunctivitis. Vasoconstrictor eye drops (e.g., naphazoline and tetryzoline) are also not indicated for allergic conjunctivitis.

Nonallergic rhinitis

Pharmacist management of nonallergic rhinitis involves:

- **non-pharmacological measures:**
    - Education regarding avoidance and minimisation of exposure to known/clinically obvious irritants where possible.
    - Nasal irrigation with saline to soothe dry nasal passages, and clear out mucous [12].
- **pharmacotherapy**:
    - A medicine mentioned in Therapeutic Guidelines: Respiratory (Rhinitis and rhinosinusitis - Nonallergic rhinitis) [9]. Intranasal corticosteroids are the mainstay of treatment for nonallergic rhinitis [9].

Confirm management is appropriate

Pharmacists must consult the *Therapeutic Guidelines*, *Australian Medicines Handbook* and other resources to confirm that management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- individual product and medicine use e.g., dosing and correct intranasal spray technique
- non-pharmacological measures
- how to manage adverse effects
- when to seek further care and/or treatment
- when to return for follow-up.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources provided to patients (and/or parents/caregivers), and to ensure these comply with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

Patient resources

- Therapeutic Guidelines: Allergic Rhinitis (Instructions for using a nasal spray)
- National Asthma Council Australia: How-to videos (Using your nasal spray and irrigation)

General advice

At the time of initial consultation, patients (and/or parents/caregivers) should be advised to see an appropriate healthcare provider if:

- the condition does not improve within 4 weeks of starting treatment
- the patient's condition worsens, including developing systemic signs and symptoms
- the patient develops new signs or symptoms (e.g. infection)
- the patient develops new, or worsening, asthma symptoms
- quality of life is significantly affected
- adverse effects of treatment cannot be managed in the pharmacy setting.

Clinical review

Clinical review with the pharmacist (or the patient's chosen healthcare provider) should occur in line with recommendations in the *Therapeutic Guidelines*.

- Patients should return for a follow-up consultation **4 weeks** after initiation of intranasal corticosteroids to assess for:
    - response to treatment and if changes to the treatment plan are required (continue, modify, stop and/or refer)
    - intranasal spray administration technique.
- If the patient's symptoms are mild and adequately managed with an oral or intranasal antihistamine after trialling a new treatment for **4 weeks,** clinical review will generally not be required, and the patient may be advised to remain on the minimum effective dose.

For patients continuing therapy, follow-up should be individualised to the patient's clinical presentation, response to treatment, and therapeutic goals.

Pharmacists should prescribe a quantity of medicine (including repeats), that aligns with the expected duration of therapy, ensuring sufficient supply until the patient's next review. The total quantity prescribed should not exceed 12 months' supply (including repeats).

Pharmacist resources

- Therapeutic Guidelines:
  - Allergic rhinitis
  - Nonallergic rhinitis
  - Asthma
- Australian Medicines Handbook:
  - Drugs for rhinitis and rhinosinusitis
  - Drugs for allergic and inflammatory eye conditions
  - Antihistamines
  - Montelukast
  - Ipratropium
- Australian Pharmaceutical Formulary and Handbook (APF): Treatment guideline for pharmacists (Allergic rhinitis (hay fever))
- Australian Society of Clinical Immunology and Allergy: Allergic Rhinitis Clinical Update
- Australian Family Physician (RACGP): Allergic rhinitis Practical management strategies
- The Royal Children's Hospital Melbourne: Allergic rhinitis hay fever

Document version number	Date	Comments
Version 1.0	01.02.2024	
Version 2.0	01.07.2025	Updated to reflect current clinical guidance, revised referral criteria, and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy

Gastro-oesophageal Reflux and Gastro-oesophageal Reflux Disease - Clinical Practice Guideline

Service restricted to patients aged 18 to 55 years of age.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
  - Presenting signs and symptoms
- Examination
  - Vital signs

Develop management plan

- Non-pharmacological measures
  - Dietary and lifestyle modification
- Pharmacotherapy

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation

Communicate agreed management plan

- Medication counselling
- General advice and patient information
- When to seek further care/treatment
- Communication with other health practitioners

Clinical review

- Response to treatment
- Continue, modify or stop treatment
- Communication with other health practitioners

Refer when

- Unclear diagnosis
- Chest pain as the primary symptom
- Symptoms of anaemia, including fatigue, shortness of breath, weakness, dizziness or an irregular heartbeat
- Difficulty or painful swallowing
- Prolonged or recurrent vomiting (longer than 24–48 hours)
- Jaundice
- Haematemesis or melaena
- Unexplained weight loss
- Severe, frequent or changing symptoms
- Inadequate response to optimal treatment (8 weeks of PPI therapy at the standard dose)

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Refer the patient to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- unclear diagnosis
- chest pain as the primary symptom
- symptoms of anaemia, including fatigue, shortness of breath, weakness, dizziness or an irregular heartbeat
- difficulty or painful swallowing
- prolonged or recurrent vomiting (longer than 24–48 hours)
- jaundice
- haematemesis or melaena
- unexplained weight loss
- severe, frequent or changing symptoms
- inadequate response to optimal treatment (8 weeks of PPI therapy at the standard dose).

Key points

- The fundamental signs of gastro-oesophageal reflux are heartburn and regurgitation [1, 2, 3].
- The *Therapeutic Guidelines* defines gastro-oesophageal reflux disease (GORD) as gastro-oesophageal reflux symptoms that are frequent (2 or more episodes per week) or severe enough to significantly impair quality of life, or if complications of gastro-oesophageal reflux are present [3].
- The causes of gastro-oesophageal reflux and GORD are multifactorial and may involve dysfunctional peristalsis, delayed gastric emptying and impaired or transient relaxation of lower oesophageal sphincter resting tone [1, 4].
- The presumptive clinical diagnosis of gastro-oesophageal reflux and GORD are primarily based on patient reported symptoms [3, 4]. However, response to a trial of proton pump inhibitors (PPIs) can help to confirm a diagnosis of GORD [2, 3]. A trial of high dose PPIs has a sensitivity of 80% as a diagnostic test for GORD [1].
- The initial management of gastro-oesophageal reflux and GORD is based on the severity of symptoms; with symptom control the aim for most patients [3]. Mild intermittent symptoms may be managed with diet and lifestyle modification only. However, on-demand ('when required') drug therapy may also be used if symptoms are significantly troublesome [3].
- Approximately one third of patients with symptoms of reflux have some form of mucosal damage, ranging from minor erosions to circumferential ulceration or metaplasia (e.g. Barrett's oesophagus) [3]. GORD may lead to oesophageal strictures in a minority of patients or on rare occasions, malignancy [3].

Patient history and assessment

Presenting signs and symptoms

The clinical symptoms of GORD are common to multiple other conditions but can often be differentiated using the patient's history.

In addition to the cardinal signs of heartburn and regurgitation other atypical symptoms attributed to reflux include:

- belching
- epigastric pain
- dyspepsia
- acid brash (saliva mixing with stomach acid in the mouth) [1, 2].

Extra-oesophageal symptoms that are associated with, but not specific to gastro-oesophageal reflux, may include:

- tooth sensitivity, enamel erosion (particularly on palatal surface of upper incisors) and halitosis
- pharyngeal and laryngeal signs and symptoms such as hoarseness and throat clearing, persistent cough, sore throat, pharyngitis and sinusitis
- chest pain
- sleep disturbance
- nausea
- asthma-like symptoms (cough, wheeze, shortness of breath)
- early post-prandial satiety [3, 4, 5].

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- weight
- pregnancy and lactation status (if applicable)
- nature, severity and frequency of symptoms (typical, atypical and extra-oesophageal), including signs and symptoms that may indicate the need for referral
- onset and duration of symptoms
- precipitating and relieving factors
- underlying medical conditions and conditions that may complicate diagnosis e.g., coronary heart disease and asthma
- dietary patterns or changes
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)
- allergies/adverse drug reactions
- pharmacological and non-pharmacological (dietary and lifestyle) strategies tried to treat current symptoms

- alcohol and drug history
- smoking status.

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination

- Where appropriate, conduct assessment of vital signs.

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

The initial management of gastro-oesophageal reflux and GORD is based on the severity of symptoms; symptom control is the aim for most patients [3].

Pharmacist management of gastro-oesophageal reflux and GORD involves:

- **non-pharmacological measures:**
    - Advice regarding dietary and lifestyle modification as per the Therapeutic Guidelines: Diet and lifestyle modification for the management of gastro-oesophageal reflux in adults [3].
- **pharmacotherapy [NB1,2]:**
    - As per the Therapeutic Guidelines: Gastrointestinal (Gastro-oesophageal reflux in adults) [3].

NB1: The efficacy and availability of PPIs has led to overuse. Long-term regular PPI therapy is only recommended for a limited number of indications and should be reviewed regularly [3, 6].

NB2: Where regular drug therapy is required, consider referral to an appropriate healthcare provider [3].

Confirm management is appropriate

Consult the *Therapeutic Guidelines*, *Australian Medicines Handbook* and other resources to confirm that management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- individual medicine use e.g., initial dosing, maintenance and step-down therapy
- non-pharmacological measures
- how to manage adverse effects
- when to seek further care and/or treatment
- when to return for follow-up.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources and information provided to patients (and/or parents/caregivers), and to ensure compliance with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

General advice

At the time of initial consultation, patients (and/or parents/caregivers) should be advised to see an appropriate healthcare provider if:

- the condition does not improve with recommended treatment (e.g., within 8 weeks of starting treatment with a PPI)
- ongoing maintenance treatment with a PPI is required
- the patient's condition worsens, including developing systemic signs and symptoms
- the patient develops new or changed signs or symptoms
- adverse effects of treatment cannot be managed in the pharmacy setting.

Clinical review

Clinical review with the pharmacist (or the patient's chosen healthcare provider) should occur in line with recommendations in the Therapeutic Guidelines.

- Patients should return for a follow up consultation **4-8 weeks** after initiation of a new therapy to assess:
    - adherence to pharmacotherapy and non-pharmacological management strategies
    - response to treatment and if changes to the treatment plan are required (continue, modify, stop and/or refer).

For patients continuing therapy, the frequency of further follow up appointments for maintenance and step-down therapy should be individualised to the patient's clinical presentation, response to treatment, and therapeutic goals, and based on recommendations in the *Therapeutic Guidelines*.

Pharmacists should prescribe a quantity of medicine (including repeats), that aligns with the expected duration of therapy, ensuring sufficient supply until the patient's next review. The total quantity prescribed should not exceed 12 months' supply (including repeats).

Pharmacist resources

- Therapeutic Guidelines: Gastrointestinal
  - Oesophageal disorders
  - Functional gastrointestinal disorders
- Australian Medicines Handbook:
  - Drugs for dyspepsia, reflux and peptic ulcers
- Australian Pharmaceutical Formulary and Handbook (APF): Treatment guideline for pharmacists (Gastro-oesophageal reflux)
- Australian Prescriber: The management of gastro-oesophageal reflux disease
- Pharmaceutical Society of Australia: Heartburn and indigestion 'Self-care' patient information card.
- Gastroenterological Society of Australia (GESA) Patient Resources: Information about Reflux

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Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy Herpes Zoster (Shingles) – Clinical Practice Guideline

Service restricted to patients aged 18 years or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
- Examination
  - Clinical features and presenting symptoms

Develop management plan

- Non-pharmacological measures
  - Care of lesions
  - Transmission precautions
- Pharmacotherapy
  - Oral antiviral therapy
  - Oral analgesia
  - Zoster vaccination

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation

Communicate agreed management plan

- Medication counselling
- General advice and patient information
- When to seek further care/treatment
- Communication with other health practitioners

Clinical review

- Response to treatment
- Progression of signs and symptoms
- Communication with other health practitioners

Refer when

- Unclear diagnosis
- The patient is pregnant and does not have a history of VZV (or uncertain history)

Treat and concurrently refer

- Immunocompromised
- Superinfection of HZ skin lesions
- Multidermatomal rash or disseminated zoster
- Complications of HZ, PHN or specific zoster syndromes
- HZ affecting the face or genitals
- Pain management for neuropathic pain or moderate to severe nociceptive pain is needed
- Pregnant with a definite history of previous VZV infection
- Previously vaccinated against HZ
- Inadequate response to optimal treatment

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Refer the patient to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- unclear diagnosis
- the patient is pregnant and does not have a past history of varicella zoster virus (VZV) (or uncertain history).

Treat (if clinically appropriate) and concurrently refer

Provide initial treatment (antiviral therapy and pain relief) and concurrently refer to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- immunocompromised
- superinfection of herpes zoster (HZ) skin lesions
- multidermatomal rash or disseminated zoster
- complications of HZ, including postherpetic neuralgia or specific zoster syndromes
- HZ affecting the face or genitals
- pain management for neuropathic pain or moderate to severe nociceptive pain is needed
- pregnant with a definite history of previous VZV infection
- previous vaccination against HZ
- inadequate response to optimal treatment.

Key points

- Herpes zoster (HZ) can occur in anyone who has previously been infected with VZV at any age, and the risk increases with age [1, 2].
- Early identification and management with antivirals and analgesia (including all immunocompetent patients who present within 72 hours of rash onset as well as all patients who are immunocompromised regardless of rash duration) reduces the acute pain, rash duration, viral shedding and potential for ocular complications of HZ [3, 4].
- Vaccination is the most effective prevention for HZ and its complications [1, 3]. The National Immunisation Program (NIP) recommends HZ vaccination for patients at risk [5].
- Under the Queensland *Public Health Regulation 2018*, VZV infection (chicken pox or shingles) is a pathological diagnosis notifiable condition; clinical notification is generally not required. See the ***Public health notification*** section for more information.

Patient history and assessment

Typical presentations of HZ can be diagnosed based on the patient history and examination for the characteristic appearance and distribution of the rash [1-3, 6, 7]. The clinical presentation will vary depending on the patient's age, general health and affected dermatome [7].

Presenting signs and symptoms

Prodromal symptoms

- Prodromal symptoms may be observed between 48–72 hours before the localised, characteristic vesicular rash becomes evident [5-8]:
    - localised nerve pain (usually described as stabbing, prickling or burning)
    - lethargy, fever, headache
    - abnormal skin sensations such as burning, itching, hyperesthesia and/or paraesthesia
    - photophobia [3, 6, 7].

Rash

- The HZ rash is typically unilateral with a dermatomal distribution and distinct anterior and posterior midline cut-offs; however, some satellite lesions may also appear [4-8].
- New lesions will continue to erupt for 3-5 days within the locality of the affected nerve, becoming pustular, before scabbing and crusting over between 7-10 days [3, 6, 7].
- The most affected areas are the chest, neck, forehead (ophthalmic) and lumbar/sacral sensory nerve supply regions [3, 6-8].
- When the ophthalmic division of the trigeminal nerve is affected (herpes zoster ophthalmicus), the patient will present with a blistering rash around the eye/eyelid with associated pain, swelling and redness [6, 8]. Urgent referral to an appropriate healthcare provider is required.
- If the facial nerve and subsequently, the vestibulocochlear nerve is affected (herpes zoster oticus/Ramsay Hunt syndrome), symptoms include earache, blistering in and around the ear canal, with or without facial paralysis [6]. Urgent referral to an appropriate healthcare provider is required.

Complications

Complications occur in approximately 13%-26% of patients with HZ (most prevalent in older people and those who are immunocompromised) and will require referral to an appropriate healthcare provider at presentation or at any stage if a complication develops [3, 5].

Postherpetic neuralgia (PHN)

- Neuropathic pain that persists for at least 3 months beyond the duration of the rash or reoccurs (occurs in 10% to >70% of patients, with the incidence increasing with age) [2, 3, 5, 8, 9].
- Postherpetic pain may be sharp/shooting and intermittent, or described as constant burning, often with extreme sensitivity to touch (allodynia) [7, 9].

Herpes zoster ophthalmicus

- HZ affecting the ophthalmic branch of the trigeminal nerve with a high incidence of eye complications [3, 10].
- It occurs in 10%–25% of cases and commonly causes keratitis (approximately two thirds of cases) as well as conjunctivitis, uveitis, retinitis and glaucoma [3].
- Vesicles on the nose have been found to be predictive of eye involvement [3].

Herpes zoster oticus (Ramsay Hunt syndrome)

- HZ affecting the facial nerve and subsequently, the vestibulocochlear nerve resulting in ear pain, taste loss, facial weakness or paralysis, and other neurological symptoms [3, 8].

Disseminated zoster (VZV dissemination)

- Whilst most individuals have some lesions external to the primary dermatome, disseminated zoster is defined as 20 lesions outside of the dermatome and may be clinically indistinguishable from varicella infection [3].
- Viral dissemination to the central nervous system and viscera (lungs, gut, liver and brain) may occur [3].
- Occurs most frequently in immunocompromised patients, although is rare overall [5].

Other complications

- Neurological complications such as meningoencephalitis and myelitis (which is more likely in the elderly).
- Secondary bacterial skin infections (referral to an appropriate healthcare provider for swab and culture is required).
- Scarring.
- Pneumonia [5].

HZ in pregnancy

- Pregnant patients who are exposed to VZV (chicken pox or HZ) for the first time (no or uncertain history of previous chicken pox infection) may develop chicken pox which can have serious consequences. These include maternal mortality and morbidity, fetal varicella syndrome and the associated abnormalities [11, 12]. Urgent referral to an appropriate healthcare provider is required as zoster immunoglobulin should be given to all seronegative women within 96 hours [5].
- Unlike chicken pox, it is thought that HZ contracted during a healthy pregnancy is not associated with intrauterine infection or an increased foetal risk [12].

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- pregnancy and lactation status (if applicable)
- nature, severity and frequency of symptoms
- nature of rash (distribution, appearance, number of lesions)
- onset and duration of symptoms
- precipitating and relieving factors
- history of VZV infection
- underlying medical conditions e.g., immunocompromise (auto-immune diseases including diabetes, rheumatoid arthritis, HIV, cancer, conditions treated with immune suppressants, or renal impairment)

- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)
- treatment and other strategies used to treat current symptoms and associated response
- drug allergies/ adverse reactions
- immunisation status as per the *Australian Immunisation Handbook* (HZ and varicella vaccinations).

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination

Examination of the rash and documentation of its characteristics and location, as well as any signs of complications is important for the diagnosis of HZ and the exclusion of other conditions with similar presentations.

Laboratory confirmation is generally not required for typical presentations and uncomplicated cases of HZ [3, 13]. Confirmatory pathology testing is required for cases where diagnosis is uncertain, or in cases of HZ in people who have been previously vaccinated against HZ [13, 14].

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Antiviral treatment can reduce acute pain, rash duration, viral shedding, and the risk of ocular complications if started within 72 hours of rash onset in immunocompetent patients, and at any stage of rash duration in immunocompromised patients [4].

Pharmacist management of HZ involves:

- **non-pharmacological measures:**
    - Education and advice regarding care for lesions (use of dressings, cleaning and appropriate clothing).
    - Education and advice regarding transmission precautions in accordance with the Queensland Health Guideline for Public Health Units [14].
- **pharmacotherapy:**
    - Oral antiviral therapy in accordance with the Therapeutic Guidelines: Antibiotic (Skin and soft tissue infections - Shingles (Herpes zoster)) [4].
    - Oral analgesia for **mild nociceptive shingles pain** in accordance with the:
        - Therapeutic Guidelines: Pain and analgesia (Pain associated with shingles (herpes zoster) - Acute pain associated with shingles (herpes zoster)) [9], or

- Therapeutic Guidelines: Pain and analgesia (Pharmacological management of acute pain - Mild, acute nociceptive pain) [15].

Zoster vaccination

Vaccination against HZ is the best way to prevent HZ and reduce the risk of complications, however it is not indicated during an acute HZ episode or to treat PHN [5].

- People who have had a previous episode of HZ can be vaccinated against a recurrence; the interval between the episode and vaccination is dependent on the patient's immune status [5].
- Refer to the Australian Immunisation Handbook for HZ vaccination information and recommendations [5].

Public health notification

Under the Queensland *Public Health Regulation 2018*, VZV infection (chicken pox or shingles) is considered a pathological diagnosis notifiable condition (as opposed to a clinical diagnosis notifiable condition) and must be notified in accordance with section 72 and 73 of the *Public Health Act 2005*.

- Clinical notification (diagnosis made on the basis of clinic evidence) is generally not required; however, if the diagnosis is uncertain or the patient has previously been vaccinated against HZ, they should be referred to a medical practitioner who may initiate pathology testing.
- Cases of HZ confirmed by pathology testing must be notified by the director of the pathology laboratory.

Confirm management is appropriate

Consult the *Therapeutic Guidelines*, *Australian Medicines Handbook*, *Australian Immunisation Handbook* and other resources to confirm that management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

Reminder

The prescribing and use of antimicrobials should be consistent with the principles described in the Therapeutic Guidelines: Antibiotic.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- individual product and medication use
- non-pharmacological measures

- how to manage adverse effects
- recommendations for vaccination against HZ (see Australian Immunisation Handbook)
- when to seek further care and/or treatment, including recognising superinfection and HZ complications
- when to return for follow-up.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources and information provided to patients (and/or parents/caregivers) and to ensure compliance with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

General advice

At the time of initial consultation, patients (and/or parents/caregivers) should be advised to see an appropriate healthcare provider if:

- signs or symptoms worsen
- new signs or symptoms develop (e.g., superinfection of HZ lesions)
- the blisters spread to the face or genitals, or a new area
- they do not respond adequately to the recommended treatment or pain management is inadequate:
    - skin lesions usually heal within 2-4 weeks although it may take longer, particularly for those that are immunocompromised or those with severe disease [2]
    - pain and systemic symptoms subside as the rash resolves with recovery complete in 2-4 weeks in most cases [3, 7]
    - pain can persist after shingles has resolved [4].
- they are experiencing complications, see *Refer when* and *Complications of HZ*
- they are having unmanageable adverse effects.

The incubation period of VZV ranges between 10-21 days (14-16 days on average) [14].

Preventing transmission

The patient (and/or parents/caregivers) should be advised to inform any high-risk contacts (pregnant persons, neonates in first month of life, premature infants born at <28 weeks gestation and immunocompromised individuals) who have had significant exposure to a person with active VZV (household contacts or where there has been direct face-to-face contact for 5 minutes or being within the same room for at least an hour) to seek medical care as soon as possible [14].

People with HZ are infectious from the onset of the rash, until vesicles have dried and scabbed [14].

- Rashes should be covered with appropriate dressings until the patient is no longer infectious and contact with high-risk contacts must be avoided [14].

Clinical review

Clinical review is generally not required. Pharmacists should prescribe a quantity of medicine that aligns with the expected duration of therapy. Advise patients (and/or parents/caregivers, if applicable) to see an appropriate healthcare provider if the condition does not improve or resolve (see *Refer when* and *General advice*).

Pharmacist resources

- Therapeutic Guidelines: Antibiotic
  - Shingles (herpes zoster)
  - Treatment of herpes zoster ophthalmicus
- Therapeutic Guidelines: Pain and analgesia
  - Acute pain associated with shingles (herpes zoster)
  - Postherpetic neuralgia
  - Mild, acute nociceptive pain
- Australian Medicines Handbook:
  - Antivirals (Guanine analogues)
  - Paracetamol
  - NSAIDs
  - Zoster immunoglobulin
- Australian Pharmaceutical Formulary and Handbook: Treatment guideline for pharmacists (Shingles (herpes zoster))
- Australian Immunisation Handbook - Zoster (herpes zoster)
- DermNet NZ:
  - Herpes Zoster
  - Herpes Zoster (images)
  - Blistering skin conditions
- MSD Manual (Professional version): Herpes Zoster
- Skin Deep: An open-access bank of high-quality photographs of medical conditions in a wide range of skin tones for use by both healthcare professionals and the public.

Document version number	Date	Comments
Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance, revised referral criteria, and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy Hormonal Contraception - Clinical Practice Guideline

Service restricted to female patients aged 16 years or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
  - Sexual (and social) history
- Sexual and reproductive health counselling
- Examination
  - Body mass index (BMI)
  - Blood pressure measurement

Contraceptive and sexual health plan

- Select appropriate contraceptive option
  - Combined oral contraceptive pill
  - Combined hormonal vaginal ring
  - Progestogen-only contraceptive pill
  - Depot medroxyprogesterone acetate (DMPA)

Confirm contraception plan is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation
- UKMEC

Communicate agreed contraceptive plan

- Medication counselling
- General advice and patient information
- Communication with other health practitioners

Clinical review

- Measure BP
- Adverse effects
- Continue, change or stop contraception
- Communication with other health practitioners

Refer when

- The patient requests or requires a method of contraception that is not available in the Hormonal Contraception Service
- The patient has unexplained vaginal bleeding, severe or painful menstrual bleeding, irregular periods, amenorrhea, or pain/discomfort with sexual intercourse
- The patient may be pregnant
- The patient has signs and symptoms of PCOS that have not been assessed by an appropriate healthcare provider
- The patient has undiagnosed hypertension, is hypertensive, or has poorly managed hypertension
- The preferred contraceptive type is contraindicated or not appropriate

Treat and concurrently refer

- The patient has been, or suspected to have been, subject to reproductive coercion, sexual abuse or sexual violence
- The patient has a BMI >35 kg/m2
- Patient has symptoms of or risk factors for sexually transmitted infection (STI), or requests STI screening

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Refer the patient to an appropriate healthcare provider (including sexual health service or emergency services, if required) in the following situations:

- the patient requests or requires a method of contraception that is not available in the Hormonal Contraception Service e.g., long-acting reversible contraception (LARCs)
- the patient has unexplained vaginal bleeding, severe or painful menstrual bleeding, irregular periods, amenorrhea, or pain/discomfort with sexual intercourse
- the patient may be pregnant (refer to *Exclude pregnancy*)
- the patient has signs and symptoms of polycystic ovarian syndrome (PCOS) that have not been assessed by an appropriate healthcare provider
- the patient has undiagnosed hypertension, is hypertensive, or has poorly managed hypertension
- the preferred contraceptive type is contraindicated or not appropriate (UKMEC 3 or 4).

Treat and concurrently refer

Provide treatment and concurrently refer to an appropriate healthcare provider (including sexual health service or emergency services, if required) in the following situations:

- patient has been, or suspected to have been, subject to reproductive coercion, sexual abuse or sexual violence (refer to *Reproductive coercion, sexual and domestic violence and abuse*)
- patient has a body mass index (BMI) >35 kg/m^2
- patient has symptoms of or risk factors for sexually transmitted infection (STI), or requests STI screening.

Key points

- The Hormonal Contraception Service allows for the prescription of four types of hormonal contraception:
  - the combined oral contraceptive pill (COC)
  - the progestogen-only Pill (POP)
  - depot medroxyprogesterone acetate (DMPA) injection (including administration of the injection), and
  - the combined hormonal contraceptive vaginal ring.
- Patients are required to have a pharmacist consultation, including blood pressure (BP) measurement, before a hormonal contraceptive method may be prescribed, regardless of whether the patient is currently using the type of contraception requested.
- Before prescribing hormonal contraception, pharmacists should discuss the use of LARCs with women seeking contraception, as per the Royal Australian and New Zealand College of Obstetricians and Gynaecologists (RANZCOG) recommendations.
- Individuals accessing the Service should be provided with appropriate resources regarding STIs and be advised to be tested if at risk.

Patient history and assessment

Undertaking a comprehensive assessment of the patient's needs is essential to determine the most appropriate contraception method.

Initial consultation

- An individualised assessment of contraceptive and sexual health needs and suitability for hormonal contraception and management within the Service should consider:
  - patient history including sexual history
  - discussion of all available contraceptive options (including options not available in the Service that may be suitable)
  - assessment of medical eligibility and contraindications
  - BP measurement and BMI
  - risk assessment for STIs.
- Selection of an appropriate hormonal method and formulation should consider:
  - factors that could affect contraceptive effectiveness (including product efficacy)
  - how long the patient thinks they will be using contraception (e.g. <1 year, <3 years, >3 years)
  - non-contraceptive benefits
  - common side effects and risk of adverse effects
  - health risks and individualised advice.
- Comprehensive counselling and advice may include:
  - the use of tailored regimens
  - avoiding contraceptive failure, including instructions on missed pills or DMPA injections.

For detailed information on each contraceptive option, including contraindications and precautions, pharmacists should consult the *Therapeutic Guidelines: Sexual and Reproductive Health*, current guidance from The Royal Australian and New Zealand College of Obstetricians and Gynaecologists (RANZCOG) and the Royal College of Obstetricians and Gynaecologists (RCOG) Faculty of Sexual and Reproductive Healthcare (FSRH) including:

- UK Medical Eligibility Criteria for Contraceptive Use (UKMEC) [1]
- FSHR Clinical guideline – Progestogen-only Pills [2]
- FSHR Clinical guideline – Combined Hormonal Contraception [3]
- FSHR Clinical guideline – Progestogen-only injectable contraception [4]
- RANZCOG Contraception clinical guideline [5]

Sexual and reproductive health counselling

Working with Aboriginal and Torres Strait Islander people

Sexual health is not openly discussed in Aboriginal and Torres Strait Islander cultures and 'shame' (a deeply internalised feeling of inadequacy, self-doubt or ostracism) may be a strong barrier to seeking sexual health care or contraception, especially in smaller communities [6, 7].

All health care providers should be cognisant of causing additional 'shame' to Aboriginal and Torres Strait Islander people while providing reproductive counselling [6].

Pharmacists should consult the Pharmaceutical Society of Australia's Guideline for Pharmacists supporting Aboriginal and Torres Strait Islander peoples with Medicines Management for guidance on understanding culture, building relationships and communication [7].

Where appropriate, Aboriginal and Torres Strait Islander health workers or health practitioners may be involved to support clients with contraception self-management and facilitate the provision of a culturally safe service by the pharmacist [7].

Working with young people

People younger than 16 should be confidentially referred to an appropriate health care provider or sexual health clinic.

People aged 16 and 17 years are generally assumed to have the capacity to consent to medical treatment (known as Gillick Competency), including for contraception, and consent from a parent/caregiver is not required (however, it is considered good practice for the parent/caregiver to be involved in decision making around health care) [8, 9]. If the minor has the capacity to consent and does not wish to involve an adult, this should be respected [9].

Pharmacists should consider whether there may be child protection concerns relating to a request for contraception and report to Child Safety accordingly. More information is available on the Queensland Government Child Abuse website.

To assess the minor's capacity to consent to treatment, healthcare providers must consider the minor's capacity to understand the risks and benefits of the contraception (and the seriousness of the risks of using hormonal contraception and not using contraception) to make an informed decision [8,9].

Assessments of capacity to consent for minors (or adults where competency may be in doubt, for example people with an intellectual disability) must be documented in the patient record.

If the pharmacist cannot determine the minor's capacity, they should refer the patient to an appropriate healthcare provider for further assessment.

Reproductive coercion, sexual and domestic violence and abuse

Pharmacists should be aware of the possibility that a woman seeking contraception may be and/or may have been subjected to reproductive coercion, sexual violence or abuse (assault or sexual coercion), either within a relationship or outside of a relationship (e.g. from a family or community member) [10].

If the pharmacist becomes aware of this during the consultation, they should provide appropriate support and assistance, including referral to support options depending on the patient circumstances.

Referral options include to the local hospital, sexual health clinic and/or community-based sexual violence support services. A list of services, including sexual assault and domestic violence helplines and local services is provided on the Queensland Government Sexual abuse and assault: getting help webpage.

If emergency contraception (EC) is required, oral EC may be supplied as per usual pharmacy care. Ulipristal is the most effective oral EC, however the most appropriate oral EC for a patient depends on several factors. Insertion of a copper intrauterine device (IUD) is the most effective

form of EC [5]. The patient may require referral to an appropriate medical practitioner or health service for IUD insertion.

Contraceptive options for transgender and non-binary people

Transgender (trans) and non-binary people may benefit from referral to specialist sexual health services for comprehensive and culturally safe sexual healthcare that is tailored to their individual needs.

Queensland Health sexual health clinics and outreach services are located across Queensland. For more information, see www.health.qld.gov.au/clinical-practice/guidelines-procedures/sex-health/services.

Current gender identity for trans or non-binary people assigned female at birth does not impose any restrictions on methods of contraception that may be used. The same considerations apply for choosing safe and effective contraception, including personal characteristics, existing medical conditions and current drug therapies [11, 12].

Information on cervical screening should be provided to trans men and non-binary people who have a uterus [13].

Women over 50 years of age

The choice of contraceptive should be reconsidered at age 50 and at menopause [12].

Contraception is required for 12 months after the final menstrual period if it occurs after the age of 50. If the final menstrual period occurs before the age of 50, 24 months of contraception is required before cessation [14].

Some methods of contraception are not suitable for certain age groups; more information is provided below.

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- gender assigned at birth and fertility status (for transgender and non-binary patients)
- plans to become pregnant, current pregnancy status (and how this has been determined)
- recent pregnancies, and lactation status
- underlying medical conditions including mental health conditions, which may:
  - be a contraindication to hormonal contraception e.g., migraine with aura, liver disease
  - impact on contraceptive effectiveness and choice.
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements, and over-the-counter medicines)
- drug allergies/adverse reactions
- prior use of contraceptives, tolerability and adverse effects including mood-related symptoms
- smoking status

- BMI
- other factors such as age of menarche, usual menstrual cycle, heavy periods, period pain or changes in menstrual cycle, irregular or post-coital bleeding, symptoms that may suggest endometriosis (e.g., pelvic pain, severe pain with menstruation and/or sexual intercourse), symptoms that may suggest PCOS (e.g., oligomenorrhea, hirsutism, acne, weight gain)
- last STI screen and Cervical Screening Test (CST) (see *Sexually Transmitted Infections*)
- presence of genitourinary symptoms that may suggest an STI:
  - changes in vaginal or urethral discharge
  - vulval, genital skin problems or symptoms
  - lower abdominal pain
  - dysuria.
- Human Papillomavirus (HPV) vaccination status.

Sexual (and social) history

In addition to a standard patient history, pharmacists should also take a social and sexual history from the patient to inform shared decision making.

When taking a sexual history:

- Start by asking open questions about the patient's social background (to enable the patient to relax and provide the appropriate context for decision-making) [15]
- More specific questions can then be asked about risk behaviours, followed by questions around sexual practices if required [15].

The following issues should be considered to determine if the patient is at risk of an STI, or if STI testing is indicated, but may not be relevant to all people [15]:

- previous use and experiences with contraception
- current relationship status
- number of recent partners and their gender
- risky behaviours including illicit and/or intravenous drug use, alcohol use
- potential for reproductive coercion, sexual and domestic abuse (refer to *Reproductive coercion, sexual and domestic violence and abuse*)
- previous STIs and risk factors for STIs (including STI history of current and/or recent partner if applicable).

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination

The pharmacist should measure BP to determine the patient's suitability for their preferred contraceptive type, and to monitor potential impacts of contraception, including hypertension.

- BP should be measured in accordance with the National Heart Foundation of Australia Guideline for the diagnosis and management of hypertension in adults [16].
- A single elevated BP reading is not enough to classify a patient as hypertensive, and a second BP reading should be taken at the end of the consultation [16, 17].

Exclude pregnancy

Pregnancy should be reasonably excluded before commencing any hormonal contraception [18-20].

A patient can be reasonably assumed not to be pregnant if:

- there are no symptoms or signs of pregnancy

AND any of the following criteria are met:

- they have not had intercourse since the start of the last normal (natural) menstrual period, or since childbirth, abortion or miscarriage
- they have been correctly and consistently using a reliable method of contraception (barrier methods can be assumed to be reliable if used correctly in this context)
- they are within the first 5 days of the onset of a normal (natural) menstrual period, within the first 5 days after abortion or miscarriage
- they are less than 21 days postpartum (non-lactating women)
- they have not had intercourse for >21 days and have a negative urine pregnancy test [4, 12].

Use of an appropriate pregnancy test is recommended – this may include referral for a blood test for serum human Chorionic Gonadotropin (hCG) levels.

If there is any doubt regarding whether the patient may be pregnant, even if a negative urine test is returned, hormonal contraception should NOT be prescribed and the patient referred to an appropriate healthcare provider.

Contraceptive and sexual health plan

Selecting the most appropriate hormonal contraceptive option

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines in accordance with Therapeutic Guidelines: Sexual and Reproductive Health (Contraception) [21]

The choice of hormonal contraception will depend on:

- **the patient's history**
- **the patient's preference:**
    - cost (short and long term)
    - ease of use
    - management of other conditions
    - potential side effects.
- **contraceptive efficacy:**
    - When used correctly, the hormonal methods available in the Hormonal Contraception Service have similar failure rates. However, the consistency of user

adherence required is variable, and this may impact on effectiveness [22]. LARC methods (IUDs and implants) are the most reliable and cost-effective reversible methods of contraception [21].

- Pharmacists should not assume effectiveness is the only factor in an individual's choice of contraception but they should explain relative efficacy to patients [10].

Patient resources

- The Faculty of Sexual and Reproductive Healthcare: Medical Eligibility Criteria for Contraceptive Use (UKMEC) [1]
- True relationships and reproductive health: Contraception Factsheets [23]
- Jean Hailes for Women's Health: Contraception fact sheets [24]
- Family Planning New South Wales website [25]
- SPHERE Centre of Research Excellence: Which contraceptive method is right for you? [26]

Combined oral contraceptive pill

Refer to: RANZCOG & FSRH Contraception Clinical Guideline and FSRH Clinical guideline: Combined Hormonal Contraception [3, 5].

COCs containing 50 micrograms or more ethinylestradiol, or preparations containing mestranol, are NOT permitted for use in the Hormonal Contraception Service.

Choice of COC

- Comparative information about COCs to assist with COC choice is available in the Australian Medicines Handbook: Combined oral contraceptives [22].
- The COC with the lowest effective dose of estrogen and progestogen that is well tolerated and able to provide acceptable menstrual cycle control for the patient should be chosen as first-line [10, 27].
- Other guiding principles for COC pill selection are: safety profile, affordability and additional non-contraceptive benefits if desired [27].
- There have been limited head-to-head trials to guide COC selection; some pills have specific indications and non-contraceptive benefits that may assist with selection use [18].
- Newer progestogens reduce the potential for androgenic, estrogenic and glucocorticoid effects [27].
- The quadriphasic pill is indicated for heavy menstrual bleeding as well as contraception but it has complex instructions for managing a missed pill [18].
- Triphasic pills are no longer commonly used because of a lack of benefits over other types [18].
- The COCs available in Australia and for use in the Hormonal Contraception Service are listed in the Therapeutic Guidelines: Formulations of combined hormonal contraception table [18].

- COCs with a high estrogen dose (50 micrograms of ethinylestradiol, or containing mestranol) are not routinely recommended for contraception because of the unacceptable risk of venous thromboembolism (VTE) [18]. These formulations are not permitted for use in the Hormonal Contraception Service.

Combined hormonal vaginal ring

Refer to: RANZCOG & FSRH Contraception Clinical Guideline and FSRH Clinical guideline: Combined Hormonal Contraception [3, 5].

Patients that require assistance with insertion of a vaginal ring should be referred to an appropriate healthcare provider.

Progestogen-only contraceptive pill

Refer to: RANZCOG & FSRH Contraception Clinical Guideline and FSRH Clinical guideline: Progestogen-only Pills [2, 5].

Choice of POP

For patients who have a contraindication to estrogen in COC pills or prefer estrogen-free contraception, POPs can be considered [2]. Drospirenone provides more consistent suppression of ovulation than levonorgestrel or norethisterone and may provide better bleeding control [5, 20].

DMPA injection

Refer to: RANZCOG & FSRH Contraception Clinical Guideline and FSRH Clinical guideline: Progestogen-only injectables [4, 5].

DMPA may be the contraceptive of choice for individuals with contraindications to estrogen, those taking drugs that induce the CYP3A4 liver enzyme (refer to the *Australian Medicines Handbook*) or if a discrete method of contraception is preferred with less user input required.

DMPA is not a first-line option for adolescents or in perimenopausal women because of the theoretical propensity to cause bone mineral density (BMD) loss and/or limit peak BMD [12].

- During each annual consultation, pharmacists should assess the user's risk factors for low BMD and cardiovascular disease [19].
- Women who wish to continue using DMPA should be reviewed by an appropriate healthcare provider every 2 years to assess any changes in benefits and potential risks of use [4].

Confirm contraceptive plan is appropriate

Consult the current editions of the *Therapeutic Guidelines*, the *Australian Medicines Handbook*, and the *UKMEC* to confirm contraceptive management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

A hormonal contraception that is classified as UKMEC category of 3 or 4 for a patient cannot be prescribed as part of the Hormonal Contraception Service.

Communicate agreed contraceptive plan

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- individual product and medicine use
  - instructions for commencing
  - deviation from schedule (e.g. a pill is missed, a DMPA injection is not administered at 12 weeks, or a new vaginal ring is inserted too late or removed before the 3-week interval).
  - how long a secondary method of contraception must be used to avoid unplanned pregnancy
  - instructions on tailored regimens
  - advice on when fertility can be expected to return after ceasing the contraceptive.
- emergency contraceptive options that are available in instances of contraceptive failure
- how to manage adverse effects
- when to seek further care and/or treatment
  - signs of VTE and what to do if it is suspected
  - the importance of reporting new or worsening mood-related symptoms
  - before ceasing the contraceptive.
- review with an appropriate healthcare provider at least every 2 years
- when to return for follow-up.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

Other considerations

Where appropriate, individuals may be provided with additional resources to support sexual health.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources provided to patients (and parent/caregiver if applicable), and to ensure compliance with all copyright conditions.

Venous Thromboembolism

- DMPA and other progestogen-only contraceptives are not associated with a clinically significant risk of VTE [12].
- There is approximately a three times increase in the risk of VTE in women using COC, however, this increased risk is still smaller than the risk of VTE during pregnancy and immediately post-partum [27].
- The risk is highest in the first year of use, and still remains higher than for non-users after the first year, however, the absolute risk is still very low [12].
- COCs containing levonorgestrel or norethisterone have a slightly lower risk of VTE [12].

- Women should be reassured of the overall safety of the contraceptive pill. Studies show that compared to non-users of the COC, users have significantly lower death rates from cancer, cardiovascular disease and other disease, and have generally lower rates of death from any cause [27].

Contraception and cancer risks

- Users of COC have a slightly increased relative risk of breast cancer (risk returns to normal 10 years after cessation of the OCP) and cervical cancer (non-causal), however, the risk of endometrial and ovarian cancers is slightly decreased [12, 18].

Sexually Transmitted Infections (STIs)

- Individuals who may be at risk of STIs should be informed of the risk factors, the availability of screening and treatment, and strategies on how to prevent STIs.
- Pharmacists should recommend STI testing for individuals who may be at risk, even if the individual does not report any symptoms.
- Individuals should be advised of the link between HPV and cervical cancer and advised of strategies that reduce the risk, including HPV vaccination and primary HPV screening, as per the current national guidelines [19, 32-34].
- Aboriginal and Torres Strait Islander People are disproportionally affected by STIs. Consider the Australian STI Management Guidelines for use in primary care - Aboriginal and Torres Strait Islander People for priority populations testing and frequency for recommendations on STI screening [35].
- Flyers and information suitable for patients on STIs include:
  - the Queensland Government's Stop the Rise of STIs webpage [29]
  - STI factsheets and other resources designed to promote sexual health for Aboriginal and Torres Strait Islander young people are available as part of the Young Deadly Free campaign (permission for use is requested) [30].
- 13 HEALTH Webtest service:
  - The 13 HEALTH Webtest service is a free and confidential urine test for chlamydia and gonorrhoea that is available to Queenslanders aged 16 years and over, without a Medicare card, ordered by the individual via 13 HEALTH Webtest webpage [31].
  - The 13 HEALTH Webtest is not a substitute for comprehensive STI testing and is not intended to replace existing testing and treatment processes, however it provides a pathway for people who may be unlikely or unable to attend a GP or sexual health clinic.
- Access the 13 HEALTH (13 43 25 84) service to find a GP or sexual health clinic, an Aboriginal Medical Service or a community-based testing site.

National Cervical Screening Program

- A cervical screening test (CST) for HPV is recommended every 5 years for women and people with a cervix aged 25-74 years of age who have had sexual contact [32, 34, 36].

Clinical review

Clinical review with the pharmacist (or the patient's chosen healthcare provider) should occur in line with recommendations in the *Therapeutic Guidelines*.

- Patients should return for a follow up consultation **3-4 months** after initiation of a new contraceptive method (i.e. the patient has no recent experience of taking that type of contraception) to screen for adverse effects including hypertension.
- Clinical review, including measurement of the patient's BP, is recommended at 12 monthly intervals for all patients on regular contraceptive [5].
- For patients continuing therapy, follow-up should be individualised to the patient's clinical presentation, response to treatment, and therapeutic goals.

Pharmacists should prescribe a quantity of medicine (including repeats), that aligns with the expected duration of therapy, ensuring sufficient supply until the patient's next review. The total quantity prescribed should not exceed 12 months' supply (including repeats).

Pharmacist resources

- Therapeutic Guidelines: Sexual and Reproductive Health
- Australian Medicines Handbook: Drugs for contraception
- Royal Australian and New Zealand College of Obstetricians and Gynaecologists (RANZCOG): Contraception clinical guideline
- Faculty of Sexual and Reproductive Health (FSRH):
  - UK Medical Eligibility Criteria for Contraceptive Use
  - UKMEC Calculator
  - Standards and guidance
- Australasian Society for HIV, Viral Hepatitis and Sexual Health Medicine (ASHM):
  - Decision Making in Contraception
  - Australian STI Management Guidelines
- Sphere resources
  - For consumers
  - News and media
- Pharmaceutical Society of Australia Essential CPE: Women's sexual and reproductive health
- World Health Organisation: Family Planning: A Global Handbook for Providers (2018)
- Australian Pharmaceutical Formulary and Handbook (APF): Treatment guidelines for pharmacists (Emergency contraception)
- Queensland Health:
  - Aboriginal and Torres Strait Islander Adolescent Sexual Health Guideline

    - Sexual health services in Queensland
    - Online chlamydia and gonorrhoea test requests
- True relationships and reproductive health (Queensland Not for Profit organisation)
- 13 HEALTH Webtest online testing service for chlamydia and gonorrhoea: Information for health professionals
- Cancer Council: Cervical screening for LGBTIQ people
- Australian Menopause Society: Contraception

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Version 1.0	01.02.2024	
Version 1.1	01.07.2024	Version updated for the Queensland Community Pharmacy Hormonal Contraception Pilot.
Version 1.2	11.09.2024	The table outlining available combined hormonal contraception options has been removed and replaced with direct reference to Therapeutic Guidelines.
Version 1.3	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance, revised referral criteria, and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy Impetigo - Clinical Practice Guideline

Service restricted to patients aged 1 year or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
- Examination
  - Clinical features and presenting symptoms

Develop management plan

- Non-pharmacological measures
- Treatment of (or referral for) co-occurring skin conditions
- Pharmacotherapy
  - Topical antibiotics
  - Empirical oral antibiotics

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation
- Appropriate antimicrobial use

Communicate agreed management plan

- Medication counselling
- General advice and patient information
- When to seek further care/treatment
- Communication with other health practitioners

Clinical review

- Response to treatment
- Communication with other health practitioners

Refer when

- Unclear diagnosis
- Concurrent condition is suspected that cannot be treated in the community pharmacy setting
- Widespread, painful rash
- Raised purple rash that doesn't blanch
- Generalised erythema that covers ≥90% of skin surface
- Blistering of skin and mucous membranes
- Systemic signs or symptoms
- Chronic sores or ulcers
- The impetigo is widespread, severe and/or has ecthyma (ulceration) present
- Immunocompromised
- At high risk of ARF (refer to Table 2)
- Patient resides, or has recently resided in an endemic community (refer to Table 2)
- Inadequate response to optimal treatment
- Infection reoccurs frequently

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Refer the patient to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- unclear diagnosis
- concurrent condition is suspected that cannot be treated in the community pharmacy setting (e.g. viral infection, eczema herpaticum, cellulitis)
- widespread, painful rash
- raised purple rash that doesn't blanch
- generalised erythema that covers ≥90% of skin surface
- blistering of skin and mucous membranes (including mouth and eyes)
- systemic signs or symptoms, including fever, lethargy, headache, rash, nausea and vomiting
- chronic sores or ulcers
- the impetigo is widespread, severe and/or has ecthyma (ulceration) present
- immunocompromised
- at high risk of acute rheumatic fever (ARF) (refer to *Table 1*)
- patient resides, or has recently resided in an endemic community (refer to *Table 1*)
- inadequate response to optimal treatment
- infection reoccurs frequently.

Key points

- Impetigo commonly affects children between the ages of 2 and 5 but may also affect older children and adults [1].
- While impetigo is usually self-limiting and can spontaneously resolve without treatment, Australian guidelines advise that all patients with impetigo require anti-infective treatment [2, 3]. Household contacts should be assessed and treated if affected.
- Rare complications of impetigo include ARF, acute post-streptococcal glomerulonephritis (APSGN) and sepsis [3, 4].
- The management of impetigo consists of pharmacotherapy in addition to general measures for the patient, family and community to reduce the spread [5].
- People of Aboriginal and Torres Strait Islander descent presenting for treatment of impetigo should be assessed on a case-by-case basis to determine the appropriate treatment pathway, and whether referral to an appropriate healthcare provider for treatment with intramuscular (IM) benzathine benzylpenicillin is required.

Patient history and assessment

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- weight (if a child)
- pregnancy and lactation status (if applicable)
- ethnicity (Aboriginal and Torres Strait Islander, Māori or Pacific Islander)
- most recent place of residence/community and/or recent travel to a place where impetigo or ARF may be endemic (a high rate of ARF within the population)
- onset, duration, nature, location and extent of lesions
- recent skin, throat or other infections (e.g. respiratory tract infection) and treatment received
- other signs and symptoms e.g., pain, lymphadenopathy, signs of sepsis or other complications (including APSGN and ARF) such as fever, confusion, tachycardia, hypotension or hypertension, clammy skin, vomiting and diarrhoea, facial or peripheral oedema, severe headache, joint pain and/or hot swollen joints, swelling beneath impetigo lesions
- potential source of infection e.g., skin trauma, contact with people with similar symptoms
- underlying medical conditions including, skin conditions or immunosuppression that may lead to complications (e.g., atopic dermatitis, scabies, herpes simplex)
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)
- drug allergies/adverse reactions
- other risk factors including poor hygiene, day care settings, crowding and malnutrition.

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination

Clinical features and presenting symptoms

Impetigo can be classified as primary or secondary form:

- the primary form is a direct bacterial infection of otherwise healthy skin
- the second form (most common) is bacterial infection of a break in the skin from trauma or pruritic conditions (impetiginisation) [6-8].

Impetigo may have a similar presentation or co-occur with other common skin conditions including contact dermatitis, thermal burns, folliculitis, dermatophytosis, candidiasis, eczema, scabies, herpes zoster, atopic dermatitis, varicella or molluscum contagiosum [4, 9-10].

Table 1. Types of impetigo [1, 2, 7, 11-13]

Type	Description
Non-bullous	• Characterised by thin-walled pustules or vesicles that may be itchy, usually not painful, rupture quickly and progress to honey-coloured crusts. Lesions have minimal or no surrounding erythema.

	• May start with a single vesicle that coalesces with others; often self-inoculation leads to multiple lesions, particularly on the face and extremities (although any body part can be affected). • Patients are generally otherwise well, although they may have regional lymphadenopathy. • Lesions may resolve spontaneously within 2–4 weeks without treatment and heal without scarring.
Bullous	• Presents as irritating, larger (diameter often > 1 cm) fluid-filled vesicles and blisters that rupture quickly to broad-based bullae, and progress to a thin, flat yellow/brown crust. • Tends to occur on moist intertriginous areas including the nappy area, axillae and neck, as well as the face, trunk and extremities. • Lesions typically have scaling on the border of the bullae (collarette). • Patients with bullous impetigo are more likely to experience systemic symptoms including fever, malaise and lymphadenopathy. • Lesions may resolve spontaneously within 2–4 weeks without treatment and heal without scarring.
Ecthyma (deep impetigo)	• Characterised by crusted sores with underlying ulcers. ○ If the crust is removed, an indurated ulcer will appear red, swollen and oozing. ○ Most commonly affects the buttocks, thighs, legs, ankles and feet. • People more likely to develop ecthyma include children, immunocompromised people, people with untreated impetigo and people living in crowded conditions with poor hygiene. • Lymphadenopathy may occur. • Lesions resolve slowly and sometimes spontaneously without treatment, although they may gradually enlarge and also scar.

Investigations

Non-bullous impetigo infection can be caused by *Staphylococcus aureus* or *Streptococcus pyogenes (Group A streptococcus),* or both. Bullous impetigo is usually caused by *Staphylococcus aureus* [2, 7].

Patients with impetigo do not require an initial skin swab prior to commencing empirical antibiotic therapy unless the diagnosis is uncertain or impetigo is recurrent [2, 7, 12].

Complications of Impetigo

Complications of impetigo, particularly bullous and ecthyma, include:

- lymphangitis, lymphadenitis
- widespread infection, cellulitis, gangrene and bacteraemia
- permanent scarring [3, 7, 8, 11, 14].

Complications arising from Group A Streptococcal skin infections (more likely within endemic settings) include:

- APSGN and chronic kidney disease

- ARF and rheumatic heart disease (RHD)
- sepsis
- osteomyelitis, septic arthritis [3, 5, 8, 10].

APSGN

- APSGN is an immune-mediated sequalae of nephritogenic strains of *S. pyogenes.*
- May occur approximately 2-3 weeks after a skin or throat infection of group A streptococcus bacteria [15, 16].
- May affect any age but most commonly affects children aged between 12 months and 17 years [15, 16].
- Common symptoms and clinical features include obvious facial and orbital swelling, particularly upon waking, elevated blood pressure, proteinuria and macroscopic haematuria with dark brown urine, and lethargy, weakness and/or anorexia [16].

ARF

- ARF is an immune-mediated sequalae of *S. pyogenes* involving multiple system organs, including the heart, joints and central nervous system [17, 18].
- May occur between 1-5 weeks post-streptococcal infection, and is most common after recurrent streptococcal infections, particularly pharyngitis and impetigo [17, 18].
- Most frequently affects children aged between 5 and 14 years, although recurrent episodes may continue into the fourth decade of life [17].
- Common symptoms and clinical features include sore and/or swollen joints, fever, increased resting heart rate facial or peripheral oedema, Sydenham chorea [10].

People considered at high risk of developing ARF are specified in Table 1.

Table 1. Individuals at high risk of developing ARF [2, 17, 19]
• Aboriginal and Torres Strait Islander people residing in a rural or remote area, or living in a household affected by household overcrowding (>2 people per bedroom) or experiencing socioeconomic disadvantage. • Māori and/or Pacific Islander person living in a household affected by overcrowding (>2 people per bedroom) or experiencing socioeconomic disadvantage. • A person with a personal history of ARF or RHD. • A person with a family or household member with a recent history of ARF or RHD.
Additional risk factors for individuals aged ≤40 years (particularly between 5-20 years)
• People living in a household affected by household overcrowding (>2 people per bedroom) or experiencing socioeconomic disadvantage. • People with current or prior residence in (or frequent or recent travel to) an area with a high rate of ARF e.g., refugees and migrants from low-middle income countries, rural and remote communities.

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Treatment of impetigo supports skin healing and prevents the development of more serious complications (e.g. sepsis, bone and joint infections, poststreptococcal complications including ARF) [2].

Pharmacist management of impetigo involves:

- **non-pharmacological measures:**
    - Advice and education to reduce the spread of impetigo as per the Therapeutic Guidelines: Antibiotic (Skin and soft tissue infections - Impetigo) [2] and the Royal Children's Hospital Melbourne: Impetigo (school sores) fact sheet [20].
- **examination and treatment of co-occurring skin conditions** (e.g. dermatitis, scabies, head lice, tinea) or referral to an appropriate healthcare provider.
- **pharmacotherapy:** NB1
    - A topical or oral medicine mentioned in the Therapeutic Guidelines: Antibiotic (Skin and soft tissue infections - Impetigo) [2].
    - Topical anti-infectives are considered sufficient treatment for patients with nonrecurrent limited non-bullous impetigo (up to 2 sores) who are not at high risk of ARF.
    - Topical antibiotics are not appropriate if the infection is widespread or multiple family/community members are infected, due to the risk of rapid resistance [1-3].
    - Pharmacists may prescribe oral antibiotics if impetigo is in a localised area with more than 2 sores, unless referral criteria apply. See '*Refer when*'.

NB1: The correct use of antibiotics is essential. Prolonged or incorrect use may increase the risk of antimicrobial resistance, particularly for topical antibiotics [5, 13]. Discussing the treatment regimen with patients (and/or parent/caregiver) reduces the risk that antibiotics will be incorrectly used [5]. Where the patient (and/or parent/caregiver) ability and/or motivation to correctly use topical treatment is in doubt e.g., when there are multiple sores, oral antibiotics should be considered, or referral to an appropriate healthcare provider for IM benzathine benzylpenicillin [2, 5]. Refer to the Primary Clinical Care Manual for further information [10].

Confirm management is appropriate

Consult the *Therapeutic Guidelines*, *Australian Medicines Handbook* and other appropriate resources to confirm that management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

Reminder

The prescribing and use of antibiotics should be consistent with the principles described in the Therapeutic Guidelines: Antibiotic. Pharmacists should carefully assess whether antibiotic therapy is clinically indicated, and if so, consider local antimicrobial resistance patterns and relevant clinical factors when selecting therapy.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- individual product and medicine use
- non-pharmacological measures
- how to manage adverse effects
- when to seek further care and/or treatment, including recognising complications of impetigo, particularly ARF and APSGN.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources provided to patients (and/or parents/caregivers) and to ensure compliance with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.



Patient resources

- The Royal Children's Hospital Melbourne Fact sheet: Impetigo (school sores) [20]

General advice

At the time of initial consultation, patients (and/or parents/caregiver) should be advised to see an appropriate healthcare provider if:

- the condition does not improve within 5–10 days of starting treatment
- the patient's signs or symptoms worsen
- the patient develops new signs or symptoms (e.g. systemic signs or symptoms)
- the condition reoccurs after treatment course is completed
- the adverse effects of treatment cannot be managed in the pharmacy setting.

Clinical review

Clinical review is generally not required. Pharmacists should prescribe a quantity of medicine that aligns with the expected duration of therapy. Advise patients (and/or parents/caregivers) to see an appropriate healthcare provider if the condition does not improve or resolve (see *Refer when* and *General advice*).

Pharmacist resources

- Therapeutic Guidelines: Antibiotic
  - Impetigo
  - Acute rheumatic fever
- Australian Medicines Handbook:
  - Antibacterials (skin)
  - Antibacterials
  - Skin and soft tissue infections
- Australian Pharmaceutical Formulary and Handbook (APF): Treatment guideline for pharmacists (Impetigo)
- Queensland Health and Royal Flying Doctors Service (Queensland branch): Primary Clinical Care Manual (12th ed. 2025)
  - Impetigo
  - Acute post streptococcal glomerulonephritis
  - Acute rheumatic fever
- DermNet NZ:
  - Impetigo
  - Ecthyma
- MSD Manual (Professional version): Impetigo and Ecthyma
- Australian Family Physician (Royal Australian College of General Practitioners): Managing skin infections in Aboriginal and Torres Strait Islander children
- Telethon Kids Institute: National Healthy Skin Guideline: for the Prevention, Treatment and Public Health Control of Impetigo, Scabies, Crusted Scabies and Tinea for Indigenous Populations and Communities in Australia
- Rheumatic Heart Disease Australia: ARF RHD Guideline and risk calculator
- Australian Journal of General Practice: A refresher on the primary care and public health management of acute post-streptococcal glomerulonephritis
- Skin Deep - An open-access bank of high-quality photographs of medical conditions in a wide range of skin tones for use by both healthcare professionals and the public.

Document version number	Date	Comments
Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy

Management for Overweight and Obesity - Clinical Practice Guideline

Service restricted to patients aged 18 years or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
- Examination and screening for common comorbidities
- Investigations/test results

Develop management plan

- Non-pharmacological measures
  - Lifestyle modification
  - Reduced energy diets and low energy diets
- Secondary weight loss strategies
  - Very low energy diets
  - Pharmacotherapy
    - Orlistat

Confirm management is appropriate

- Contraindications and precautions and interactions
- Pregnancy and lactation

Communicate agreed management plan

- Weight management plan including goals
- Medication counselling
- General advice and patient information
- Communication with other health practitioners

Clinical review

- Response to strategies
- Continue, modify or stop treatment
- Communication with other health practitioners

Refer when

- The patient is pregnant or lactating
- BMI > 40 kg/m2
- Comorbidity not being managed by an appropriate healthcare provider or that requires specialist medical management for weight loss
- Previous bariatric surgery
- Inadequate response to Management Plan B (<5% loss after 12 weeks)
- Taking a medicine associated with weight gain

Treat and concurrently refer

- Patients with diagnosed (or suspected) psychiatric disorders, including psychosis, schizophrenia, bipolar disorder, major depression, anxiety or eating disorder (binge eating, anorexia nervosa, bulimia), or a history of eating disorders
- Patients with clinically significant nutritional (vitamin or mineral) deficiencies
- Patients under active medical treatment and supervision for weight reduction
- Patients with CVD, asthma or COPD
- Patients aged >65 years
- Investigations indicate further investigation of a possible comorbidity is required

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Refer the patient to an appropriate healthcare provider in the following situations:

- Patient is pregnant or lactating
- BMI > 40 kg/m^2
- Comorbidity not being managed by an appropriate healthcare provider* or that requires specialist management for weight loss
- Previous bariatric surgery
- Taking a medicine associated with weight gain
- Inadequate response to Management Plan B (<5% loss after 12 weeks).

Treat (if clinically appropriate) and concurrently refer

Provide initial treatment and concurrently refer to an appropriate healthcare provider for collaborative management in the following situations:

- Patients with diagnosed (or suspected) psychiatric disorders, including psychosis, schizophrenia, bipolar disorder, major depression, anxiety or eating disorder (binge eating, anorexia nervosa, bulimia), or a history of eating disorders
- Patients with clinically significant nutritional (vitamin or mineral) deficiencies
- Patients under active medical treatment and supervision for weight reduction
- Patients with CVD, asthma or COPD*
- Patients aged >65 years
- Findings indicate further investigation of a possible comorbidity is required*.

Reminder

*Patients with hypertension, dyslipidaemia, mild COPD and mild to moderate asthma may be suitable for management as part of the Queensland Community Pharmacy Chronic Conditions Management Pilot – refer to the respective clinical protocols for information and eligibility criteria if in a participating pharmacy.

Key points

- Human behaviour, genetics, environment and lifestyle all play a role in managing body weight [1, 2].
- The social determinants of health influence a patient's risk of being overweight and obese, of which social inequality and disadvantage contribute to unfair and avoidable differences in health outcomes within groups in society [3].
- Excess weight (overweight and obesity), generally defined in adults as a body mass index (BMI) ≥ 25 kg/m^2, is a risk factor for metabolic and chronic conditions, including type 2 diabetes, cardiovascular disease (CVD) and some cancers, leading to higher rates of premature morbidity and mortality [4, 5].
- Body mass index (BMI) is an internationally recognised standard for classifying overweight and obesity in adults, particularly in a population health context [4, 6]. BMI does not directly

measure body fat and may be insensitive to body composition, particularly in older people and very muscular people/athletes; however, it is strongly correlated to other direct measures of body fat and obesity-related comorbidities (CVD) [6, 7].

- Excess body weight for height is associated with osteoarthritis, poor mental health and conditions affecting the reproductive and gastrointestinal systems [7].
- While many people can achieve short-term weight loss, sustained weight loss/maintenance over the long-term (over 2 years) is required for cardiovascular benefits [7].
- Effective and long-term management of overweight and obesity requires a multidisciplinary team care approach tailored to the individual [4, 8-10].
- Lifestyle modification is the foundation of weight management and for some patients, lifestyle modification alone may be sufficient to achieve weight loss; evidence suggests that interventions that address nutrition, physical activity and psychological approaches to behavioural change are more effective for weight loss than single interventions [2, 7]. Social prescribing to local non-clinical services may support lifestyle modification.
- Additional strategies may include behavioural and psychological therapy, pharmacotherapy and bariatric surgery [11].
- Weight and weight loss are sensitive issues for many people and most individuals with overweight and obesity will have attempted multiple weight loss interventions [12]. it is important to communicate and provide care to patients in a way that is person-centred, culturally sensitive, non-judgmental, and private [13].

Patient history and assessment

Each patient's needs regarding weight management and suitability for pharmacist management should be assessed on a case-by-case basis.

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- pregnancy and lactation status (if applicable)
- weight history including highest and lowest weights (and when)
- previous weight loss and strategies used, including the degree of success and levels of medical/health practitioner support (e.g., previous drug therapy, very-low energy diets (VLED), exercise)
- recent relevant biochemical markers (see *Investigations*)
- medical conditions and comorbidities, including weight and non-weight related
- known, suspected or previous eating disorder (see *Appendix 1*)
- surgical history, including any bariatric surgery
- current and recently ceased treatments (including prescribed medicines, over-the-counter medicines, vitamins, herbs, powders (like protein powders), other supplements)

- drug allergies/adverse drug reactions
- alcohol and drug history
- smoking status
- diet history (usual intake over the last 6-month period) and fluid intake. Use objective measures (e.g. food diary, food tracking app, measure portions)
- current level of physical activity (structured and incidental activity)
- Other factors impacting upon successful weight management:
    - cooking/shopping skills
    - health and nutritional literacy
    - cooking/shopping responsibilities, including the number of people in the household
    - stage of readiness to change
    - patient support networks.

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination and screening

All patients should undergo basic screening and examination for common weight-related comorbidities and eating disorders. Information to guide screening has been provided in Appendix 1. However, it is also recommended that the patient be comprehensively screened, either as part of the Queensland Community Pharmacy Pilot Cardiovascular Disease (CVD) Risk Reduction Program (if eligible) or by referred to their usual healthcare provider.

If the patient is enrolled in the Queensland Community Pharmacy Pilot CVD Risk Reduction Program, examination does not need to be repeated (relevant screening that has not previously been undertaken should still occur).

Investigations

Recommended investigations are outlined in Table 2.

If available, recent test results may be used (e.g., provided by a medical practitioner with referral or available on My Health Record) if there have been no changes to the patient's medication or health status and the test was conducted within the past 12 months.

Consider investigation for secondary cause of weight change if clinically suspected based on patients presenting signs and symptoms, e.g. thyroid dysfunction.

Table 2. Recommended investigations/tests (based on personal and clinical circumstances)	
Glycated haemoglobin (HbA1c) or fasting blood glucose (FBG) [14] NB1	
Testing requirements	**Referral requirements**
• People **without** a previous diagnosis of diabetes **AND** • no HbA1c or FBG test results from the previous 3 years **AND** • at least one of the following criteria are met ○ AUSDRISK score of ≥12 ○ Aboriginal and Torres Strait Islander people aged ≥18 years ○ Pacific Islander, Indian sub-continent, Southern European, North African, Latin American, Middle Eastern or Asian background ○ history of gestational diabetes ○ polycystic ovary syndrome ○ taking antipsychotic medicines ○ aged ≥40 and overweight or obese ○ aged 18–40, overweight or obese and hypertension ○ aged 18–40, overweight or obese and clinical evidence of insulin resistance ○ history of a cardiovascular event ○ history of impaired glucose tolerance (IGT) or impaired fasting glucose (IFG) ○ has a first degree relative with type 2 diabetes ○ 1 or more classical symptoms of diabetes (weight loss, polyuria, polydipsia, blurred vision).	Concurrent referral to an appropriate healthcare provider is required for further investigation of diabetes when: • HbA1c of ≥6.5% (48 mmol/mol) • Fasting blood glucose ≥7.0 mmol/L. **Immediate/emergency referral:** HbA1c >10% (86 mmol/mol) or any blood glucose level ≥ 20.0 mmol/L is considered to be severe hyperglycaemia and is a medical emergency in patients who are unwell (signs of ketosis, dehydration, vomiting, signs or symptoms of underlying infection, confusion and/or delirium/altered level of consciousness) [15].
Serum lipids (fasting) (including LDL-C, HDL-C and triglycerides) [16-18]	
Testing requirements	**Referral requirements**
• If lipid testing has not been performed in the previous 5 years **AND** ○ Aboriginal and Torres Strait Islander people aged ≥30 years **OR**	Concurrent referral to an appropriate healthcare provider for further investigation of dyslipidaemia when: • total cholesterol (TC): ≥5.5 mmol/L (or >4.0 mmol/L for people on lipid modifying drug therapy)

○ non-Aboriginal or Torres Strait Islander people aged ≥45 years.	• LDL-C: ≥2.0 mmol/L • HDL-C ≤1.0 mmol/L • Triglycerides: ≥2.0 mmol/L.
BP measurement [19]	
Testing requirements	**Referral requirements**
All patients: • BP measurement should be conducted in accordance with the National Heart Foundation of Australia: Guidelines for the diagnosis and management of hypertension in adults 2016 [19].	Concurrent referral to an appropriate healthcare provider for further investigation of hypertension and CVD: • BP ≥140/90 mmHg. **Immediate/ emergency referral:** Systolic BP ≥180 and/or diastolic BP ≥110 mmHg is considered to be severe hypertension and is a medical emergency in patients with signs or symptoms of a hypertensive urgency (headache or dizziness).
Thyroid stimulating hormone (TSH) [20-22]	
Testing requirements	**Referral requirements**
• If clinically suspected based on patient signs and symptoms. • Weight gain may be a symptom of hypothyroidism.	Concurrent referral to an appropriate healthcare provider for further investigation of thyroid function if: • TSH >4 mlU/L.

NB1: Pharmacists should consider the limitations of HbA1c tests and factors that may cause hyperglycaemia and misleading test results, as per the Therapeutic Guidelines: Tests to diagnose diabetes, and the RACGP: Management of type 2 diabetes [14, 15].

Develop management plan

The management of overweight and obesity, in accordance with the Australian Obesity Management Algorithm is centred around lifestyle interventions (initial weight-loss strategies) that are tailored to relevant factors for the individual [12].

Management may include more intensive interventions (secondary weight-loss strategies) if initial weight loss strategies have not been effective (if clinically indicated and appropriate) [12].

Lifestyle interventions are indicated for all patients with overweight or obesity. The Australian Obesity Management Algorithm provides guidance on the use of pharmacotherapy for patients with BMI ≥30 kg/m^2 [12]. Orlistat is indicated for patients with BMI≥27 kg/m^2 and other cardiovascular risk factors [23].

Pharmacists should also consider referral to, and collaboration with, other members of the multidisciplinary health team, depending on the patient's needs, location and financial means.

Management Plan A (Initial weight loss strategies)

Lifestyle modification (for relevant factors):

- Counselling and education for lifestyle modification[NB1] as per the 5As framework detailed in the RACGP Smoking, nutrition, alcohol and physical activity (SNAP) guide [24] the

Handbook of Non-Drug Interventions (HANDI) [25] and the Australian Obesity Management Algorithm [12].

- It is recommended that consenting patients are referred to an appropriate healthcare provider for physical activity and psychosocial interventions, where required.

Reduced energy diets (RED) and low energy diets (LED):

- In accordance with the Australian Obesity Management Algorithm [12].
- It is recommended that consenting patients are referred to a dietitian for specialised and individualised advice and a tailored eating plan.

Management Plan B (Secondary weight loss strategies)

Management Plan B can be considered for individuals who have not adequately responded to initial weight loss strategies as described in Management Plan A, including where these may have been attempted outside of this Service as part of another structured weight loss program [12].

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Very low energy diets (VLEDs)[NB2]:

- For individuals who have not adequately responded to a RED or LED, in accordance with individual product instructions and indications, and the Australian Obesity Management Algorithm [12].
- It is recommended that consenting patients are referred to a dietitian for specialised and individualised advice and a tailored eating plan.

Pharmacotherapy:

- For patients following Management Plan A showing less than a 5% decrease in weight after 3 months **AND** with a BMI >27 kg/m^2 (if other cardiovascular risk factors are present) or BMI >30 kg/m^2 (no other CVD risk factors or comorbidities):
    - prescribe Orlistat, in accordance with the Australian Medicines Handbook: Orlistat [23, 26]. Liver function should be monitored. If Orlistat is unsuitable, refer to an appropriate healthcare provider for other pharmacological management.

NB1:

- For smoking cessation, refer to the Queensland Community Pharmacy Smoking Cessation Clinical Practice Guideline for management.
- For recommendations of alcohol consumption, refer to the Australian guidelines to reduce health risks from drinking alcohol [27].
- For advice regarding physical activity focusing on increasing energy expenditure, refer to the Physical activity and exercise guidelines for all Australians [28].
- For general diet/nutrition advice focusing on reducing energy intake and optimising diet quality as per the Australian Dietary Guidelines [29].

NB2: VLEDs are recommended to be used in collaboration with a dietitian and medical practitioner for periodic monitoring of biochemistry (electrolytes, creatinine, liver function tests, fasting glucose, lipids, uric acid) and haematology (full blood count, iron studies).

Setting appropriate weight management goals

Weight management is challenging for most people. Weight loss prompts compensatory physiological responses that oppose further weight loss and promote weight gain.

Weight loss goals should be set in collaboration with the patient and reviewed regularly. Targets and goals should be patient-centred and individualised, and encourage sustainable, weight loss that can be maintained over the long-term [7, 12].

A weight loss of between 5% and 10% is achievable for most overweight adults and will provide cardiovascular and other health benefits.

Encourage patients to focus on their achievements and progress, rather than what they have not achieved [30, 31]. In some cases, it may be more useful to encourage them to focus on preventing weight gain than trying to lose weight [32].

Confirm management is appropriate

Consult the *Australian Medicines Handbook*, the Australian Obesity Management Algorithm and other relevant references to confirm that management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information if required), should be provided to the patient regarding the weight management plan. Relevant information may include:

- instructions for individual product and medicine use
- non-pharmacological measures
- how to manage adverse effects, including serious adverse effects requiring medical care
- when to return for follow up.

A Weight Management Plan template has been provided in Appendix 2 for guidance.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources and information provided to patients (and/or parents/caregivers), and compliance with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

Patient resources

- Queensland Government websites:
  - Staying healthy
  - Healthier Happier
- Queensland Health: Nutrition Education Materials
- Baker Heart and Diabetes Institute: Health Hub Fact sheets

- Australian Government: Eat for health calculators

Clinical review

Follow up and clinical review is an important component of overweight and obesity management. Clinical review with the pharmacist (or the patient's chosen healthcare provider) should occur in line with recommendations in the *Australian Obesity Management Algorithm*.

Clinical review on a monthly basis is recommended to monitor progress towards the weight loss target, efficacy of lifestyle modification, and to monitor for adverse effects.

Clinical review appointments may include:

- weight and waist circumference measurements, and BMI calculation
- blood pressure monitoring
- brief advice and reinforcement of lifestyle modification
- review of adverse effects.

Pharmacists may also consider repeating laboratory testing (where indicated).

For patients continuing therapy, follow-up should be individualised to the patient's clinical presentation, response to treatment, and therapeutic goals.

Pharmacists should prescribe a quantity of medicine (including repeats), that aligns with the expected duration of therapy, ensuring sufficient supply until the patient's next review.



Pharmacist resources

- Therapeutic Guidelines:
  - Cardiovascular
  - Diabetes
  - Respiratory
  - Psychotropic
- Australian Medicines Handbook:
  - Orlistat
  - Liraglutide
  - Naltrexone with bupropion
  - Phentermine
  - Semaglutide
  - Tirzepatide
- Australian Pharmaceutical Formulary and Handbook (APF): Treatment guideline for pharmacists (Weight management)
- The Australian Obesity Management Algorithm: A simple tool to guide the management of obesity in primary care
- Royal Australian College of General Practitioners:
  - Smoking, nutrition, alcohol, physical activity (SNAP): A population health guide to behavioural risk factors in general practice

    - Guidelines for preventive activities in general practice
    - Handbook of non-drug interventions (HANDI)
- National Health and Medical Research Council:
    - Australian Dietary Guidelines
    - Australian Guidelines to reduce health risks from drinking alcohol
- Australian Government Department of Health and Aged Care: Physical activity and exercise guidelines for all Australians
- Australian CVD risk calculator: CVDCheck
- StatPearls: Obesity and Comorbid Conditions
- Australian Family Physician:
    - Pharmacotherapy for obesity
    - Obesity and weight management at menopause (includes detailed information on VLEDs)
    - Bariatric-metabolic surgery: A guide for the primary care physician
    - The bariatric surgery patient: nutrition considerations
- Australian Journal of General Practice: Bariatric surgery
- The Australian and New Zealand Obesity Society
- Australian peak body for obesity: The Obesity Collective
- Diabetes, Metabolic syndrome and Obesity: Medications that cause weight gain and alternatives in Canada: A narrative review
- Butterfly Foundation – Eating disorders and body image support
- InsideOut Institute for Eating Disorders

Appendix 1 – Screening for weight related co-morbidities

Co-morbidity screening (based on personal and clinical circumstances)

Obstructive Sleep Apnoea [12, 33, 34]

Screening using the STOP-BANG questionnaire

STOP	YES (=1)	NO (=0)
*Do you **SNORE** loudly (louder than talking or loud enough to be heard through closed doors)?*		
*Do you often feel **TIRED**, fatigued, or sleepy during daytime?*		
*Has anyone **OBSERVED** you stop breathing during your sleep?*		
*Do you have or are you being treated for high blood **PRESSURE?***		
BANG		
***BMI** more than 35kg/m^2?*		
***AGE** over 50 years old?*		
***NECK** circumference >16 inches (40cm)?*		
***GENDER**: Male?*		
Total Score		

Referral requirements

Concurrent referral to an appropriate healthcare provider for further investigation is recommended for all patients suspected of having sleep apnoea or another weight-related respiratory condition, for example:

- Patients with at least one risk factor or comorbidity + multiple symptoms of sleep apnoea
- A score of ≥3 on the STOP-Bang Questionnaire.

BMI and waist circumference [2, 4, 7, 35]

Requirements	Referral requirements
All patients: • Measurement of BMI and waist circumference at commencement and regular reviews (recommended monthly) is required to inform the management plan and assist with patient education regarding risk factors for obesity comorbidities. • A larger waist circumference indicates visceral fat deposits and central obesity: ○ Males with a waist circumference ≥94 cm are at increased risk of CVD and other comorbidities (≥102 cm = a greatly increased risk, especially if BMI >25 kg/m^2). ○ Females with a waist circumference ≥80 cm are at increased risk (≥88 cm = a greatly increased risk, especially if BMI >25 kg/m^2).	Referral to an appropriate healthcare provider or specialist weight loss service is recommended for people with: • BMI > 40 kg/m^2.

Mental health/psychological distress [12, 36, 37]

Screening requirements	Referral requirements

All patients: • Pharmacists may use a validated screening tool such as the Kessler Psychological Distress Scale (K10) [38].	Concurrent referral to an appropriate healthcare provider is recommended for patients suspected of having a mental health disorder or in psychological distress, for example: • Patients who score ≥20 on the K10.
Eating disorders [39, 40]	
Screening requirements	**Referral requirements**
All patients: Pharmacists may use a validated screening tool such as the SCOFF Questionnaire or InsideOut Screener	Concurrent referral to an appropriate healthcare provider is recommended for patients with or suspected of having an eating disorder.
Diabetes [12, 14, 15]	
Screening requirements	**Referral requirements**
Patients aged ≥40 years: • Risk of developing type 2 diabetes should be estimated every 3 years using the validated Australian type 2 Diabetes Risk Assessment Tool (AUSDRISK). All patients: • Clinical signs and symptoms of insulin resistance including acanthosis nigricans, skin tags, central obesity, hirsutism.	Not required for screening, proceed with recommended investigations/tests in Table 2.
Dyslipidaemia [7, 41, 42]	
Screening requirements	**Referral requirements**
All patients: • Family history of familial hypercholesterolaemia • Clinical signs and symptoms including tendon xanthomata, arcus cornealis before 45 years of age and xanthelasma	Concurrent referral to an appropriate health care provider is recommended for patients with a history of familial hypercholesterolemia.

Appendix 2 – Weight Management Plan

Page 1

Weight Management Plan

Plan date:

Name:		**Date of birth:**	
Date of commencement:			
Pharmacy details			
Pharmacist name		**Phone number**	
Pharmacy name and address		**Opening hours**	

Baseline weight and measurements

Weight:	kg	**BMI:**		**Waist circumference:**	cm

Weight loss goal tracker:

Week beginning	Goal weight	Actual weight	BMI	Waist circumference

My lifestyle prescription

Diet and nutrition	• Referral to a dietitian for an individualised eating plan and education • Summary of nutritional advice for a balanced diet from the *Australian Dietary Guidelines* ○ more of/ increase… ○ less of/ limit… • Specific advice and food recommendations and strategies tailored for each individual's barriers and risks • Education and advice on interpreting nutritional labels and referral to resources to support cooking and shopping skills
Exercise and physical activity	• Enter recommendations for physical activity for patient's age and capability based on the *Physical activity and exercise guidelines for all Australians*: ○ Informal exercise e.g. building exercise into everyday activities ○ Formal exercise e.g., walking (moderate intensity) for 30 minutes 5 days of the week, strength building 2 days per week ○ individualised guidance for building up to recommendations • If required: Referral to a GP, exercise physiologist, physiotherapist or other supports for safe exercise
Other lifestyle modification strategies	• If required: Smoking cessation ○ Refer to Service smoking cessation program and/or other supports e.g. Quitline or GP • If required: Referral to a GP, psychologist or other clinician for mental health support • Specific advice for sleep and if required referral to a GP for management of obstructive sleep apnoea • If required: summary of advice regarding alcohol consumption based on the *Australian guidelines to reduce health risks from drinking alcohol*

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Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy

Mild, Acute Musculoskeletal Pain - Clinical Practice Guideline

Service restricted to patients aged 18 years or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
  - Pain history and assessment
- Examination
  - Vital signs
  - Examine area

Develop management plan

- Non-pharmacological measures
- Pharmacotherapy
  - Oral analgesia

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation

Communicate agreed treatment plan

- Medication counselling
- General advice and patient information
- When to seek further care and/or treatment
- Communication with other health practitioners

Clinical review

- Response to treatment
- Communication with other health practitioners

Refer when

- Acute swelling, erythema and significant reduction in range of motion of a joint
- Headache and acute onset of visual or auditory disturbances
- Digital ischaemia
- Suspected nerve damage or mononeuropathy
- Neurological signs including altered bladder or bowel function
- Systemic symptoms suggestive of serious pathology including fever, malaise, weight loss, nausea and vomiting
- Pain cannot be attributed to acute mechanical MSK pain (chronic pain or inflammatory pain)
- History of chronic inflammatory MSK pain or complex underlying rheumatological conditions (including acute flares of these conditions)
- The patient is at high risk of acute rheumatic fever
- The patient rates their pain as moderate to severe
- The patient is unable to accurately rate their pain due to cognitive impairment or other special needs
- The patient requests treatment with an opioid
- The patient is opioid-tolerant or recovering from an opioid use disorder
- Inadequate response to optimal treatment

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Refer the patient to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- acute swelling, erythema and significant reduction in range of motion of a joint
- headache and acute onset of visual or auditory disturbances
- digital ischaemia
- suspected nerve damage or mononeuropathy (e.g., loss of feeling, weakness, pain or burning and/or paraesthesia ('pins and needles'))
- neurological signs including altered bladder or bowel function
- systemic symptoms suggestive of serious pathology including fever, malaise, weight loss, nausea and vomiting
- pain that cannot be attributed to acute mechanical MSK pain (chronic pain or inflammatory pain)
- history of chronic inflammatory MSK pain or complex underlying rheumatological conditions (including acute flares of these conditions)
- the patient is at high risk of acute rheumatic fever
- the patient rates their pain as moderate to severe
- the patient is unable to accurately rate their pain due to cognitive impairment or other special needs
- the patient requests treatment with an opioid
- the patient is opioid-tolerant or recovering from an opioid use disorder
- inadequate response to optimal treatment.

Key points

- Acute musculoskeletal (MSK) pain is generally mechanical, caused by damage to tissue or structural changes to joints, vertebrae and muscles/soft tissue [1-3].
- While mechanical pain can lead to inflammation, it is different to inflammatory pain caused by an underlying chronic inflammatory disease such as arthritis, that results in chronic pain and is treated differently [1].
- Injury is the most common cause of acute mechanical MSK pain. Acute MSK pain is expected to be self-limiting (last less than 3 months), with a return to usual function as the underlying injury resolves [2, 4].
- Non-pharmacological therapies and lifestyle measures (where there is a proven benefit) and pharmacotherapy (paracetamol and non-steroidal anti-inflammatory drugs (NSAIDs)) should be considered in the overall management plan for acute, mild MSK pain [5-8].
- Opioids are not required for the management of acute, mild MSK pain and cannot be prescribed by pharmacists [9].

Patient history and assessment

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines provided.

The patient history should consider:

- age
- pregnancy and lactation status (if applicable)
- underlying or co-existing medical conditions, including rheumatological and autoimmune conditions such as inflammatory arthritis and psoriasis
- family history of inflammatory pain, rheumatological and/or autoimmune conditions e.g., ankylosing spondylitis or inflammatory arthritis
- non-musculoskeletal symptoms that may indicate serious pathology, including weight-loss or gain, fever, malaise, nausea, vomiting and sweating
- risk factors for developing acute rheumatic fever as per the Therapeutic Guidelines: Antibiotic (Acute rheumatic fever) [11]
- alerting symptoms and features as per the Therapeutic Guidelines: Rheumatology (Clinical assessment of musculoskeletal symptoms in adults) [12]
- response to any previous treatments
- impacts on quality of life including sleep, ability to self-care, mobility, work, leisure, emotional health and relationships
- precipitating, aggravating and relieving/alleviating factors
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)
- weight, dietary patterns, levels of physical activity
- allergies/adverse drug reactions
- alcohol, tobacco and other drug history/status
- if the injury occurred in a workplace
- if the patient has been, or suspected to have been, subject to domestic abuse
- other psychosocial factors.

Pain history and assessment

Where appropriate, conduct assessment of vital signs.

Pain history and assessment should be conducted in accordance with the Therapeutic Guidelines: Pain and analgesia (General principles of acute pain management) and Therapeutic Guidelines: Pain and analgesia (Assessing a patient with pain) [4, 13].

Pain severity is subjective and should be interpreted in the context of the patient's presentation as well as additional information gained through the pain history [14].

The Therapeutic Guidelines: Pain and analgesia (General principles of acute pain management) includes information to assist with determining pain severity [4].

If the patient's self-reported pain severity is not consistent with other assessment findings e.g., physical function or examination findings, the patient should be referred to an appropriate healthcare provider for comprehensive assessment including socio-psycho-biomedical factors [4, 14]

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination

Examine the area of pain in accordance with the Therapeutic Guidelines: Rheumatology (Clinical assessment of musculoskeletal symptoms in adults) [12].

To determine appropriate management or referral, it is important to identify the aetiology of the pain (nociceptive, neuropathic, nociplastic, or a combination (mixed pain)) and to distinguish pain associated with inflammation from mechanical pain. See Therapeutic Guidelines: Pain and analgesia (Understanding Pain) [14].

MSK pain is predominantly nociceptive, with a superficial or deep origin and localisation of pain from the bone, joint or muscles (somatic) [14].

Acute pain is defined by a duration of less than 3 months, usually with a nociceptive pain component arising from actual or threatened tissue damage [4, 14].

Table 1. Clinical features/characteristics of mechanical versus inflammatory pain [1, 15]

Inflammatory pain	Mechanical pain (musculoskeletal)
• improves with exercise/movement • does not improve with rest • lasts more than 3 months • patient may experience morning stiffness (for greater than 30 minutes) • pain waking patient during second half of the night with improvement on getting up.	• may worsen with movement/exercise • often improves with rest • most cases have acute onset lasting less than 3 months • precipitating physical injury may be identifiable.

Investigations

Referral to an appropriate healthcare provider for further investigation may be required for acute pain that does not respond to treatment, or if indicated by patient history and/or pain history e.g., patients presenting with traumatic injuries that may require radiological examination.

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Pharmacist management of acute mild MSK pain involves:

- **non-pharmacological measures:**
    - Advice regarding non-pharmacological pain management strategies for all patients in accordance with the Therapeutic Guidelines: Pain and analgesia (Nonpharmacological management of acute pain) [6].
- **pharmacotherapy:**
    - Oral analgesia in accordance with the:
        - Therapeutic Guidelines: Pain and Analgesia (Pharmacological management of acute pain - Mild, acute nociceptive pain) [16] or
        - Therapeutic Guidelines: Rheumatology (Overview of limb conditions - Analgesia for acute soft-tissue limb conditions) [17].

Confirm management is appropriate

Consult the *Therapeutic Guidelines*, *Australian Medicines Handbook* and other resources to confirm that management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- management options and expectations for resolution
- individual product and medicine use e.g., treatment regimens with paracetamol and NSAIDS and application instructions for topical NSAIDs
- non-pharmacological measures
- how to manage adverse effects
- when to seek further care and/or treatment.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources and information provided to patients (and/or parents/caregivers if applicable), and to ensure compliance with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

General advice

General advice to be provided to the patient regarding strategies to manage acute MSK pain includes:

- Physical interventions that may provide adequate pain relief on their own e.g., hot and cold therapy
- Other interventions based on the reported injury e.g., ice, immobilisation and compression therapy [6, 10]

- Evidence for the use of complementary medicines and possible adverse effects and interactions with conventional medicines.

At the time of initial consultation, patients (and/or parents/caregivers) should be advised to see an appropriate healthcare provider if:

- the pain relief provided is not adequate
- symptoms do not improve with recommended treatment
- the pain or other symptoms worsen
- new pain or symptoms develop.

Advise patients that analgesia (NSAIDs or paracetamol) should be discontinued if there is no benefit, or treatment causes adverse effects [7, 8]. Patients prescribed NSAIDs should be advised to use the minimum effective dose for the shortest duration possible [7].

Clinical review

Clinical review is generally not required. Pharmacists should prescribe a quantity of medicine that aligns with the expected duration of therapy. Advise patients (and/or parents/caregivers, if applicable) to see an appropriate healthcare provider if the condition does not improve or resolve (see *Refer when* and *General advice*).

Pharmacist resources

- Therapeutic Guidelines: Rheumatology
  - Clinical assessment of musculoskeletal symptoms in adults
  - Principles of NSAID use for musculoskeletal pain
  - Principles of paracetamol use for musculoskeletal pain
  - Overview of limb conditions
- Therapeutic Guidelines: Pain and analgesia
  - Understanding pain
  - Assessing a patient with pain
  - General principles of acute pain management
  - Nonpharmacological management of acute pain
  - Pharmacological management of acute pain
  - Mild, acute nociceptive pain
- Australian Medicines Handbook:
  - Drugs for pain relief
  - NSAIDs
- Australian Pharmaceutical Formulary and Handbook (APF) Treatment guideline for pharmacists (Musculoskeletal pain)
- Queensland Ambulance Service - Pain assessment
- National Centre for Complementary and Integrative Health:

- Pain information for Health Professionals
- Herbs at a glance

Document version number	Date	Comments
Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy
Mild to Moderate Acne - Clinical Practice Guideline

Service restricted to patients aged 12 years or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
- Examination
  - Clinical features
  - Severity

Develop management plan

- Non-pharmacological measures
- Pharmacotherapy
  - Topical over-the-counter products, retinoids and antibiotics (and combinations)
  - Oral antibiotics

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation
- Appropriate antibiotic use

Communicate agreed management plan

- Medication counselling
- General advice and patient information
- When to seek further care and/or treatment
- Communication with other health practitioners

Clinical review

- Response to treatment
- Continue, modify or stop treatment
- Communication with other health practitioners

Refer when

- Unclear diagnosis
- The patient is planning a pregnancy or is pregnant
- The patient is taking a medicine that can cause or aggravate acne
- A female patient has PCOS or presents with signs of androgenisation including hirsutism, obesity and/or menstrual irregularity
- Quality of life is significantly affected
- Acne is classified as severe, cystic or scarring, or the patient has a family history of scarring acne
- The patient requires >6 months of oral antibiotic therapy
- Inadequate response to optimal treatment

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Queensland Government

Refer when

Refer the patient to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- unclear diagnosis
- the patient is planning a pregnancy or is pregnant
- the patient is taking a medicine that can cause or aggravate acne
- a female patient has polycystic ovarian syndrome (PCOS) or presents with signs of androgenisation including hirsutism, obesity and/or menstrual irregularity
- quality of life is significantly affected
- acne is classified as severe, cystic or scarring, or the patient has a family history of scarring acne
- inadequate response to optimal treatment
- the patient requires >6 months of oral antibiotic therapy.

Key points

- Acne is common in adolescents (typically more severe in males) but may affect older adults, particularly women in their 30s and 40s [1-3]. The onset of acne often peaks during puberty [1, 2, 4].
- Adult female acne may be different from typical adolescent acne, suggesting possible endocrinological abnormalities including PCOS [5].
- Acne can have significant impacts on a person's social, emotional and mental wellbeing, and can result in permanent physical scarring. Early and effective treatment is important, even for mild presentations [2, 4, 6].

Patient history and assessment

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- pregnancy and lactation status (if applicable), including plans to become pregnant
- onset, duration, nature, location and severity of lesions
- underlying medical conditions, including PCOS
- hormonal changes and signs of androgenisation in women, including hirsutism, obesity and menstrual irregularity
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)

- drug allergies/ adverse reactions and/or sensitivities to treatments
- current management strategies for acne, including hygiene, skin care regimes and products (cosmetic, complementary, over-the-counter and scheduled medicines/products)
- response to prior therapy or management strategies
- aggravating factors (e.g., occupation and exposure to comedogenic factors)
- emotional and social impacts of acne.

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination

Clinical features

Acne is characterised by a mixed eruption of lesions on the face, neck and/or trunk, where severity can be classified according to the Therapeutic Guidelines: Acne (Diagnosis and classification of acne) [2].

Other conditions with similar signs and symptoms that may require consideration and referral for investigation and/or treatment are included in the Therapeutic Guidelines: Dermatology (Acne - Diagnosis and classification of acne) [2].

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Pharmacist management of mild to moderate acne involves:

- **non-pharmacological measures:**
  - Advice regarding skin care and minimising exposure to aggravating factors, in accordance with the Therapeutic Guidelines: Dermatology (Acne - General measures for acne) [2].
- **pharmacotherapy:**
  - In accordance with the Therapeutic Guidelines: Dermatology (Acne) [2], specifically [NB1]:
    - over-the-counter topical products
    - topical retinoids (and combination products with benzoyl peroxide)
    - topical antibiotics (and combination products with benzoyl peroxide or retinoids)
    - oral antibiotics.

NB1: Isotretinoin is not able to be prescribed by pharmacists.

Additional information for consideration

Antibiotic therapy should not be used long-term (topical or oral) [2].

- The *Australian Medicines Handbook* recommends limiting treatment with oral antibiotics to 3–6 months when possible [7].
- The use of topical clindamycin should be limited to 3 months [8].
- Oral and topical antibiotics should be discontinued when inflammatory lesions have adequately resolved but may be resumed if inflammation reoccurs [2,7].
- Antibiotics should not be switched in a patient who is responding to therapy [7].

Women of childbearing age being treated with teratogenic topical or oral medicines for acne should be advised to take appropriate measures to prevent pregnancy. Refer to the Queensland Community Pharmacy Hormonal Contraception Service Guideline.

Confirm management is appropriate

Consult the *Therapeutic Guidelines*, *Australian Medicines Handbook* and other resources to confirm that management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

Reminder

The prescribing and use of antibiotics should be consistent with the principles described in the Therapeutic Guidelines: Antibiotic. Pharmacists should carefully assess whether antibiotic therapy is clinically indicated, and if so, consider local antimicrobial resistance patterns and relevant clinical factors when selecting therapy.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- individual product and medicine use
- non-pharmacological measures and strategies to assist adherence to therapies
- how to manage adverse effects
- when to seek further care and/or treatment
- when to return for follow-up.

Patients may have been exposed to confusing, conflicting and/or misleading information related to acne management and products. Pharmacists should be judicious in their recommendation of supplements, cosmetics and hygiene products [4].

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources and information provided to patients (and/or parents/caregiver) and to ensure compliance with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

General advice

At the time of initial consultation, patients (and/or parents/caregivers) should be advised to see an appropriate healthcare provider if:

- inadequate response to treatment, including if >6 months of oral antibiotic treatment is required
- the acne worsens or repeatedly relapses
- new signs or symptoms develop (e.g., scarring)
- quality of life is significantly affected
- adverse effects cannot be managed in the pharmacy setting
- they become pregnant.

Inconsistent use of medication and insufficient duration of treatment is a common reason for treatment failure. The patient should be advised that most treatments take between 6-12 weeks of consistent and correct use to start to be visibly effective [4].

Clinical review

Clinical review with the pharmacist (or the patient's chosen healthcare provider) should occur in line with recommendations in the *Therapeutic Guidelines*.

- Patients should return for a follow up consultation **6 weeks** after commencing a new therapy.
- For patients continuing therapy, follow-up should be individualised to the patient's clinical presentation, response to treatment, and therapeutic goals.

Pharmacists should prescribe a quantity of medicine (including repeats), that aligns with the expected duration of therapy, ensuring sufficient supply until the patient's next review. The total quantity prescribed should not exceed 12 months' supply (including repeats).

Pharmacist resources

- Therapeutic Guidelines: Dermatology (Acne)
- Australian Medicines Handbook:
  - Drugs for acne
  - Antibacterials
  - Antibacterials (skin)
- Australian Pharmaceutical Formulary and Handbook (APF): Treatment guideline for pharmacists (Acne)
- DermNet NZ: Acne
- The Australasian College of Dermatologists – A-Z of Skin: Acne Vulgaris
- All about acne: Health Professionals

Document version number	Date	Comments
Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy

Mild to Moderate Atopic Dermatitis - Clinical Practice Guideline

Service restricted to patients aged 6 months or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
- Examination
  - Clinical features and severity
  - Assessment (see Appendix 1 for template)

Develop management plan

- Eczema action plan
- Non-pharmacological measures
- Pharmacotherapy
  - Topical treatments

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation

Communicate agreed management plan

- Medication counselling
- General advice and patient information
- When to seek further care and/or treatment
- Communication with other health practitioners

Clinical review

- Response to treatment
- Continue, modify or stop treatment
- Communication with other health practitioners

Refer when

- Unclear diagnosis
- Widespread rash with painful skin
- Raised purple rash that doesn't blanch
- Generalised erythema that covers ≥90% of the skin
- Blistering of skin and mucous membranes
- Systemic signs or symptoms
- Chronic sores or ulcers
- Severe symptoms (SCORAD score >51 or EASI score >21.1)
- Child with a history of immediate or delayed-type hypersensitivity to food, poor feeding or sleep, concerns about failure to thrive
- The patient presents with eczema herpeticum or cellulitis
- Inadequate response to optimal treatment

Treat and concurrently refer

- The patient is planning a pregnancy or is pregnant
- The patient is immunocompromised or has diabetes
- Quality of life is significantly affected

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Refer the patient to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- unclear diagnosis
- widespread rash with painful skin
- raised purple rash that doesn't blanch
- generalised erythema that covers ≥90% of the skin
- blistering of skin and mucous membranes (including mouth and eyes)
- systemic signs or symptoms, including fever, lethargy, nausea, vomiting, headache
- chronic sores or ulcers
- severe symptoms (SCORAD score >51 or EASI score >21.1)
- child with a history of immediate or delayed-type hypersensitivity to food, poor feeding or sleep, concerns about failure to thrive
- presentations with eczema herpeticum or cellulitis
- inadequate response to optimal treatment.

Treat (if clinically appropriate) and concurrently refer

Provide initial treatment and concurrently refer to an appropriate healthcare provider in the following situations:

- the patient is planning a pregnancy or is pregnant
- quality of life significantly affected
- the patient is immunocompromised or has diabetes.

Key points

- Both pruritus and a rash must be present for a diagnosis of atopic dermatitis (AD) [1].
- AD imposes a significant financial, psychological and social burden on the lives of patients and their families, and is associated with poor sleep, depression, anxiety, poor self-esteem and reduced quality of life [2, 3].
- Secondary bacterial infection is a common complication of AD e.g., secondary impetigo or cellulitis [4]. If impetigo is present, it can be treated by the pharmacist as per the Queensland Community Pharmacy Clinical Practice Guideline for Impetigo. If cellulitis is present the patient should be referred to an appropriate healthcare provider.
- Patients with AD are more likely to have an immediate or family history of atopic conditions e.g., allergic rhinitis and asthma [5].
- The mainstay of treatment for active AD is topical corticosteroids (TCS) in combination with emollients, identification and avoidance of triggers, and the early treatment of infection [6].

Patient history and assessment

AD is usually diagnosed clinically, based on patient history and physical findings [7, 8].

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- weight
- pregnancy and lactation status (if applicable)
- onset, duration, nature, location, severity and extent of patches and plaques, including recent or previous relapses or flare ups
- response to any previous treatment/s
- impacts on quality of life and psychosocial wellbeing including sleep and learning
- dietary history and changes in diet
- underlying and associated medical conditions including asthma, allergic rhinitis, allergic conjunctivitis, allergic contact dermatitis, food allergy and depression
- exposure to potential triggers or irritants
- other factors including family history, environmental factors (e.g., exposure to smoking and airborne pollution) and infectious factors
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)
- drug allergies/adverse drug reactions.

A comprehensive assessment template to assist with patient history is provided at Appendix 1.

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Examination

Clinical features

- AD is characterised by dry, scaly erythematous patches with the primary hallmark being itch [9]. While AD can affect any area of skin, patches typically occur on the:
    - face
    - inside of the elbow/arm and back of the knee (cubital and popliteal fossae)
    - wrists and ankles.
- For a diagnosis of AD to be made, both signs of itching and a rash must be present [7].

- Further information on clinical features is available from the Therapeutic Guidelines: Dermatology (Atopic dermatitis) [10] and Dermnet NZ: Guidelines for the diagnosis and assessment of eczema [11].
- The skin signs of AD may vary depending on age and ethnicity [7, 12].
- People with AD are at higher risk of allergic contact reactions (e.g., nickel is a common contact allergen) and are prone to other viral skin infections (e.g., common warts and molluscum contagiosum) [4].
- For patients without an existing diagnosis of AD, other conditions with similar clinical features, symptoms and presentations that may be considered are outlined in Dermnet NZ: Guidelines for the diagnosis and assessment of eczema [11].

Severity

For the purposes of the Service, the SCORing Atopic Dermatitis (SCORAD) index [5] or the Eczema Area and Severity Index (EASI) [5] can be used to assess the severity of the AD to determine whether the patient requires referral, to inform the treatment plan and to monitor treatment effectiveness:

- mild AD is indicated by a score of <25 (SCORAD) or <7 (EASI)
- moderate AD is indicated by a score 25–50 (SCORAD) or 7.1–21 (EASI)
- severe AD is indicated by a score >50 (SCORAD) or >21.1 (EASI) [2, 13].

Conventional scoring systems may underestimate AD severity and erythema in people with darker skin tones [8]. Consider the skin tone when assessing erythema and where there is uncertainty, severity can be assessed by using the EASI and upgrading the score by 1 grade or more for heavily pigmented skin (e.g., from mild to moderate) [5, 13].

Complications of AD

- Secondary bacterial infections (impetiginisation) are the most common complication of AD [4, 12].
    - Treatment of concurrent infection is important for successful management of active AD [6, 10].
    - Suspected impetigo can be assessed and treated as per the Queensland Community Pharmacy Clinical Practice Guideline for Impetigo or referred to an appropriate healthcare provider.
    - Suspected cellulitis must be referred to an appropriate health care provider.
- Eczema herpeticum is an infection of the AD with herpes simplex virus (HSV) [4, 12].
    - Vesicles usually develop in areas of active or recent AD, followed by the onset of high fever and adenopathy [4].
    - Painful corneal lesions will develop if the eye is involved and if the HSV infection becomes systemic, it may be fatal [4].
    - Refer to an appropriate healthcare provider if eczema herpeticum is suspected.

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Pharmacist management of mild to moderate AD involves:

- **development of an Eczema Action Plan:**
  - Based on the Australasian Society of Clinical Immunology and Allergy (ASCIA) Action Plan for Eczema template [14].
- **non-pharmacological measures:**
  - Advice regarding skin care and minimising aggravating factors as per the Therapeutic Guidelines: Dermatology (Atopic dermatitis) [10] and the Australian Medicines Handbook: Eczema [15].
- **pharmacotherapy** [NB1]**:**
  - A topical treatment for the management of **mild to moderate atopic dermatitis** in accordance with the Therapeutic Guidelines: Dermatology (Atopic dermatitis) [10].

NB1: Very potent TCS (class IV), and potent TCS (class III containing betamethasone dipropionate 0.05%, betamethasone valerate 0.1%, methylprednisolone aceponate 0.1% or mometasone furoate 0.1%) should not be used for children aged 12 months or younger [15].

Additional information for consideration

- Selecting the correct potency and formulation of TCS is important to avoid both undertreatment and fears around TCS use [16]. Pharmacists should consult:
  - the Therapeutic Guidelines: Dermatology (Considerations in the use of topical corticosteroids) [9]
  - the Australian Medicines Handbook: Corticosteroids (skin) [15]
  - the Australasian College of Dermatologists:
    - Consensus statement: Management of atopic dermatitis in adults [1]
    - Consensus statement: Topical corticosteroids in paediatric eczema [6]
  - other sources e.g., Australian Journal of General Practice: Selection of an effective topical corticosteroid [16].
- Therapeutic regimens for AD can be expensive and complicated, making long-term compliance difficult and leading to poor outcomes. Choice of treatment should consider the impact of factors such as age, socioeconomic status, cost and literacy on the patient's ability to adhere to prescribed therapies [4].

Confirm management is appropriate

Consult the *Therapeutic Guidelines*, *Australian Medicines Handbook* and other resources to confirm that management is appropriate, including for:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- individual product and medicine use (e.g., application instructions for emollients, topical treatments and wet dressing use)
- non-pharmacological measures
- how to manage adverse effects
- when to seek further care and/or treatment, including recognising secondary infection
- when to return for follow-up.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources and information provided to patients (and/or parents/caregivers if applicable) and to ensure compliance with copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

Patient resources

- Eczema Association Australasia: Managing Eczema
- American Academy of Dermatology Association: Eczema Resource Centre
- Therapeutic Guidelines: Dermatology (Atopic dermatitis - How to apply a wet dressing) and (Atopic dermatitis - How to use the soak and smear technique)

General advice

At the time of initial consultation, patients (and/or parents/caregivers if applicable) should be advised to see an appropriate healthcare provider if:

- signs or symptoms worsen
- new signs or symptoms occur (including systemic signs or symptoms)
- there is inadequate improvement of symptoms within 1–2 weeks
- symptoms recur within 2–4 weeks of initial resolution
- stable, long-term control of the condition is not achieved
- ongoing impairment is present e.g. pruritus, pain, loss of sleep, learning difficulties
- quality of life is significantly affected.

Treatment may be unsuccessful when there is poor adherence with therapy, skin infection, allergy or severe dermatitis [4, 8, 15].

Common adverse effects of TCS, such as transient burning, stinging or pain on application, can generally be reversed by stopping the medicine. Refer to an appropriate healthcare provider when adverse effects cannot be managed in the pharmacy setting.

When recommending TCS for AD, pharmacists should reassure patients (and/or parents/caregivers) who have concerns about the safety of TCS that they are safe when used appropriately [6].

Patients (and/or parents/caregivers) should be counselled on the role of emollients in both the management of AD, and the ongoing maintenance of remission after the flare is clear and TCS treatment is ceased [10]. Advice regarding skin care and minimising aggravating factors should be reiterated [10].

Clinical review

Clinical review with the pharmacist (or the patient's chosen healthcare provider) should occur in line with recommendations in the *Therapeutic Guidelines*.

Patients should return for a follow up consultation **7-14 days** after initiation of a new therapy and/or changes to the treatment plan. Clinical review should consider:

- response to treatment (using the SCORAD index or EASI)
- if changes are required to the treatment plan. Decisions to continue, modify or stop treatment should be reflected in the patient's eczema action plan.

If a good response has been achieved, the medication can be stopped, and the patient can continue using regular emollients to maintain remission [9]. If there has been an inadequate response, the pharmacist should check the patient's compliance with the medication and skin care measures, ensuring enough medication is applied [9].

For patients continuing therapy, follow-up should be individualised to the patient's clinical presentation, response to treatment, and therapeutic goals.

Pharmacists should prescribe a quantity of medicine (including repeats), that aligns with the expected duration of therapy, ensuring sufficient supply until the patient's next review. The total quantity prescribed should not exceed 12 months' supply (including repeats).

Pharmacist resources

- Therapeutic Guidelines: Dermatology
  - Atopic dermatitis
  - Application and quantity of topical corticosteroids
- Australian Medicines Handbook:
  - Drugs for Eczema
    - General principles: topical treatment of skin conditions
    - Corticosteroids (skin)
    - Topical steroids – how much do I use?
- Pharmaceutical Society of Australia: Australian Pharmaceutical Formulary and Handbook (APF) Treatment guideline for pharmacists (Dermatitis)
- DermNet NZ:
  - Atopic dermatitis
  - Atopic dermatitis (images)
  - Guidelines for the diagnosis and assessment of eczema

    - Fingertip unit
    - Eczema area and severity index (EASI score)
    - SCORAD
- SCORAD calculators:
    - MDApp SCORing Atopic Dermatitis (SCORAD)
    - SCORing Atopic Dermatitis Calculator
- MSD Manual (Professional version): Atopic dermatitis (Eczema)
- Australian Journal of General Practice: Selection of an effective topical steroid
- Australasian Society of Clinical Immunology and Allergy: Action Plan for Eczema
- The Australasian College of Dermatologists:
    - Consensus Statement: Management of atopic dermatitis in adults
    - Consensus Statement: Topical corticosteroids in paediatric eczema
    - A-Z of skin – Atopic dermatitis
    - Taking care of skin: How to recognise and respond to health issues in Aboriginal and Torres Strait Islander Health Peoples (an online course/ teaching resource for Aboriginal Health Workers)
- Skin Deep: An open-access bank of high-quality photographs of medical conditions in a wide range of skin tones for use by both healthcare professionals and the public.
- Cardiff University: The Children's Dermatology Life Quality Index (CDLQI)
- University of Nottingham: Patient-Orientated Eczema Measure (POEM)

Appendix 1 - Assessment template

Atopic Dermatitis Assessment Template [17]
Identification of triggers: • irritants   ○ soaps and detergents   ○ abrasive clothing   ○ chlorinated swimming pools   ○ emollients containing sodium lauryl sulphate. • skin infections   ○ *Staphylococcus aureus*   ○ *Streptococcus pyogenes*   ○ HSV (eczema herpeticum)   ○ molluscum contagiosum. • contact allergens • stress • food and inhalant triggers[NB1].
Assessment of current and previous treatments: • bathing/showering frequency and temperature • use of soap, soap-free cleansers, shampoos and/or bath additives • emollient/moisturiser use, application frequency and quantity applied • topical corticosteroids (TCS) use, types, sites of application and quantity applied[NB2] • adverse reaction to topical agents e.g., stinging • antihistamine and antibiotic use.
Impact of eczema[NB3]: • psychosocial impact • frequency of skin infections • frequency of days off employment, school/activities, learning • sleep.

NB1: AD is associated with an increased risk of immediate or delayed-type hypersensitivity reactions to food proteins. However, food allergies are more likely to cause urticaria than dermatitis. Modified diets have little benefit. Irritation around the mouth from food is not a food allergy. Applying a greasy barrier to the face and hands prior to eating irritant foods can protect skin and minimise irritation [9].

NB2: Underuse of TCS is a common cause of previous treatment failure.

NB3: Formal measures may be used e.g., The Children's Dermatology Life Quality Index (CDLQI) [18] or Patient-Orientated Eczema Measure (POEM) [19].

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Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 2.0	1.07.2025	Updated to reflect current clinical guidance, revised referral criteria, and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy

Oral Health Risk Assessment and Fluoride Application - Clinical Practice Guideline

Service recommended for patients aged 18 months and over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Patient history
  - Dental history
- Examination
  - Risk assessment
  - Visual screening

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation

Explain treatment and apply fluoride varnish

- Discuss treatment
  - Application procedure
  - Recommended frequency
  - What to expect
  - Adverse effects
  - Post-application instructions
- Duraphat® application

Provide general and preventative advice

- General advice and patient information
- Communication with other health practitioners

Refer when

- The patient is screened at high or moderate risk of tooth decay/poor oral health, but the application of fluoride varnish is contraindicated
- Immediate (same day) referral required:
  - Recent orofacial trauma (not yet seen by a dental and/or medical practitioner)
  - Facial swelling or fever from an oral infection
  - Difficulty swallowing or breathing
  - A toothache that is worsening despite taking regular analgesia
- Urgent (within 3 days) referral required:
  - Advanced decay (discolouration and cavitation)
  - A visible abscess
  - A toothache that is not impacting daily functioning and can be managed with over-the-counter analgesics

Treat and concurrently refer

- The patient is screened at high or moderate risk of tooth decay/poor oral health
- The patient presents with white spot lesions or advancing decay (brown spot lesions – no cavitation evident)
- The patient has not seen a dental practitioner in the previous 12 months

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Refer the patient to an appropriate healthcare provider (including emergency services, if required) in the following situations:

- the patient is screened at high or moderate risk of tooth decay/poor oral health but the application of fluoride varnish is contraindicated
- immediate (same day) referral required:
  - recent orofacial trauma (not yet seen by a dental and/or medical practitioner)
  - facial swelling or fever from an oral infection
  - difficulty swallowing or breathing
  - a toothache that is worsening despite taking regular analgesia.
- urgent (within 3 days) referral required:
  - advanced decay (discolouration and cavitation)
  - a visible abscess
  - a toothache that is not impacting daily functioning and can be managed with over-the-counter analgesia.

Treat (if clinically appropriate) and concurrently refer

Apply fluoride varnish and concurrently refer to an appropriate healthcare provider:

- if the patient is screened at high or moderate risk of tooth decay/poor oral health
- if the patient presents with white spot lesions or advancing decay (brown spot lesions – no cavitation evident)
- if the patient has not seen a dental practitioner in the previous 12 months.

Key points

- Tooth decay (caries), one of the most prevalent chronic diseases, is caused by acid-producing bacteria (commonly *Streptococcus mutans)* present in the thin biofilm on teeth (plaque) [1, 2]. Acidophilic and cariogenic bacteria grow in plaque and produce acid that demineralises the tooth, starting with the enamel [1].
- If left untreated, the demineralisation of the tooth structure will progress into the dentine layer until it reaches the nerve/pulp of the tooth to cause localised infection [1]. Cavitation (holes) in the outer surface of the tooth enamel may occur [3].
- An established body of evidence suggests that the application of fluoride varnish to teeth every six months can reduce dental decay and prevent progression of early-stage carious lesions [4-6].
- Patients may either be referred by a dental practitioner (an oral health therapist, dental therapist, oral hygienist or dentist), or assessed by the pharmacist as at moderate or high risk of tooth decay or poor oral health and be eligible for fluoride varnish application.

- Fluoride varnish contains 22.6 mg/mL (22,600 ppm) fluoride ion suspended in an alcohol and resin base [5-7]. The only fluoride varnish currently registered in Australia to prevent caries is Duraphat® 5% w/v sodium fluoride varnish [5, 8].

Obtain patient history and complete assessment

To assess the patient's suitability for an application of fluoride varnish, and/or whether a referral to a dental practitioner or another appropriate healthcare provider is required, an oral health risk assessment, patient history and dental history must be undertaken by the pharmacist. Fluoride varnish should be considered for patients 18 months of age and older.

Appendix 2 provides descriptions and clinical photographs of common oral presentations.

Review for other signs and symptoms that may require the attention of another healthcare provider (see *Refer when)* and consider standard pharmacy treatment and referral where appropriate (e.g., for mouth ulcers).

Risk assessment and screening

Risk assessment

Refer to Appendix 1.

There is no arbitrary scoring system for risk status due to the variability of the interaction between risk factors and protective factors. Pharmacists must use their professional judgement to evaluate a child's risk status.

- A child with no obvious clinical signs of decay, only a few moderate level risk factors and multiple protective factors may be deemed low risk.
- If the child has at least one high risk factor, regardless of the number of protective factors, they would be at least at moderate risk of tooth decay and poor oral health. Fluoride varnish application should be considered.
- For patients aged 6 years of age and over a risk assessment questionnaire is completed with the patient, and/or parent/caregiver.

Visual screening using the 'lift the lip' technique

Visual screening is recommended for patients aged 18 months to 5 years.

- Depending on age, the child may either sit or stand in front of the pharmacist or lay in the parent/caregiver's lap (see Figure 1).
- The parent/caregiver should lift the upper lip of the child in front of the pharmacist so that the front teeth and gums are clearly visible to look for signs of decay [4].
- A torch may be used to check around the mouth and back teeth if desired. A soft toothbrush with no toothpaste may be used to clean visible plaque and debris off teeth for better visibility [9]. Consider the use of plastic sunglasses to not expose the patient's eyes to excessive light.
- During the screening, the pharmacist should look for:
    - signs of tooth decay (white spot lesions, brown spot lesions, cavitation (holes))
    - facial trauma (loose or broken teeth, lacerations on gums)
    - oral infection, facial swelling/asymmetry due to spreading infection

- gum disease (red, swollen gums)
- other relevant findings including mouth ulcers and possible signs of fluorosis.

- Consider referring to a dental practitioner if the child is not able to co-operate with visual screening (e.g., unwilling/unable to show you their mouth or stay still enough for assessment).

Figure 1. Young child positioned sitting in parent/caregivers lap for lift the lip screening



Image source: Metro South Health [10].

Visual screening may not be required for patients aged 6 years of age and over, as signs of poor oral health may be easily observable, such as plaque accumulation, decay, abscess, trauma, gingivitis or bad breath.

Patient history

Obtain sufficient information from the patient and/or their parent/caregiver to assess the safety and appropriateness of any recommendations.

Consider:

- age
- pregnancy and lactation status (if applicable)
- current medical conditions or special healthcare needs that may impact on oral health status and the safe use of fluoride varnish, e.g.:
  - asthma (severity and current level of symptom control)
  - history of hospitalisation for allergic reactions
- current illness e.g., colds, influenza, current infection of herpes simplex virus (HSV)
- current and recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)
- drug allergies/adverse drug reactions
- hypersensitivity/allergic reaction to colophony (natural resin) e.g., sticking plasters or medical adhesives
- smoking status (including vaping).

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Dental history

Consider the following:

- Does the patient have a dental practitioner that they visit routinely?
- When did the patient last see a dental practitioner? What was the reason e.g., emergency treatment, toothache, routine check-up?
- When was the last fluoride varnish application? Any previous adverse experiences with fluoride varnish?
- History of recent orofacial trauma (within the last 6 months), oral infections, toothaches or other oral pain (including use of analgesics), soft tissue pathology in or around the mouth (ulcerative gingivitis, herpetic stomatitis or aphthous ulcers), tooth discolouration
- Impacts of oral health on quality of life including sleep and school attendance
- Oral hygiene routines including type of toothbrush used, type of toothpaste (fluoride strength), toothbrushing frequency, use of fluoridated products (toothpaste, mouthwash), use of other oral hygiene aids (e.g. mouthwash, flossing)
- Consumption of fluoridated water.

Confirm management is appropriate

Refer to the Queensland Health Guideline for Fluoride Varnish (QH-GDL-410:2013) **and** specific product information provided by the manufacturer to confirm the application of fluoride varnish is appropriate, including:

- contraindications and precautions
- drug interactions
- pregnancy and lactation.

General information only is provided within this guideline.

Where there is any doubt about the suitability of a patient, the pharmacist should not undertake the procedure and refer the patient to their preferred dental practitioner, closest public dental service or Aboriginal and Torres Strait Islander Community Controlled dental clinic.

Explain treatment and apply fluoride varnish

Advise patients and/or parents/caregivers that fluoride varnish is a preventative treatment and cannot reverse advanced tooth decay (although it may halt further development), treat dental infections or relieve dental pain.

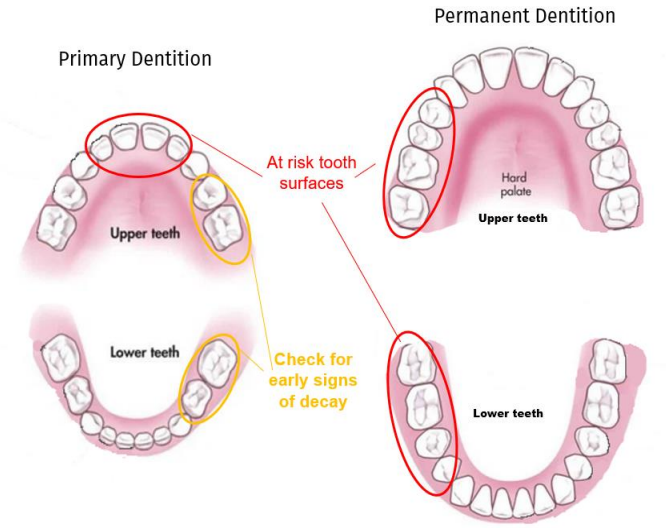
Fluoride varnish is applied to at-risk tooth surfaces where it sets rapidly on contact with saliva to form a coating that is subsequently worn off over a 24 hour period through chewing and delayed tooth brushing. In the context of this guideline an 'at-risk' tooth surface is defined as:

- a tooth surface that has signs of early decay (white spot or brown spot lesions)

- a tooth surface directly abutting a decayed tooth surface (in between two teeth)
- the four front primary teeth in the upper jaw (present in children up to age 6–7 years) (as per Figure 2) – apply to outside (front), inside (back) and biting surfaces [11]
- chewing (occlusal) surfaces and interproximal surfaces (in between) of permanent molars and pre-molars (if present) (as per Figure 2).

The recommended frequency of fluoride varnish application is twice per year. More frequent applications (up to four applications per year) are recommended for patients at higher risk of dental caries [5].

Figure 2. Common areas of decay



Advise patients and/or parents/caregivers that fluoride varnish is safe, easy to apply, and generally well tolerated, even in children [6]. In rare cases swelling of the oral mucosa, ulcerative gingivitis and stomatitis has been observed in sensitive individuals and/or those with a tendency to allergic reactions. If necessary, the varnish layer can easily be removed from the mouth by brushing and rinsing [7]. Teeth may appear yellow for up to 24 hours due to the tint of the varnish, before it wears off and tooth colour returns to normal.

There is no published evidence that professional application of fluoride varnish is a risk factor for enamel fluorosis in children of any age [4].

Guidance regarding Duraphat® application is provided in Appendix 4.

Post application instructions

Provide patients and/or parents/caregiver with post-application instructions and advice before the procedure is undertaken.

The efficacy of fluoride varnish depends on the length of time it can remain undisturbed on the teeth surfaces. As such, patients and/or parents/caregivers should be advised:

- eating and drinking should be avoided for at least 30 minutes after application, after which time soft food and fluids can be consumed
- not to chew food (i.e. harder foods) for at least 4 hours after application (it is important that the varnish is left undisturbed for as long as possible)

- not to brush teeth for at least 4 hours after application [5].

Provide general and preventative advice

Patients and/or parents/caregivers should be reminded of the importance of regular check-ups with a dental practitioner and provided with appropriate information to enable effective home care. Anticipatory guidance should include instruction on toothbrushing, flossing and the use of other oral hygiene aids tailored to the patient's age.

Children aged 18 months to 5 years:

- Teeth should be cleaned twice daily by a parent/caregiver with a pea sized amount of toothpaste (containing 500-550 ppm fluoride) using a small, soft bristled toothbrush.
- The child should be instructed to spit out, and not swallow the toothpaste, and not to rinse after brushing [6, 12].

People aged ≥6 years:

- Teeth should be cleaned twice daily (or more frequently) with a soft bristled toothbrush and a toothpaste containing 1000-1500 ppm fluoride (a standard toothpaste readily available at supermarkets or the pharmacy).
- Toothpaste should not be swallowed, and the patient should avoid rinsing after brushing [6, 12]. Floss should be used to clean in between teeth.
- Plaque disclosing tablets may be used to demonstrate effective methods of plaque removal.

Patient resources

- National Institute of Dental and Craniofacial Research:
  - The Tooth Decay Process: How to Reverse it and Avoid a Cavity
  - Tooth Decay
- Australian Dental Association: Brushing Teeth
- NSW Health: Resources for Aboriginal and Torres Strait Islander People

Pharmacist resources

- Queensland Health Guidelines:
  - Fluoride Varnish (QH-GDL-410:2013)
  - Fluoride – Prevention of Dental Caries and Maintenance of Oral Health (QH-GDL-411:2013)
  - Lift the Lip (Metro North Health)
- Remote Primary Health Care Manual: Clinical Procedures Manual

- Queensland Health and Royal Flying Doctors Service (Queensland branch): Primary Clinical Care Manual 12th edition 2022
- Northern Territory Department of Health: Healthy Smiles: Oral Health and Fluoride Varnish Information for Health Professionals
- NSW Health: Health Mouths Healthy Living
- Australian Dental Association: Practical Guidelines for the use of Fluorides
- MIMS Online: Duraphat
- NHRMC: Water fluoridation and human health in Australia: questions and answers

Appendix 1 - Risk assessment tool (18 months to 5 years)

Risk assessment and screening process – children aged 18 months to 5 years (Adapted from Metro North Health [13] and CAMBRA risk assessment [14])	
Risk Factors	**Risk level**
Individual or family history of decay (including parents/caregivers and siblings with fillings, teeth extracted or active untreated decay). ***Routine referral to a dental practitioner (including public dental clinic) is required if the child has not had a dental check-up (not just emergency treatment) in the past 12 months.***	Moderate to high
Child has a bottle or sippy cup with a fluid other than water, plain milk or plain formula during the day.	Moderate
Child takes a bottle or sippy cup to bed with fluid other than water, including on-demand breastfeeding. ***Routine referral to a dental practitioner (including public dental clinic) is required.***	High
Snacking between meals, especially high sugar and carbohydrate intake, e.g., lollies, chips, biscuits, muesli bars, flavoured milk, juice, soft-drinks, cordial, jams etc. Consider frequency of snacking and child's diet as a whole.	Moderate to high
Special health care needs and/or regular medications that have a high sugar content or may cause dry mouth. ***Routine referral to a dental practitioner (including public dental clinic) is highly recommended.***	Moderate
Clinical signs	
White spot lesions visible on any teeth or obvious decay (brown spots, holes/cavitation) (see Appendix 2). ***Routine referral to a dental practitioner (including public dental clinic) is highly recommended.***	High
Visible plaque accumulation. ***Routine referral to a dental practitioner (including public dental clinic) is highly recommended.***	Moderate
Gingivitis (bleeding, swollen or red (as opposed to pink) gums). ***Routine referral to a dental practitioner (including public dental clinic) is highly recommended.***	Moderate
Toothache, recent trauma (not yet seen by a dental practitioner), facial swelling from infection or visible abscess (see Appendix 2). **Immediate referral to the nearest dental practitioner or a medical practitioner/ emergency department is required.**	High
Protective factors that reduce risk of poor oral health and caries	
• child drinks fluoridated water (tap water in communities with fluoridated water supply) • child/primary caregiver has a usual dental practitioner • child has seen a dental practitioner for a check-up (not just emergency care) in the past 12 months • fluoride varnish has been applied in previous 6 to 12 months • parents/caregiver brushes the child's teeth twice daily with an appropriate fluoride toothpaste.	

Appendix 2 - Clinical photographs

Clinical photographs to assist with visual screening [15]

Healthy teeth and gums:

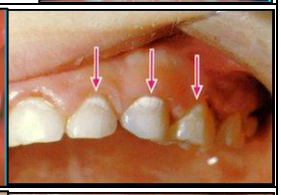
- no (or minimal) plaque on teeth and gums
- gums are pink with no evidence of swelling or bleeding.



White spot lesions:

White, chalky/frosty lines/spots, usually along the gumline (this differentiates white spot lesions from similar presentations such as dental fluorosis)

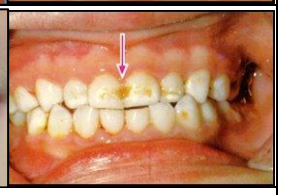




Advancing decay:

Yellow or brown spots on teeth (possibly in combination with white spots) that don't wipe off, sometimes with visible cavitation.





Abscess:

- Tooth decay that has progressed to the nerve causing periapical infection (at the root tip).
- Presents as a lump on the gums (usually seen in the presence of decay).
- May be pus-filled and painful, or draining (usually without pain).
- Can usually be differentiated from mouth ulcers and other mucosal conditions by location (ulcers are more likely to occur on the inner lip, cheek or tongue).



Gingivitis:

Reversible gum inflammation, often observed in the presence of plaque and as red 'puffy' gums, particularly on the gum/tooth line that may bleed when touched/during brushing.

Mouth ulcers:

- Painful sores, often with a yellow or white centre, surrounded by a red border/halo on oral mucosal surfaces and tongue (occasionally palate and gums), usually as a result of minor trauma (accidental biting, scratching or poking at gums with sharp objects e.g., fingernails).
- Recurrent severe or multiple ulcers may require a medical or dental referral to investigate the possibility of herpes simplex virus (HSV), or other infections [16].

Dental fluorosis:

- The observable effect of higher fluoride intake in early childhood as teeth are developing (before age of 8). There is usually no effect on dental or enamel function [17].
- Prevalence of dental fluorosis in Queensland is low (8.2% for all grades of fluorosis) and the vast majority of dental fluorosis observed is classed as very mild (almost imperceptible) to mild (0.1% cases were moderate or severe [18]).







Image source: *American Academy of Pediatrics* [19]

Appendix 3 - Risk assessment and screening tool ≥6 years of age

Risk assessment and screening process – patients aged ≥6 years of age [13, 20]	
Risk Factors/Disease indicators - *Routine referral to a dental practitioner / public dental clinic is highly recommended.*	**Risk level**
• Patient comes from a low socio-economic background or has low health literacy.	High
• Patient has had dental restorations placed (e.g. fillings, implants, crowns, bridges) in the last 3 years.	High
• Frequent snacking and consumption of sugar-containing snacks or drinks in-between meals (>3 times a day).	High
• Patient is a recent immigrant	Moderate
• Patient has special health care needs and/or uses medications that may cause dry mouth.	Moderate
• Patient has orthodontic or other oral/dental appliances (may make cleaning teeth difficult).	Moderate
Clinical signs	
• Visible decay (e.g. white spot lesions, brown spot lesions, holes/cavitation) (see Appendix 2). ***Routine referral to a dental practitioner/public dental clinic is required.***	High
• Visible plaque accumulation. ***Routine referral to a dental practitioner/public dental clinic is highly recommended.***	High
• Patient has low saliva production/flow ***Routine referral to a dental practitioner/public dental clinic is highly recommended.***	High
• Gingivitis (bleeding, swollen or red (as opposed to pink) gums) [14]. ***Routine referral to a dental practitioner/public dental clinic is highly recommended.***	Moderate
• Patient has dental restorations (fillings, implants, crowns, bridges) that are defective. ***Routine referral to a dental practitioner/public dental clinic is required.***	Moderate
• Toothache, recent trauma (not yet seen by a dental practitioner), facial swelling from infection or visible abscess (see Appendix 2). **Immediate referral to the nearest dental practitioner or a medical practitioner/ emergency department is required.**	High
Protective factors that reduce risk of poor oral health and caries	
• Patient drinks fluoridated water (tap water in communities with fluoridated water supply) • Patient has seen a dental practitioner for a check-up (not just emergency care) in the past 12 months • Fluoride varnish has been applied in previous 6 to 12 months • Patient brushes teeth twice daily with an appropriate fluoride toothpaste.	

Appendix 4 – Duraphat® application

Equipment

- hand sanitiser or hand washing station
- Duraphat® varnish (single dose (0.4ml) pack or multi dose tube)
- Duraphat® dosage pad
- cotton gauze
- disposable microbrush/applicator
- appropriate personal protective equipment (PPE):
    - gloves
    - protective eyewear
    - single use surgical mask
    - apron (optional).
- single use, self-adhesive bib to protect patient clothing
- clean dry soft-bristle toothbrush (optional).

Dose

As per the manufacturer's instructions for use, the dose of Duraphat® varies based on age (and subsequently, dentition type) [5, 21]:

- Young children with primary teeth only: up to 0.25 ml Duraphat ® (5.6 mg fluoride)
- Older children with mixed dentition: up to 0.4 ml Duraphat® (9.04 mg fluoride)
- Adolescents and adults with permanent dentition: up to 0.75 ml Duraphat® (16.95 mg fluoride)

A single use, Duraphat® Dosage Pad that indicates the amount of fluoride to be applied, or Duraphat® Varnish Single Dose (0.4ml) must be used to ensure the correct amount of varnish is dispensed and applied.

Application procedure

The following application procedure is based on the Queensland Health Fluoride Varnish Guideline (QH-GDL-410:2013) and the Remote Primary Health Care Manual – Clinical Procedures Manual and has been contextualised for the community pharmacy setting [5, 11]:

1. Explain the procedure to the patient and what they can expect e.g., teeth will feel sticky, keep mouth open and tongue away from teeth during application.
2. Don appropriate PPE, perform appropriate hand hygiene.
3. Position the patient; young children may be positioned as per the 'lift the lip' technique, otherwise the patient should sit facing the pharmacist with their head tilted slightly back.
4. Dispense the correct amount of fluoride for the age of the patient using a single use dosage pad or a Duraphat® Varnish Single Dose pack.
5. Remove any visible plaque from teeth using cotton gauze, or by brushing with a wet toothbrush (without toothpaste).
6. Dry the teeth using a clean piece of cotton gauze.

a. One quadrant (quarter of the mouth) should be dried and isolated at a time for application; although thorough drying is not necessary as the varnish sets on contact with saliva.
b. Isolate the teeth by putting a finger and thumb on each sides of the tooth for application. Older children and adults may tolerate rolled up gauze tucked into the space between the gum and cheek. Change the gauze periodically as it becomes wet.

7. Apply a thin layer of the varnish to 2–3 teeth (as required) in 1 quadrant at a time using a disposable microbrush/applicator, starting with the upper front teeth, moving to the back, then lower teeth. Do not overload the applicator and periodically check the tongue and cheeks for varnish and wipe away as required.
    a. The varnish can be applied to the chewing surfaces of the back teeth and around the gumline on the outside of front teeth if there are white spots indicating early tooth decay.
    b. For restless young children who may struggle to sit still, prioritise the upper front teeth (apply to outside, inside and biting surfaces).
8. Finish the procedure by checking the tongue and soft tissues for varnish residue and wipe away.

Note: Duraphat® is indicated as a spot application fluoride treatment for at-risk tooth surfaces. It should not be applied to the whole dentition in one session (systemic treatment) [6].

Document version number	Date	Comments
Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy Smoking Cessation - Clinical Practice Guideline

Service restricted to patients aged 18 years or over.

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- 5As approach
- Patient history including smoking history
- Examination and screening
  - Blood pressure measurement
  - Common co-morbidities

Develop smoking cessation management plan

- Non-pharmacological measures
- Pharmacotherapy
  - Nicotine replacement therapy
  - Bupropion
  - Varenicline

Confirm management is appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation

Communicate agreed management plan

- Medication counselling
- General advice and patient information
- When to seek further care and/or treatment
- Communication with other health practitioners

Clinical review

- Response to treatment
- Blood pressure monitoring
- Behavioural support
- Communication with other health practitioners

Refer when

- The patient is pregnant
- The patient is taking a medicine that interacts with smoking and may require review and dose adjustment due to changing smoking patterns
- Inadequate response to optimal treatment

Treat and concurrently refer

- The patient has a chronic disease or medical condition that requires additional support or can complicate smoking cessation interventions (e.g. diabetes, cardiovascular disease, mental illness, substance use disorder, multiple comorbidities)

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Refer the patient to an appropriate healthcare provider and smoking cessation support service (e.g. Quitline) in the following situations:

- the patient is pregnant
- the patient is taking a medicine that interacts with smoking and may require review and dose adjustment due to changing smoking patterns
- inadequate response to optimal treatment.

Treat (if clinically appropriate) and concurrently refer

Provide initial treatment and concurrently refer to an appropriate healthcare provider and smoking cessation support service (e.g. Quitline) in the following situations:

- The patient has a chronic disease or medical condition that requires additional support or can complicate smoking cessation interventions (e.g. diabetes, cardiovascular disease, mental illness, substance use disorder, multiple comorbidities).

Reminder

This guideline covers tobacco smoking cessation - the management of vaping cessation is not within the scope of this guideline.

Key points

- People who smoke have a higher risk of cardiovascular disease (CVD) and other comorbidities including asthma, chronic obstructive pulmonary disease (COPD) and chronic kidney disease (CKD).
- General population rates of smoking have been reducing over time; however, several specific populations retain high smoking rates, including Aboriginal and/or Torres Strait Islander People, older people, people with mental illness, and people who live in areas of socioeconomic disadvantage [1, 2].
- Pharmacological interventions for smoking cessation are most effective when combined with support, cognitive and behavioural therapies that are readily available through services such as Quitline [2, 3].

Patient history and assessment

The 5As approach is recommended to identify an individual's support needs, barriers and other relevant cultural, population-specific, or medicine-associated factors, to develop a comprehensive smoking cessation plan [2, 4].

- Refer to the Therapeutic Guidelines: Addiction (Tobacco smoking and nicotine dependence) [3] and Royal Australian College of General Practitioners (RACGP) Supporting smoking cessation: A guide for health professionals [2].
- Refer to Therapeutic Guidelines: Addiction (Tobacco smoking and nicotine dependence - Screening and assessment of tobacco smoking and nicotine dependence) to assist with determining nicotine dependence and whether pharmacotherapy may be appropriate [3].

Patient history

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines.

The patient history should consider:

- age
- pregnancy and lactation status (if applicable)
- underlying medical conditions and chronic diseases including current (or history of) psychiatric illness (including but not limited to psychosis, schizophrenia, bipolar disorder, clinically diagnosed depression and anxiety, eating disorder, substance use disorders and history of suicidal ideation, suicide related behaviour or self-harm)
- current or recently ceased treatments (including prescribed medicines, vitamins, herbs, other supplements and over-the-counter medicines)
- drug allergies/adverse drug reactions.

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Smoking history

Consider:

- age when smoking first commenced
- type of smoking e.g., cigarettes, pipes
- time to first cigarette in the mornings
- number of cigarettes smoked per day
- cravings for a cigarette, urges to smoke or withdrawal symptoms upon stopping smoking
- duration, recording past and current pack years
- previous attempts to quit and periods of not smoking, including support required
- previous responses to pharmacotherapy for smoking cessation.

Examination and screening

Take an initial blood pressure measurement to screen for hypertension.

It is recommended that the patient be screened for common comorbidities associated with smoking, including CVD, asthma and COPD, either as part of a chronic disease program in the Service (if eligible), or with or an appropriate healthcare provider.

Develop smoking cessation management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

A comprehensive approach to smoking cessation should include counselling and behavioural support.

The type of assistance should be based on the patient's willingness to quit, needs, preferences and level of nicotine dependence [2].

Pharmacist support for smoking cessation involves:

- **non-pharmacological interventions and support:**
  - Develop a cessation plan, e.g., Your quit smoking plan available on the Quit website; including identifying barriers to stopping, strategies to address barriers, and identifying and engaging social support.
  - For patients that are not ready to quit, use motivational approaches, explore barriers, and encourage harm minimisation such as reducing the number of cigarettes smoked each day and working towards a quit day [2, 5].
  - Counselling and behavioural support and referral to Quitline and/or other support programs [NB1].
- **pharmacotherapy [NB2]:**
  - Guided by the Pharmacotherapy treatment algorithm in the RACGP Supporting smoking cessation: A guide for health professionals [2]:
    - A medicine[NB3] mentioned in the Therapeutic Guidelines: Addiction (Tobacco smoking and nicotine dependence) [3].

NB1: Resources to support counselling and behavioural support are available on the Queensland Government QuitHQ website for health professionals [6]. Referral to Quitline or a relevant health professional with expertise specific to smoking cessation is recommended [2, 3].

NB2: This guideline covers tobacco smoking cessation, the use of pharmacotherapy for vaping cessation is not within the scope of this guideline.

NB3: Nortriptyline or nicotine vaping products are not permitted for use by pharmacists[3].

Confirm management is appropriate

Consult the *Therapeutic Guidelines*, *Australian Medicines Handbook* and other resources (including relevant product information) to confirm the smoking cessation plan is appropriate, including for:

- contraindications and precautions
- drug interactions [NB1]
- pregnancy and lactation.

NB1: Both smoking itself and pharmacotherapy to assist with smoking cessation are associated with a number of drug interactions that require consideration when someone is deciding to reduce and/or cease smoking [7].

Communicate agreed management plan

Comprehensive advice and counselling (including supporting written information if required), should be provided to the patient regarding the smoking cessation plan. Relevant information may include:

- instructions for individual dosage form and medicine use e.g., medicine dosage and NRT product selection and use
- how to manage adverse effects, including serious adverse effects requiring medical care e.g., mood disturbances
- when to return for follow-up.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources and information provided to patients (and/or parents/caregivers if applicable) and to ensure compliance with all copyright conditions.

The agreed management plan should be shared with members of the patient's multidisciplinary healthcare team, with the patient's consent.

General advice

The most successful approach to smoking cessation for nicotine dependent people is counselling and behavioural support, combined with first-line pharmacotherapy and follow-up [2].

While pharmacotherapy for smoking cessation has good efficacy, relapse is not uncommon. People should be advised not to be discouraged by unsuccessful attempts and reassured that most people that have previously smoked have progressed through a number of quitting attempts and relapses before achieving long-term cessation [3].

Patients should be advised to seek support from an appropriate healthcare provider and a smoking cessation service (e.g. Quitline) if they relapse, or at any time they require extra assistance or have any concerns.

Patient resources

- Quit.org.au: including quit plan, quitting tips and tactics, cost of smoking calculator
- Queensland Government QuitHQ: including smoking cost calculator and support program (My Quit Journey 12 week program)
- Commonwealth Government My QuitBuddy app

Clinical review

Clinical review with the pharmacist (or the patient's chosen healthcare provider) should occur in line with recommendations in the *Therapeutic Guidelines*, the RACGP Supporting smoking cessation: A guide for health professionals and the Australian Medicines Handbook: Nicotine dependence [2, 3, 7].

- Patients prescribed NRT should return for a follow up consultation **2 weeks** after starting treatment to review for tolerability and effectiveness, and additionally at regular intervals during the minimum 12-week course [3].
- Patients prescribed varenicline or bupropion should return for a follow up consultation **2 weeks** after commencing treatment to monitor for progress and adverse effects of pharmacotherapy, to measure blood pressure, and to provide additional support (particularly for relapse prevention) [3]. An additional follow up consultation should occur at the end of the treatment course (9 weeks for bupropion, 12 weeks for varenicline) [3].
    - An additional 12 weeks of treatment with varenicline may be considered if required, to reduce chances of relapse.
    - People who have already undergone 2 courses of varenicline (24 weeks) within the previous 12 months, or those who require additional treatment with varenicline after 2 courses should be referred to an appropriate healthcare provider.
- It is recommended that the patient also receive follow up and ongoing support from a smoking cessation service such as Quitline.

For patients continuing therapy, follow-up should be individualised to the patient's clinical presentation, response to treatment, and therapeutic goals.

Pharmacists should prescribe a quantity of medicine (including repeats), that aligns with the expected duration of therapy, ensuring sufficient supply until the patient's next review.

Pharmacist resources

- Therapeutic Guidelines: Addiction
    - Tobacco smoking and nicotine dependence
- Australian Medicines Handbook:
    - Drugs for nicotine dependence
- Royal Australian College of General Practitioners:
    - Supporting smoking cessation: A guide for health professionals
    - Supporting smoking & vaping cessation: A guide for health professionals – Guidance on smoking and vaping cessation support related to changes to Australia's vaping regulation
- Pharmaceutical Society of Australia: Professional practice guidelines for pharmacists: Nicotine dependence support
- Australian Pharmaceutical Formulary and Handbook (APF): Treatment guideline for pharmacists (Nicotine dependence)
- Queensland Government Quit HQ:
    - For Health professionals
    - Refer a patient to Quitline
- Quit Centre: Clinical Tools & Guidelines
    - Ask, Advise, Help chart
    - Drug interactions with smoking

- Quit:
  - Resources
  - Your quit smoking plan
  - Shared care – helping our community quit smoking

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Version 1.0	01.02.2024	
Version 1.1	11.11.2024	Administrative update.
Version 2.0	01.07.2025	Updated to reflect current clinical guidance and improved usability across sections.

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Queensland Health

Queensland Community Pharmacy

Travel Health - Clinical Practice Guideline

Service restricted to patients aged 2 years or over (diarrhoea)

Service restricted to patients aged 8 years or over (Malaria, Altitude sickness)

Ineligibility for treatment as part of a prescribing service does not preclude a patient from accessing services provided as part of usual pharmacy care.

Guideline overview

Meet Service and professional obligations

- Initial patient eligibility and suitability for management within the scope of the Service
- Patient informed consent (including consent for communication with other health practitioners if relevant)

Patient history and assessment

- Comprehensive pre-travel risk assessment
  - Patient history
  - Travel history
  - Itinerary and travel details

Develop management plan

- Travel-related conditions
- Pharmacotherapy
  - Altitude illness, traveller' diarrhoea, malaria, altitude illness
  - Rabies vaccination
- Medical kit
  - Travelling with medicines

Provide travel health counselling and education

- Tailored counselling for safe travel, risk minimisation and common issues
- Medication counselling
- General advice and patient information
- Post travel issues and medical care

Confirm recommendations are appropriate

- Contraindications, precautions and interactions
- Pregnancy and lactation
- Appropriate antimicrobial use

Refer when

- Children aged ≤8 years travelling to areas of altitude above 2,500 m or who require malaria prophylaxis or stand-by treatment
- Travellers visiting areas where malaria is endemic for longer than 8 weeks
- Pregnant travellers or travellers planning to become pregnant while travelling
- Travellers that require a vaccination that is not able to be administered
- Immunocompromised travellers
- Travellers with conditions that may be worsened by air travel or high altitudes by hypoxaemia
- Travellers with chronic diseases or medical conditions that are not well controlled or severe Recent surgery, trauma or injury
- At moderate or high risk of VTE
- Official medical clearance to travel/fitness to fly certification is needed
- Travellers whose regular medication(s) are not permitted in the planned destination
- Travellers undertaking high risk activities including:
- Travel in areas of very high or extreme altitude Moderate to high-risk ascents in high/very high-altitude areas
- Extreme sports, mountaineering or activities in remote locations
- International aid work or military posting.
- People travelling for medical treatment overseas
- Travellers returning (or arriving) with post-travel issues/symptoms (immediate referral is required if malaria is suspected)

When applying the information contained within this clinical practice guideline, pharmacists are advised to exercise professional discretion and judgement. The clinical practice guideline does not override the responsibility of the pharmacist to make decisions appropriate to the circumstances of the individual, in consultation with the patient and/or their parent/caregiver.

Refer when

Refer the patient to an appropriate healthcare provider in the following situations:

- children aged ≤8 years travelling to areas of altitude above 2,500 m or who require malaria prophylaxis or stand-by treatment
- travellers visiting areas where malaria is endemic for longer than 8 weeks
- pregnant travellers or travellers planning to become pregnant while travelling
- travellers that require a vaccination that is not able to be administered in the community pharmacy setting or contraindicated
- immunocompromised travellers, as per the CDC Yellow Book 2026: immunocompromised travellers
- travellers with conditions that may be worsened by air travel or high altitudes by hypoxaemia
- travellers with chronic diseases or medical conditions that are not well controlled or are severe (e.g., renal, liver, cardiovascular, respiratory and pulmonary diseases, diabetes, epilepsy, autoimmune and rheumatologic diseases, psychiatric illness, mental health conditions)
- recent surgery, trauma or injury (consult individual airlines and travel insurers for specific rules regarding to fitness to fly, including post-operatively)
- moderate or high risk of VTE
- official medical clearance to travel/fitness to fly certification is needed
- travellers whose regular medication(s) are not permitted in the planned destination
- travellers undertaking high risk activities including:
    - travel in areas of very high or extreme altitude (>3,500 m)
    - moderate to high-risk ascents in high/very high-altitude areas
    - extreme sports, mountaineering or activities in remote locations
    - international aid work or military posting.
- people travelling for medical treatment overseas e.g., dental treatment, cosmetic surgery, overseas-based clinical trials
- travellers returning (or arriving) with post-travel issues/symptoms (immediate referral is required if malaria is suspected).

Key points

- The primary purpose of the pre-travel health consultation is to minimise the traveller's risk of injury and illness by providing advice, education, vaccinations, medicines and/or medical kits tailored to each individual's personal and itinerary risks [1-4].
- Pre-travel consultations should be conducted as far in advance of departure as possible (ideally 6-8 weeks) to allow sufficient time for vaccination courses to be undertaken and commencement of malaria chemoprophylaxis if required [2, 5].
- Practitioners providing travel medicine services should have a working knowledge (and know how to access current and accurate information) of the global epidemiology of infectious and non-infectious diseases, health risks, trends in drug-resistant infections, and destination specific regulations regarding immunisation and the importation and use of medications [1, 2, 4].

- While conducting the pre-travel health consultation, pharmacists should be aware of the requirements for referral to other healthcare providers as these may become apparent at various stages throughout the consultation.

Patient history and assessment

Travel health consultations should be tailored to the needs of each patient. An effective pre-travel consultation should include a comprehensive pre-travel risk assessment based on the traveller's medical history, travel history, itinerary and travel details, as per Table 1 [3].

Pre-travel risk assessment

Obtain sufficient information from the patient to assess the safety and appropriateness of any recommendations and medicines. Table 1 outlines the components that should be covered in the pre-travel risk assessment.

Table 1. Components of a comprehensive pre-travel risk assessment[1-4]

Patient history

- age
- pregnancy (current or planned) and lactation status (if applicable)
- medical history including past and current conditions:
    - conditions that influence susceptibility/severity of infections or immunocompromise e.g., splenectomy, steroid therapy, active or recent cancer
    - conditions that may cause acute illness or require emergency treatment while overseas e.g., asthma, diabetes, severe allergies and past allergic reactions
    - conditions increasing susceptibility to hypoxia e.g., cardiorespiratory disease, severe anaemia
    - immunocompromise
    - mental illness and current psychological state
    - chronic conditions e.g., rheumatologic disorders
    - colostomy or ileostomy
    - disability or impairment
    - conditions that may impact on effectiveness or safety of malarial prophylaxis e.g., previous malaria, central nervous system disease or cardiac conditions
    - ongoing hormone treatment e.g., estrogen therapy, oral contraceptive pill, menopausal hormone therapy
    - thrombophilic conditions
    - history of venous thromboembolism (VTE)
    - recent surgery or injury
        - individual airlines and travel insurers have rules regarding fitness to fly post-operatively; these should be reviewed on a case-by-case basis
    - recent cardiopulmonary and/or cerebrovascular event

- current treatments (including prescribed medicines, vitamins, herbs, other supplements, and over-the-counter medicines), particularly medications that may be prohibited in other countries, require supporting documentation, or require advice to be taken safely across time zones
- allergies/adverse drug reactions, including allergies to vaccines
- alcohol and drug history (considering destination regulations)
- smoking status
- immunisation history (routine and travel vaccinations)
- obesity (BMI).

Travel history

- Previous travel experience, including:
  - experience with chemoprophylaxis for malaria
  - experience at high altitudes (if applicable to destination)
  - VTE prophylaxis
  - experience with travel-related and other relevant illnesses.

Itinerary and travel details

- Countries and regions in itinerary, considering:
  - remoteness
  - access and quality of medical facilities
  - hygiene
  - epidemiology of infectious diseases
  - season and likely weather conditions
- duration of total trip and duration of stay in each area
- reason for travel e.g., leisure, 'health tourism', international development or aid work
- planned activities e.g., snow sports, scuba diving, moped/vehicle use, disaster relief, long distance walking or cycling
- risk of acquiring sexually transmitted infections (STIs)/sexual history
- travel style, considering:
  - mode of transport and travel times e.g., cruise ships, long-haul flights
  - backpacking, guided tours
  - traveller risk tolerance
  - accommodation type.

High risk travellers

Specific personal characteristics or complex itineraries may increase a traveller's risk of illness or require special consideration, including for travel insurance and certification to fly. Referral may be required (see *Refer when* section), including for [4]:

- **travellers with chronic conditions**

- **travellers with risk factors for VTE**
- **immunocompromised travellers** detailed in the CDC Yellow Book 2026: Immunocompromised travellers [3]
- **pregnant women**
- **young children** (referral to an appropriate healthcare provider is required for all children <2 years, and may be required for other children, depending on itinerary)
- **older people** (medical practitioner assessment may be required for travel insurance purposes for adults aged ≥65 years)
- **travellers with disabilities**
- **travellers on extended trips**
- **backpackers** and **travellers visiting friends and relatives** (risks are often underestimated in these groups).

Pharmacists must use their professional judgement and consider the traveller's itinerary, personal characteristics and medical history when determining the need for referral.

Reminder

Pharmacists can access a range of clinical information in a patient's My Health Record, including details about current and past medication history, allergies and current medical conditions.

Develop management plan

Pharmacists must ensure that they are prescribing medicines according to the specified sections of the current online version of the Therapeutic Guidelines.

Vaccination

Pharmacists are responsible for confirming the specific entry and transiting vaccination requirements, risk levels and vaccine recommendations for each country and traveller.

Prepare an immunisation plan for the traveller specific for their needs:

- Routine vaccines should be updated (if required), in addition to required and recommended travel vaccines [1, 3, 4, 6].
- Laboratory testing to determine immunity as a result of previous vaccination or infection may be requested for hepatitis A, hepatitis B, measles, mumps, rubella and varicella [6].
- Other vaccines that may be required include those used for response to a declared public health emergency, e.g., COVID-19 and monkeypox.

All vaccinations must be administered in accordance with:

- the Australian Immunisation Handbook

- the individual vaccine information provided by the Therapeutic Goods Administration (TGA) and aligned with current Australian Technical Advisory Group on Immunisation (ATAGI) advice.

Travellers who require proof of vaccination should be provided with an up-to-date International Certificate of Vaccination (ICV), commonly known as a yellow card, as outlined in Annex 6 of the WHO International Health Regulations (2005) [7]. The ICV can be completed by any practitioner authorised to give vaccinations [7].

Rabies

Pre-exposure prophylaxis is recommended for people travelling to rabies-enzootic regions, based on a risk assessment; this should consider a patient's risk of rabies exposure and the patient's access to emergency medical attention in that location (refer to Table 2) [3, 6].

Table 2. Risk assessment and vaccination for Rabies[3, 6, 8, 9]	
Patient risk factors	• Travellers who may interact with animals that could be rabid. • Travellers whose occupational or recreational activities may involve contact with animals, particularly dogs.
Itinerary risk factors	• Travellers to Central and South America, Eastern Europe, Africa and Asia. The CDC conducts an annual assessment of individual countries rabies status worldwide that provides information related to rabies presence and treatment availability. See Rabies Status: Assessment by Country. • Access to emergency medical care and post-exposure prophylaxis at the destination. • Sufficient time to complete recommended pre-exposure prophylaxis prior to departure.
Preventive measures and advice	• Advise travellers, especially those with young children, to avoid contact with wild and domestic animals including stray dogs and cats. • Avoid contact with bats anywhere in the world. • Provide education about treatment for potential rabies exposure including first aid and requirement for post-exposure prophylaxis vaccine and, in some cases, human rabies immunoglobulin.
Vaccination	• Where pre-exposure prophylaxis is recommended, the pharmacist must prescribe the rabies vaccine in accordance with the Australian Immunisation Handbook: Rabies and other lyssaviruses and the Australian Medicines Handbook [6, 9]. The dispensed medicine may then be administered. • Pre-exposure prophylaxis is recommended as a 3-dose schedule, administered on days 0, 7 and 21–28 [6, 9].

Travellers' Diarrhoea

Diarrhoea is commonly experienced by travellers from industrialised countries travelling to developing countries without good sanitation; it is estimated that between 20-50% of these short-term travellers will experience diarrhoea [10].

Travellers' diarrhoea can be caused by a range of pathogens, including bacterial and viral infections, depending on the location of acquisition [10].

The information in Table 3 is provided as a summary only. Pharmacists should consult relevant resources to provide tailored information for each traveller:

Table 3. Risk assessment and management for travellers' diarrhoea[3, 4, 11, 12]

Patient risk factors	• Travellers with comorbidities such as diabetes, heart failure, bowel abnormalities, cancer, HIV or immunosuppression and older travellers are at increased risk. • Young children are more likely to have severe disease. • Travellers aged between 15–29 years and backpackers are at increased risk predominantly due to travel style.
Itinerary risk factors	• The highest risk to travellers is reported in South Asia, Sub-Saharan Africa and Latin America. • Most other developing countries pose a moderate risk.
Preventive measures and advice	• Preventive advice for travellers: ○ hand hygiene with soap and water or antibacterial hand wash before eating and food preparation ○ careful food and drink choices. • Travellers' diarrhoea is generally self-limiting and lasts between 1–3 days. • Rehydration is the mainstay of therapy for travellers' diarrhoea: ○ recognising symptoms of dehydration, appropriate fluid intake, the use of oral rehydration products, diet of light, carbohydrate-rich foods until recovered ○ symptoms may persist for protracted periods with pathogens that exist in chronic phase (e.g. Giardia). • Early self-treatment with standby medication is preferred to antimicrobial prophylaxis for most patients due to the risk of antimicrobial resistance. Travellers at risk of severe disease who require antimicrobial prophylaxis, separate to standby treatment, must be referred to an appropriate healthcare provider.

Pharmacotherapy (standby treatment)	• No treatment is required for mild, short episodes. ○ Antidiarrhoeal drugs may be considered for self-treatment for adults and children >12 years (loperamide is preferred) who **do not** have bloody stools, systemic symptoms or fever. • Most cases of travellers' diarrhoea do not require antibiotics. Self-initiated treatment with antibiotics may be acceptable for acute, **moderate to severe diarrhoea only** (diarrhoea that is distressing, incapacitating, interferes with or completely prevents planned activities or dysentery (bloody stools)) and reserved for those who are at increased risk of complications, and those in whom dehydration may exacerbate underlying comorbidities **NB1** (12). • Stand-by emergency treatment and/or standby management of associated symptoms may be provided as part of a medical kit: ○ Antibiotics in accordance with the Therapeutic Guidelines: Antibiotic (Infectious diarrhoea - Travellers' diarrhoea) **NB1** (10). Recommended treatments include azithromycin, norfloxacin and ciprofloxacin as single-dose therapy. **and** ○ Antidiarrhoeal drugs in accordance with the Therapeutic Guidelines: Gastrointestinal (Acute gastroenteritis – Other supportive therapy and considerations for acute gastroenteritis - Antidiarrhoeal drugs for acute gastroenteritis) (11). **NB1**: The benefits of antibiotic therapy for travellers' diarrhoea must be weighed against the risks of adverse effects. • Antibiotic use may lead to acquisition of resistant organisms that may cause urinary tract infection, *Clostridioides difficile* infection, and post-travel irritable bowel syndrome. • Emerging resistance to quinolones, particularly in South Asia, should be considered.
Post travel	Persistent diarrhoea (>2 weeks duration) in returned travellers may have infective or non-infective causes and must be referred to an appropriate healthcare provider for investigation.

Reminder

The prescribing and use of antibiotics should be consistent with the principles described in the Therapeutic Guidelines: Antibiotic. Pharmacists should carefully assess whether antibiotic therapy is clinically indicated, and if so, consider local antimicrobial resistance patterns and relevant clinical factors when selecting therapy.

Malaria

In humans, malaria is caused by one of five species of Plasmodium parasites, most often transmitted through the bite of an infected mosquito, although transmission occasionally occurs through needle sharing, nosocomially or from mother to foetus [3, 4, 13].

- Illness usually occurs within a few weeks of the infection, although it may be delayed by several months [4, 13].
- Almost all cases of malaria in Australia are overseas acquired. However, parts of Northern Australia are malaria-receptive; the last local acquired case was reported in the Torres Strait Islands in 2013 [4].
- The risk of malaria can vary considerably between travellers and regions within the same country [3].
- Modifiable risk factors associated with malaria fatalities include inappropriate use (or no use) of chemoprophylaxis and delay in seeking care (diagnosis and treatment) [4].

The information in Table 4 is provided as a summary only. Pharmacists should consult relevant resources to provide tailored information for each traveller.

Table 4. Risk assessment, management and prophylaxis for malaria [3, 4, 13]	
Patient risk factors	• Travellers at risk of severe disease must be referred to an appropriate healthcare provider.
Itinerary/ destination risk factors	• For Australian travellers, the most common destinations of acquisition are India and Papua New Guinea, followed by Africa [4]. • Travellers visiting friends and relatives in endemic countries (risk may be underappreciated by these travellers). • Travel in malaria endemic areas for longer than 8 weeks.
Preventive measures and advice	• All travellers to countries where malaria is endemic should be provided with advice on mosquito avoidance and the risks of malaria, including: ○ prevention through use of effective insect repellent, other insecticide products and mosquito nets, remaining indoors from dusk to dawn where possible, wearing long trousers and long-sleeved shirts in the evening and avoiding strong fragrances ○ no chemoprophylaxis regimen is guaranteed to be 100% effective. Emergency treatment must be sought for febrile illness (even if stand-by treatment has been taken) ○ treatment for malaria may be available without a prescription overseas. However, counterfeit products are common.
Pharmacotherapy (prophylactic and standby emergency	• To determine whether chemoprophylaxis is required and the appropriate medicines to provide, pharmacists should consider:

treatment of uncomplicated malaria)	○ destination(s), including regions and seasons within endemic countries and current government recommendations including patterns of antimalarial resistance in the destination(s) ○ the individual's risk of mosquito bites ○ the individual's risk of severe malarial illness ○ the individual's risk of adverse effects and drug toxicity, and ability to swallow the prescribed prophylaxis ○ the individual's other medical conditions, allergy history, other medications ○ the potential for resistance to chemoprophylactic drugs ○ the cost and efficacy of each drug [3, 4, 13]. • If required, antimalarial chemoprophylaxis may be prescribed in accordance with the Therapeutic Guidelines: Antibiotic (Parasitic infections – Malaria) [13]. • Stand-by emergency treatment should be prescribed to people travelling to malaria-endemic areas who choose not to use chemoprophylaxis as part of a medical kit (for when emergency medical care cannot be obtained within 24 hours) in accordance with the Therapeutic Guidelines: Antibiotic (Parasitic infections – Malaria) [13].
Post travel	• All people who fall ill during or after travel and have persisting symptoms should be referred to an appropriate healthcare provider.

Reminder

The prescribing and use of antibiotics should be consistent with the principles described in the Therapeutic Guidelines: Antibiotic. Pharmacists should carefully assess whether antibiotic therapy is clinically indicated, and if so, consider local antimicrobial resistance patterns and relevant clinical factors when selecting therapy.

High altitude travel and altitude illness

Altitude is assumed to start at 1,200 m. Elevations above 1,500 m are considered high altitude [14]. While the highest risk of altitude illness is in environments >2,500 m above sea-level, it may also occur at altitudes from 1,500 m.

Normal physiological changes due to hypobaric hypoxia will occur in all people at high altitudes, including shortness of breath on exertion (normal if it resolves quickly on rest), increased urination, disturbed sleep with periods of hyperpnoea and dyspnoea at night [4].

Altitude illness describes a set of pathological syndromes that may be experienced when a person ascends to a high altitude environment too quickly without allowing for sufficient time to acclimatise [14, 15].

- Acute mountain sickness (AMS) (most common type) is characterised by a headache and at least one of gastrointestinal symptoms, fatigue or weakness, or dizziness/light-headedness.
- High-altitude cerebral oedema (HACE) (usually a progression of AMS but not always) is rare below 4300 m. It is a medical emergency and requires immediate descent [3, 14].
- High-altitude pulmonary oedema (HAPE) may occur on its own or with AMS and HACE. It is a medical emergency and requires immediate descent [3, 14].

The information in Table 5 is provided as a summary only. Pharmacists should consult relevant resources to provide tailored information for each traveller.

Table 5. Risk assessment, prevention and management of altitude sickness [3, 4, 14, 15, 16]

Patient risk factors	• There can be significant variation in how individuals with pre-existing conditions may react to exposure to high altitude. • Travellers with the following pre-existing medical conditions may be at increased risk of altitude illness and must be referred to an appropriate healthcare provider: ○ cardiovascular disorders (hypertension, coronary artery disease, heart failure, cardiac arrhythmias, congenital heart disease, recent myocardial infarction or stroke) ○ chronic kidney disease ○ cirrhosis ○ coagulopathy and people taking warfarin ○ cystic fibrosis ○ damage to carotid bodies ○ diabetes ○ gastrointestinal conditions (active peptic ulcer, active inflammatory bowel disease) ○ haematological conditions (iron deficiency anaemia, sickle cell disease) ○ neurological disorders (epilepsy, cerebrovascular disease, migraine) ○ obesity ○ ophthalmic conditions (radial keratotomy, LASIK) ○ Raynaud's Phenomenon ○ respiratory disorders (chronic obstructive pulmonary disease, bronchial asthma, obstructive sleep apnoea, pulmonary hypertension, pleural and interstitial lung disease) ○ pregnancy ○ psychiatric conditions. • The symptoms of altitude illness may impact other existing medical conditions (cough, disturbed sleep, cold related illnesses, photokeratitis, corneal thickening and retinal haemorrhages). • Children do not appear to be at higher risk of altitude illness, however, there is limited evidence for specific recommendations [14].

<table>
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- Children aged <8 years travelling to elevations >2,500 m should be referred to a suitable healthcare provider for specialist advice.
- Travellers with any of the above pre-existing medical conditions planning to travel to high altitudes (> 1,500 m) must be referred to a suitable healthcare provider for specialist advice and detailed risk assessment.

</td></tr>
<tr><td>Itinerary and environmental risk factors</td><td>

- Environmental factors that influence the risk of altitude illness include:
  - the speed and maximum height of ascent (risk of ascent)
  - sleeping elevation
  - strenuous exertion at altitude e.g., skiing or trekking [15, 17].
- Altitude illness is a risk in a range of common destinations for multiple kinds of travellers, including:
  - tourists with relatively common itineraries with elevations above >2,500 m above sea-level.
    - consider the potential for rapid passive ascent to high altitudes (driving, train travel, flying, hot-air ballooning, cable cars and ski lifts etc).
  - people who live below 900 m and engage in strenuous exertion or activities on arrival in high altitude areas e.g., trekking or skiing.

Low, medium and high-risk ascents for AMS (for unacclimatised individuals) [14, 17]

Low risk	Moderate risk	High risk
• no history of AMS ascending to ≤2,800 m • ascents taking ≥2 days to reach 2,500 m–3,000 m, with further increases in sleeping elevation to a maximum of 500 m per day and an extra day for acclimatisation for every 1,000 m of ascent.	• no history of AMS when ascending to >2800 m **in one day** • history of AMS when ascending to 2,500 m–2,800 m **in one day** • increases in sleeping elevation >500 m per day when above 3,000 m with an extra day for acclimatisation every 1,000 m.	• history of AMS when ascending to >2800 m **in one day** • history of HAPE or HACE (at any altitude) • ascents to elevations >3,500 m **in one day** • increases in sleeping elevation >500 m per day when above 3,000 m **without** extra days for acclimatisation • very rapid ascent e.g., ascent to areas of extreme altitude (>5,800 m) **in <7 days**.

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</table>

	Travellers with a history of altitude illness or at moderate or high risk of AMS must be referred to an appropriate healthcare provider for management.
Preventive measures and advice	• There is no guarantee of the absolute effectiveness of any preventative strategies; vigilance is required. • All people planning to travel to elevations above 2,500 m should plan for a gradual, low risk ascent to enable acclimatisation and prevent altitude illness, specifically: ○ take at least 2 days to ascend to 3,000 m ○ do not increase sleeping altitude by more than 500 m per day if above 3,000 m ○ allow an extra day for acclimatisation every 1,000 m of sleeping altitude gain. • If gradual ascent is not possible (e.g., flying into a high-altitude airport), take extra rest days before and after. • Advice should also be provided to the traveller regarding: ○ the importance of appropriate travel insurance ○ physical fitness, an appropriate medical kit, clothing and eye wear ○ travelling/trekking with an experienced guide at high altitudes ○ how to recognise the signs of altitude illness (AMS, HACE and HAPE) and that it can be potentially fatal, specifically: ▪ all symptoms at high altitude should be considered altitude sickness until determined otherwise ▪ seek medical care as soon as signs or symptoms are experienced ▪ never ascend to a higher altitude to sleep with symptoms of altitude sickness ▪ do not leave a person with suspected altitude illness alone ▪ always attempt to descend if symptoms are severe or worsen. ▪ other conditions that may be experienced at altitude including sun exposure, hypothermia and cold injury.
Pharmacotherapy (standby)	• Travellers at low risk of altitude illness generally do not require pharmacological prophylaxis. • General treatment for altitude illness includes stopping, resting and rehydrating; **the most effective treatment is descent**. • Oral standby treatment may be prescribed for adults for the initial management of altitude illness as part of a medical kit (with appropriate instructions for use in the event the person cannot seek immediate medical care or descend) to people at low risk of altitude illness travelling to elevations > 2,500 m in accordance with the Therapeutic Guidelines: Wilderness Medicine (Altitude illness) [15]. • Oral standby treatment may be prescribed for management of nausea and vomiting associated with altitude sickness for patients who cannot

	seek immediate medical care and/or descend, in accordance with the Therapeutic Guidelines: Gastrointestinal (Nausea and Vomiting) [18].

Venous Thromboembolism (VTE)

Venous thromboembolism is a rare, but serious, complication of long-distance travel. The information in Table 6 is provided as a summary only. Pharmacists should consult relevant resources to provide tailored information for each traveller.

Table 6. Risk assessment, prevention and management of VTE [3, 4, 19–21]

Patient risk factors	• In addition to long-distance travel, risk factors for VTE (that may indicate the need for pharmacologic prophylaxis) include: ○ previous history of VTE ○ reduced mobility (at least 3 days) ○ older age (≥60 years) ○ medical conditions including thrombophilia, rheumatological disorders, active malignancies, or heart or respiratory failure ○ chemotherapy or radiotherapy within the previous 6 months ○ recent trauma/surgery (within previous month) ○ obesity ○ pregnancy and postpartum period ○ ongoing hormonal treatment [4, 21]. • Travellers with one or more additional risk factors for VTE (as above) should have their risk estimated using a VTE risk calculator, such as the Padua Prediction Score for Risk of VTE and be referred to an appropriate healthcare provider if VTE prophylaxis is indicated. Patients at moderate to high risk of VTE must be referred to a specialised travel medicine clinic or their treating medical practitioner for specialist advice and detailed risk assessment.
Itinerary and environmental risk factors	• Long-distance travel (travel by air, road or rail of 4 hours duration or greater) has been associated with VTE, particularly air travel ≥8 hours, although the absolute risk is low [21]. • The risk of VTE may remain for up to 8 weeks post travel [21].
Preventive measures and advice	• General advice to minimise the risk of VTE for all long-distance travellers includes: ○ remaining ambulant before and after travel ○ regular movement during travel such as thigh and calf stretching exercises ○ ensuring there is adequate legroom wherever possible (by sitting in an aisle seat or emergency exit seat) ○ avoiding alcohol and sedatives before and during travel

	○ wearing loose clothing [4, 6]. • The use of graduated compression stockings is not routinely recommended for travellers at low risk of VTE, however they may reduce asymptomatic deep vein thrombosis and assist with preventing leg swelling [6]. All compression stockings should be professionally fitted [6].
Pharmacotherapy (prophylaxis)	• Prophylaxis options include aspirin, rivaroxaban, apixaban or enoxaparin [19, 20]. Travellers requiring pharmacotherapy for VTE prophylaxis must be referred to an appropriate healthcare provider.

Medical Kits

Medical kits should be considered by all travellers, including those visiting low risk destinations [4].

- The contents should be tailored to each traveller based on destination and patient-specific factors [4].
- The contents of a medical kit should be documented and include written instructions for use.

Things that should be considered for each traveller when compiling a medical kit include:

- documentation:
  - immunisation record
  - documentation of pre-existing conditions
  - copies of prescriptions and letter from the pharmacist for travelling with medications, including stand-by medications
  - travel insurance details
- basic first aid kit
- common over-the-counter medications:
  - analgesics
  - antacids
  - antidiarrheals
  - antiemetics and medication for motion sickness
  - antifungal cream
  - antihistamines
  - antispasmodics
  - cough suppressant or expectorant
  - decongestants (consider regulations in destination)
  - laxatives

    - hydrocortisone cream
    - rehydration products
- standby treatments for malaria, diarrhoea and altitude sickness (as appropriate) with instructions for use
- usual medications and medical equipment
- other preventative items, for example:
    - contraception
    - condoms
    - insect repellents and bed netting
    - water purification tablets
    - hand sanitiser and/or wipes
    - face masks
    - sunscreen.
- medical alert bracelet or other identifiers
- spare glasses or contact lenses.

Travelling with medications

People planning to take any type of medication or supplements overseas (including prescribed and non-prescribed medicines and medical equipment), particularly controlled substances, must confirm any restrictions and requirements/permits with the destination country's embassy or consulate.

- All medicines should be carried in their original packaging, clearly labelled with the contents, patient name and dose; travellers should be advised not to use pillboxes [3].
- Travellers may also consider carrying copies of their prescriptions and a letter from the prescriber detailing all medicines they have been prescribed, the reasons they have been prescribed, doses, and any necessary medical equipment including syringes. Translated versions may also assist [3].
- Medications should be packed in carry-on luggage and an extra supply should be included in the event the trip is extended [3].

Provide travel health education and counselling

Comprehensive advice and counselling (including supporting written information when required) should be provided to the patient regarding:

- individual product and medicine use, including dosing and indications
- non-pharmacological management
- how to manage adverse effects
- accessing medications, treatment and health advice

- seeking medical care and/or treatment post travel for ongoing issues.

Travellers should be counselled and provided with advice to enable safe travel and risk minimisation [4]. The topics and level of detail will vary between travellers, depending on the personal and itinerary risks identified during the pre-travel risk assessment [2]. Travellers should be advised to sign up for Smartraveller subscriber alerts for up-to-date information about health risks.

Common issues to be discussed for all travellers may include [2, 4]:

- emerging health issues and diseases relevant to the traveller
- appropriate travel insurance
- personal safety.

Other topics (as applicable) [4]:

- choosing safe food and drinks
- insect and other pest avoidance
- avoiding environmental exposures (contaminated soil and water)
- dog or other animal bite management
- injuries including aquatic and traffic accidents
- alcohol and drug use
- weather conditions and potential for natural disasters
- prevention of VTE
- availability of counterfeit medications or supplements
- safe sex practices
- blood borne infections (acquired through cosmetic/medical procedures, tattoos, piercings and intravenous drug use)
- altitude illness and the importance of planned gradual ascent
- staying safe in conflicts or situations of civil unrest
- motion sickness
- international driving licences.

It is the pharmacist's responsibility to ensure the suitability and accuracy of any resources provided to patients, and to ensure compliance with all copyright conditions.

Confirm management is appropriate

Consult the *Therapeutic Guidelines*, *Australian Medicines Handbook* and other resources to confirm the treatment recommendation is appropriate, including for:

- contraindications and precautions
- drug interactions

- pregnancy and lactation.

Clinical Review

Clinical review with the pharmacist post travel is not required. Patients should be instructed as to any required follow-up vaccination doses.

All patients should be advised during the pre-travel consultation to see their usual healthcare provider as soon as they return if they have health issues post-travel.

Pharmacist resources

- Therapeutic Guidelines:
  - Antibiotic: Travellers' Diarrhoea
  - Gastrointestinal: Antidiarrhoeal drugs for acute gastroenteritis
  - Gastrointestinal: Nausea and vomiting
  - Antibiotic: Malaria
  - Wilderness Medicine: Altitude Illness
  - Cardiovascular: Venous thromboembolism prophylaxis for long-distance travel
- Australian Medicines Handbook
  - Vaccines
  - Anti-infectives
  - Gastrointestinal drugs
- Centers for Disease Control and Prevention (CDC) Yellow Book 2026
  - Vaccination & Immunoprophylaxis: General Principles
  - Medication and Vaccine interactions in Travel Medicine
  - Yellow Fever Vaccine and Malaria Prophylaxis Information, by Country
  - Rabies Status: Assessment by Country
  - Travelers' diarrhea
  - High-Altitude Travel and Altitude Illness
  - Deep Vein Thrombosis and Pulmonary Embolism
- Australian Immunisation Handbook
  - Handbook
  - Vaccination for international travellers
- World Health Organisation
  - Travel Advice
  - Yellow Fever

- Faculty of Travel Medicine of the Royal College of Physicians and Surgeons of Glasgow: Good practice guidance for providing a travel health service
- Manual of Travel Medicine (Torresi et al.)
- Padua Prediction Score for Risk of VTE
- Wilderness Medical Society Clinical Practice Guidelines for the Prevention and Treatment of Acute Altitude Illness (2019 update)
- The Department of Health and Aged Care:
    - Immunisation for travel
    - Yellow fever – general fact sheet
- Smart Traveller
- Travel Health Pro:
    - Altitude illness
    - Country information

Patient resources

- Smart Traveller
- Travel Health Pro

Altitude illness

- Therapeutic Guidelines: Go slow and sleep low is the safe way to ascend
- Travel Health Pro: Altitude illness

VTE

- Thrombosis Australia: Travel and Thrombosis
- CDC Travelers' Health: Blood Clots

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Version 1.0	01.02.2024	
Version 2.0	01.07.2025	Updated to reflect current clinical guidance and improved usability across sections.

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